



Reaching carbon-neutrality in Amsterdam; A study of barriers to addressing residual emissions

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Abstract:

To limit global warming to 1.5°C, it is crucial for cities to achieve net-negative emissions by the first half of the century. Despite the urgency, current studies predict that the majority of cities, will not achieve net-zero emissions by 2050. The emissions left over in their budgets are known as residual emissions. This thesis investigates how the policy landscape in the Netherlands and the European Union may hinder Amsterdam's efforts to address these residual carbon-dioxide emissions and achieve its carbon neutrality goals.

The research discusses the municipality of Amsterdam's plans to mitigate residual emissions and examines the existing, incoming, and intended policies and mechanisms at the national and EU levels. This is achieved through a content analysis of the Amsterdam Municipality climate action plan, Dutch Policy, and EU policy, assessing both implicit and explicit extensions of these policies to cities. Additionally, semi-structured interviews with key actors in Amsterdam provide insights into the practical aspects of policy implementation.

This study contributes to the field by offering a comprehensive overview of the current policy framework and identifying critical needs for municipalities to implement Carbon Dioxide Removal (CDR) strategies effectively. It highlights gaps in policy and legislation at both national and EU levels and provides recommendations for bridging these gaps through policy adjustments.

In conclusion, the policy environment is found to be extensive and complex but insufficient in its current form. Key gaps include inadequate policies for integrated infrastructure development, biomass availability, financial support, and the limited influence of Amsterdam on the implementation of CDR and policy development. Despite these challenges, the city has opportunities to effect change and contribute to the broader goal of climate change mitigation through horizontal interactions with the multi-level governance structure. This thesis underscores the need for more stringent policies and a cohesive approach at various governmental levels to support cities like Amsterdam in their crucial role in combating climate change.

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Chapter 1: Introduction

The European Union (EU) announced in 2021 that it will be the first 'carbon neutral continent' by 2050 (Leyen, 2020). The move marks a strengthening of its own ambition in response to pressure from environmental groups, unfolding geo-political events and growing evidence of the consequences of inaction. In 2021, the fit for 55 policy package was approved by EU parliament increased emissions reduction ambition from 40% by 2030 to 55% for the entire continent (Erbach et al., 2022; European Parliament, 2022). Included in the package was the establishment of other significant policy instruments, and amendments to strengthen existing legislation. However, despite the increased ambition, the EU's commitment to emissions reductions is still short of what member states have collectively committed to under the Paris Agreement (Climate Action Tracker, 2023). Current policy and action will lead to an additional 2000 MTCO_{2eq} / year (Millions of Tons of carbon-dioxide or equivalent per year) above the 1.5C of warming committed to under the Paris agreement. This is a significant challenge for the union as its emissions continue to grow (Friedlingstein et al., 2022; UNEP, 2022; UNIPCC, 2022).

The greatest portion of the European union's greenhouse gas (GHG) emissions are attributable to cities. In total, urban centers are responsible for 72% of the world's carbon equivalent emissions (Kovac et al., 2021). This figure includes emissions tied to energy, and material use and waste that at some point in their lifecycle are part of activities in a city (Jim Penman et al., 2006). Without coordinated action this share will increase as urban centers continue to grow from 50% in 2020 to 58% in 2050 (UN-Habitat, 2022a, 2022b). If urban centers cannot decouple spatial and population growth from emissions growth the effects will exacerbate already worsening climate induced disasters that disproportionately cities in the global south. Cities need to be front and center to the mitigation and adaptation measures to address climate change. However, cities have struggled to prioritize reducing emissions over other more pressing issues. For effective action, cities especially smaller ones will require supported from the governance structures around them (Reckien et al., 2018).

In response to inaction, a growing movement of bottom-up action by local activists and municipal actors is developing into coordinated action at the urban scale. A growing number of cities globally are becoming central actors in climate action and governance through independent action and ambition (Kern, 2009, 2019; Pincetl, 2017) by committing themselves to progressive and radical emissions reduction plans and sustainable transition pathways with emphasis on a just transition. This may due to the intense personal contact that local governance (in this case the municipality of Amsterdam) has with local actors and so creates powerful social norms (Flatt, 2007). Proximity of local governance to society makes them best suited to directly address citizens demands and operationalize action at the local scale (UN-Habitat, 2020). These factors have led cities to become embedded in multi-level governance structures with national, supranational, international, and non-governmental actors.

Following suit, Amsterdam has committed itself to reaching carbon-neutrality by 2050 aligned with targets by the Dutch national government and the European Union (Gemeente Amsterdam, 2022). The aim is to reach 95% reduction in emissions as compared to 1990 levels, such that total emissions are virtually zero, or 'neutral' (Gemeente Amsterdam, 2022). To reach this target the city emphasizes the role of emissions reduction measures and carbon capture technologies as main contributors. While necessary, emissions reduction in transportation and industry particularly those tied to the local economy and energy procurement will be difficult (Kern, 2019;

Priester Roland, 2014). This points to the potential risk that Amsterdam may not be able to reach 95% reduction by 2050, or that it may with no plan to address the residual 5%.

The primary solution proposed for cities to address residual emissions is Carbon dioxide removal (CDR) and Carbon capture and sequestration (CCS). These two terms often confused refer to two distinct concepts where CDR refers only to the removal of carbon from the atmosphere, and CCS is both the capture and storage in stable forms. CCS is primarily used in the context of heavy industry whereas CDR can refer to removal by many means including from growth of forests. Both processes will be required in the future depending on emissions reduction action taken now (Smith et al., 2023; UNIPCC, 2022).

The feasibility and total capacity of CDR's are variable, depending heavily on enabling support by multiple-scales (Huovila et al., 2022; Kreibich & Hermwille, 2021; Service, 2021; Tamme & Beck, 2021). Under all pathways, limiting warming to 1.5C modeled by the IPCC (UNIPCC, 2022) CDR deployment is modeled with total storage ranging between 450 to 1,100GtCO₂/ per year. Globally, only 1.5 – 3.5GtCO₂/year of CDR has been pledged so far, indicating a substantial gap in planned action (Smith et al., 2023, pg 86). If the goal is a stable climate, the EU, the Netherlands, and Amsterdam will need to accelerate action to establish robust CCS capacity.

This study explores this potential rift and discusses Amsterdam's plans to address residual carbon emissions and the policy environment surrounding these technologies. Amsterdam is a valuable case to study for three reasons. First, because it is leading as a city for climate ambition with a roadmap to reach 95% GHG emissions reduction by 2050. Additionally, the cities strategy includes CDR as action toward neutrality and they are likely to run into issues in CDR implementation sooner than laggards. Second, lessons learned from Amsterdam may be able to support laggards in terms of action and ambition as they may rely more heavily on CDR and CCS technologies. Finally, because it is an opportunity to examine how local actors interact with national and intranational legislative processes in the European union.

This thesis will be structured as follows. First, I outline my research question and important concepts and terminology in the CDR and CCS field and multi-level governance. I relate them to the study of innovation policy to develop a conceptual framework for categorizing the policy landscape. Second, I will examine the municipality of Amsterdam's climate action plan and how much CDR it will require. Then, I will explore the policy landscape of the EU and the Netherlands relevant to CDR and CCS from Amsterdam's perspective. With these results I will identify gaps between Amsterdam's needs and what the policy environment supports. Finally, I will discuss potential solutions to some of these gaps if they exist and conclude the discussion with an eye to future research.

1.1 Research questions:

To support the research's internal objective, I develop research questions to focus research activities:

RQ: How do policy barriers at the national, and EU levels affect Amsterdam's ability to address residual CO₂ emissions in its carbon budget through implementation of Carbon Dioxide Removal technologies.

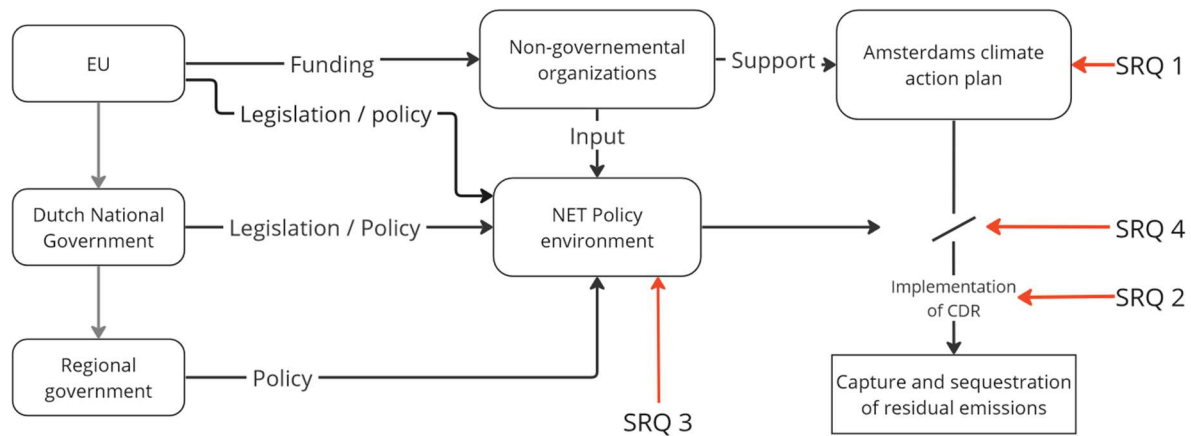


Fig 1. Research questions

To operationalize the concepts of the main research question, I deconstruct it into three sub-research questions. The questions are as follows:

SRQ 1: *What are Amsterdam plans to address residual emissions in its carbon budget?*

Amsterdam has recently published (2020) several documents regarding its strategy, and vision for the city. Included is a climate action roadmap communicating its goals and plan of action to reach carbon neutrality. I examine these plans, and ask how they include CDR technologies?

SRQ 2: *What can Amsterdam do to implement its plan to address residual emissions in its budget?*

If the city intends to address its residual emissions, it will require planning and implementation across actors and scales of governance. The city must allocate resources and initiate projects and policies to support those approaches. What are those needs, and can Amsterdam implement them?

SRQ 3: *What policies at the municipal, national and EU level are relevant to Amsterdam's climate action roadmap?*

Climate change is highly complex due to its interconnected nature that spans spatial boundaries and is only more so by including the social dimension. This arrangement requires partnership and cooperation across scales to solve, but the complexities of multi-level governance may create barriers where support was intended. Identifying the barriers is the first step to proposing a solution.

SRQ 4: *How do policy at the municipal, national, and EU level conflict with Amsterdam's climate action roadmap?*

From readings of the policy landscape, barriers and gaps can be analyzed to determine severity and opportunities for solutions.

1.2 Theoretical Framework:

In order to answer the general research questions this paper explores the nexus of multi-level governance, carbon dioxide removal strategies, and innovation policy.

To understand policy, regulation and law being put forward regarding CDR it is first important to untangle the terminology and lay the theoretical groundwork. Much of the confusion surrounding the discussions stems from differing interpretations of the physical science and terminology (Climate Social Science Network, 2022). A reasoned discussion of policy action cannot happen if first there is no agreement to what is meant by carbon neutrality.

I outline the main actors involved in the European union through the lens of multi-level governance (MLG) including the main bodies of the EU, the Dutch government, and Amsterdam. Each level of government has differing goals, purposes, needs and tools afforded to it to achieve them. By understanding how each level operates and interacts the policy landscape can be understood not just as a list of laws but as a dynamic system. I combine this with a discussion of innovation policy to provide clarity on how this dynamic system affects innovation. Additionally, innovation policy will provide a base to define a framework for examining how the landscape does so. I can then identify how and where the MLG system and policy landscape create barriers for the municipality of Amsterdam.

1.2.1 Understanding carbon neutral and negative

Critical to carbon neutrality is the existing problem of emission reductions. Emission from sectors including transportation, heating and energy production is often the greatest portion of cities emissions (Ritchie et al., 2023). While shifting these sectors to net-zero is not guaranteed, there are a growing number of both green and gray solutions available to drive emissions reduction (Beceiro et al., 2022; Green Gray Community of Practice, 2020). Heavy industry, however, faces greater difficulty in reducing emissions as alternatives to heavily polluting fuels or processes don't exist or are not yet commercially available. These industries thus have emissions that are 'hard to abate' will require significant work to eliminate or may simply be impossible to eliminate (Oreane Edelenbosch et al., 2022). Emissions that cannot be cut are referred to as residual emissions (CSSN, 2022). Cities with industrial zones thus must share responsibility with those firms to reach carbon neutral through reductions or address the residual emissions in other ways.

To address residual emissions, first an accurate accounting of a city's emissions 'budget' as it is known, is needed. A growing awareness of the importance of system boundaries is present in the expansion of cities 'carbon-footprint' (Lombardi et al., 2017). A city's footprint is tied to three scopes. Energy production and production of materials and products that end up in cities defined as scope 2 and 3 emissions. Scope 1 only refers to emissions being emitted directly from activities within city limits. Scope 2 represents all emissions from energy & heating that go to cities and 3 from materials part of any activity in municipal borders but that have indirect emissions outside (Huovila et al., 2022). Thus, whether a city has reduced its entire carbon emissions budget depends on which scopes are considered in emissions accounting (see fig.2).

When an emissions budget is zero, this is known as net-zero or carbon-neutral. Carbon-neutrality (also referred to as net-zero) is defined as a balance between the emissions from and removal of anthropogenic sources of carbon over a period (a year) such that net emissions are zero (Climate

Social Science Network, 2022). Climate-neutral then refers to all GHG's instead of specifically carbon dioxide.

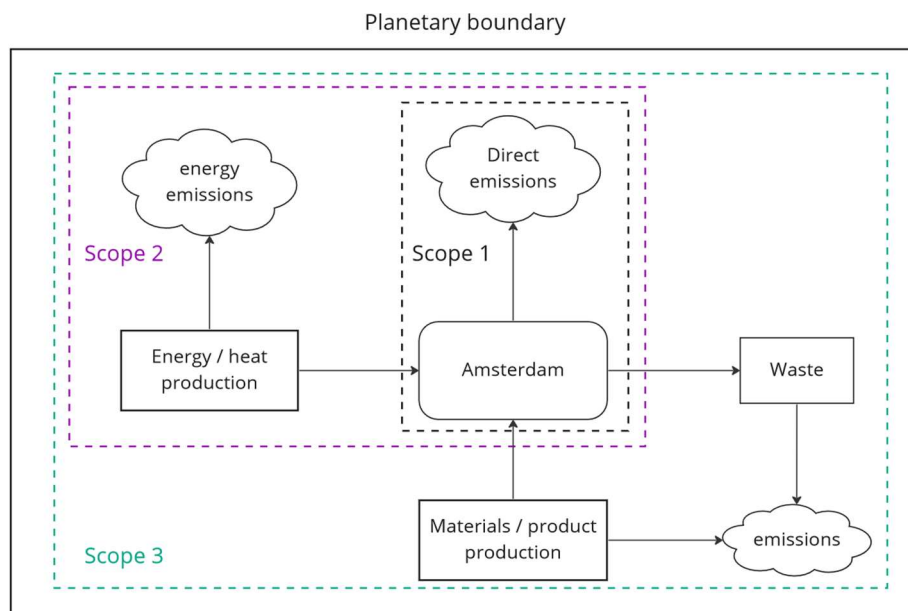


Fig 2. Emissions scope for cities (own elaboration)

A step further than neutral is net negative emissions which require that total stores of carbon dioxide in a system are net-negative after a time period (IPCC, n.d.). Achieving carbon negative in a system requires the use of Carbon Dioxide Removal (CDR) technologies also referred to as Negative Emissions technologies (NET's). CDR technologies are seen as a method for balancing these industries' output with carbon-dioxide removal. Three major forms exist: enhancement or establishment of natural carbon sinks, bioenergy with CCS or direct air capture of CO₂ from the atmosphere (DACCS).

Enhancement of natural sinks through human intervention aims to increase the carbon sequestration capacity of natural system such as forests, soil and wetlands through afforestation and other management techniques. The application of this categorization to forestry and agriculture is however highly variable as the capacity of the system may change due to climate change and other pressures (Agostini et al., 2014; Bright et al., 2020). There are also projections that use of these methods for CO₂ removal would require an unfeasible amount of land (Land Gap, 2022).

Anthropogenic sinks utilize human engineering to remove carbon-dioxide from a source (ocean and atmosphere) through natural or human direct chemical processes (Tanzer & Ramírez, 2019). Examples of this form of CDR include enhanced weathering, ocean capture and sequestration and direct air capture and sequestration (DACCS). Proponents argue that CDR could potentially become a substantial method for creating negative emissions (European Commission, 2018; IPCC, 2022; Pachauri et al., 2014). However, the technology faces steep cost, a significant learning and implementation curve, and public obstruction (Climate Social Science Network, 2022; Minx et al., 2018; Tanzer & Ramírez, 2019). Last, is Bio-energy carbon capture and sequestration (BECCS). This method utilizes the natural ability of plants to capture and store carbon, which is then burnt to

produce energy. The carbon is captured as an output of energy production and stored to create neutral or negative emissions. The removal is only negative if it removes more CO₂ than emitted over the timeframe, most often a year. Otherwise, this is known as carbon-removal offsets (CRO), as it simply balances an equivalent amount of carbon emitted (Climate Social Science Network, 2022).

Capture from industrial sources is often referred to as Carbon Capture and Sequestration (CCS). Referring to capture directly at the source of emission, for example in the smokestacks of an aluminum refinery. If this carbon-dioxide, then becomes input for a different process it is called captured carbon utilization. Another term for this is carbon capture, sequestration, and utilization (CCSU) referring to the two options for managing the captured carbon. These technologies are distinct as they have been specifically designed for use in industry (Service, 2021). When deployed they can contribute to negative emissions, but only under specific circumstances.

<i>Name</i>	<i>Technique</i>	<i>Potential Permanence</i>	<i>Potential storage (Gt per year)</i>
<i>Afforestation & reforestation</i>	Planting and enhancing of forest ecosystems to store carbon	Medium term (hundreds of years)	1-1.8 Gt
<i>Artificial upwelling</i>	Moving deep ocean waters to increase carbon intake	Long-term (thousands of years)	Unclear, some resources cite 20 Gt by 2100
<i>Biochar</i>	Burning biomass in a low-oxygen system and adding to soils for storage	Medium-term	0.3 – 35 Gt
<i>Bio-energy with Capture & storage (BECCS)</i>	Burning biomass and capturing biogenic CO ₂ for storage (BECCS)	Permanent (thousands of years)	1 – 85 Gt
<i>Soil carbon sequestration</i>	Land management which increases carbon content in soils	Medium term (depending on land use)	1 – 11 Gt
<i>Direct air-capture and storage (DACCS)</i>	Capture of carbon from the atmosphere using various techniques for storage (DACCS)	Permanent (thousands of years)	0.5 – 5 Gt (in 2050)
<i>Enhance ocean alkalinity</i>	Enhancing ocean surface alkalinity to increase Carbon intake	Long-term	3,500 Gt
<i>Enhanced weathering</i>	Adding minerals to large land areas that react with atmospheric Carbon to sequester it	Long-term	0.72 – 92 Gt
<i>Macroalgae cultivation</i>	Cultivation of Algae for bio-genic CO ₂ for storage	Potentially long-term	Maximum ~34 Gt

<i>Carbon capture and storage by Mineralization (CCSM)</i>	Storing Carbon in alkaline rich mineral by-products for storage	Long-term	Potentially 27.5 Gt (Sanna et al., 2012)
<i>Carbon capture from oceans and storage (OCCS)</i>	Separation of carbon dioxide from ocean water for storage	Permanent (thousands of years)	Data not available
<i>Ocean fertilization</i>	Use of iron to encourage plankton growth to store carbon from ocean surfaces	Potentially long term	3.7 – 44 Gt. Recent reports cite lower ranges.
<i>Ocean fertilization with macro-nutrients</i>	Use of nutrients to encourage plankton growth to store carbon	Potentially long-term	Uncertain
<i>Ecosystem restoration</i>	Restoration and enhancement of ecosystems with high carbon storage capacity (wetlands & seagrass)	Medium to long-term	Uncertain. Depends on additionality of carbon capture and sequestration.

Table 1. CDR processes under NET's. source: (IPCC, 2022; Samaniego et al., 2022) & own elaboration

In discussions and actions on reaching carbon-neutrality a codified definition of what constitutes real negative emissions is lacking. A major sticking point is what is included in rigorous carbon accounting (European Commission, 2021a) as without such a framework there is risk of double-counting, under-counting, internal leakage, and misattribution of carbon flows and stores. A strict framework for accounting is required to reach more targeted emissions reductions and net-negative emissions overall (Yin et al., 2022).

Tanzer & Ramirez (2019) define four requirements for negative emissions to be accurately labeled as such. One, physical greenhouse gases are removed from the atmosphere. Two, the removed gases are stored in a manner intended to be permanent. Natural cycles of carbon release, capture and storage exist on a spectrum from short (day to day) to geological time frames (millennia). It is crucial that any removal of CO₂ can store the gas in a stable form over geological timescales. Three, all upstream and downstream emissions within a defined system are accounted for. To cities this means that all emissions tied to inputs (upstream) and end of life processes (downstream) are included in balance calculations of scope 1 and 2 emissions. When both aspects are considered, it is known as a "cradle-to-grave" system. This is critical as poorly defined boundaries or calculations can lead to misallocation of carbon removals. Four, the balance of emissions is net-negative, such that in total sequestered emissions are 'additional'. Only when these 'minimum requirements' are met then there can be considered negative emissions.

There exists a gap between the conceptual solution and implementation of carbon-dioxide removal technology. To date there has been no full-scale implementation of any carbon-dioxide project capable of substantial removal and storage (Scott & Geden, 2018), though this is rapidly changing (Global CCS Institute, 2022). Despite this, cities emphasize these technologies as a

major pathway towards carbon-neutrality (Samaniego et al., 2022). National governments and the EU support this ambition, but policy instruments to support its development and implementation are lagging (Scott & Geden, 2018; Tamme & Beck, 2021).

1.2.2 Multi-level governance:

Multi-level governance, developed by Hooghe et al, (1993) is a conceptual framework used to describe the particularities of how governance evolved in the EU post Maastricht Treaty. It provides descriptive power for how the European union's governance has organized itself snapshots across time. Hooghe identified that the EU was characterized by a growing network of actors, a decentralization of responsibilities and competencies from national to other levels and divided to perform more functions at greater efficiency (Bran et al., 2019; Hooghe et al., 2003).

The European Union as an entity was formed by the treaty of Maastricht ratified in 1992, defining its 3-pillar structure and principles of operation in the *Treaty of the European Union* (TEU). This was later changed in the treaty of Lisbon (2009) to its current structure. It is comprised of the EU Commission, EU Council, Parliament, European Court of Justice, and European Central Bank along now many smaller bodies. The EU acts by right of the *Treaty of the functioning of the European Union* (TFEU) ratified in Rome in 1957 and is currently in force under the Lisbon treaty (2009). It sets out the competences of the union through the principle of conferral, internal markets, and monetary and fiscal policy (Consolidated Version of the Treaty on European Union, 2012).

From the TFEU, the union is able to exercise significant coercive and legal powers to define strategies and coordinate action across the local, regional, national, supra-national scale (e.g., the EU). It can also be argued that it has influence on a 5th, international scale through networks and organizations such as the United Nations. As defined in the TFEU the European Commission (EUC) acts as the implementing arm of the EU, the EU Parliament (EUP) as the administrative arm, and the Council as the monitoring body of the Parliament and Commission. Together the EU is empowered to enact binding instruments including directives, regulations, budgets for various sub-bodies, and act as a singular entity to negotiate treaties with international bodies. Additionally, the EUC or EUP may publish non-binding communications, recommendations of action to member states, visions, and guidelines.

Following the principles of decentralization, conferral, subsidiarity, proportionality, and partnership between levels (Bran et al., 2019) to develop and implement shared goals, spatial planning is located at the local level within member states of the EU. Decentralization is an arrangement where actors in a system occupy different positions (in hierarchy, geospatial position, goals) but communicate to create coordinated action. Coupled with subsidiarity, which refers to the principle of responsibility being transferred to the lowest level, sub-national actors thus become responsible for operational action on a wide variety of topics (Consolidated Version of the Treaty on European Union, 2012). This principle is based on the notion that an issue is most efficiently handled at the lowest governance rung until it requires significant coordination by a level above it.

The Netherlands operates on a parliamentary system similar to that of the United Kingdoms (Government of Netherlands, 2022). Three main bodies exist: the parliament, government, and legal system. The administrative aspects of governing are organized and implemented by ministries (ex: Ministry of the interior and kingdom relations) and agencies. They are empowered by the constitution to adopt laws and define responsibility according to the same principles as

the EU, passing responsibility and decision-making power to sub-government where appropriate. Two levels of administrative division are made, first to regions (ex: Noord-Holland), and then Municipalities (ex: Amsterdam). Under these principles, urban and regional planning has remained the responsibility of the municipal which sits closest to local affairs.

Cities interact with the policy environment they are embedded as active participants in two directions shown in Fig 2 (Hughes et al., 2018; Kern, 2019). Legislation set by the sub-national and national government poles are known as hard governance as they are legally binding by the state's judicial institutions. For example, the target of carbon-neutrality by 2050 is as of 2021 is a legally binding target by the EU and by extension all member states (European Parliament, 2021). Another form of vertical governance is soft governance and refers to the use of subsidies or other incentives to influence actors. The practice of *co-funding* in the EU supports environmental action for items that might otherwise not be done due to costs (Kemmerzell, 2018). Allowing EU bodies to work with national government to encourage action by local government if national government cannot provide that support. Cities can also influence and be influenced by soft-governance approaches. Networks like the Covenant of Mayors are soft governance in some regard in that financial incentive is built into the initiative. However, it is more accurate to use the term horizontal governance as these organizations are not public institutions (Hughes et al., 2018; Kemmerzell, 2018). Such organizations are a collection of actors organizing around a collective goal and facilitate the exchange of knowledge to improve the coordination of local actors on connected issues like mobility, gender equality, and environmental management (Nielsen & Papin, 2020).

Horizontal networks can connect actors across levels of governance and non-state actors including knowledge centers and independent actors, creating a 'horizontal' hierarchy that they interact on. Horizontal networks like the covenant of mayors have been shown to increase commitment and action by local governance on environmental issues through several mechanisms (Bellinson, 2018; Huovila et al., 2022; Kemmerzell, 2018). Most important is policy learning, where actors at the city level can connect with knowledge networks (Cortes et al., 2022). In addition to *safeguarding* where membership of a city network may confer professional status or place branding to the local government. The ultimate result being increased compliance of local government to wider environmental goals. Additionally, networks like the G4 municipalities provide a platform for cities to voice their concerns and lobby to the EU directly making cities critical actors in these networks (Kundu et al., 2020)

Several factors have contributed to the rise in prominence of cities in goal setting and climate action. They include proximity to society, the growth of city networks, and advocacy by global agreements like the Paris Agreement (Paris Agreement, 2015; Hughes et al., 2018). This sentiment is exemplified by the growing perception that cities influence people and processes beyond their municipal borders (Tosics, 2011) where their 'economic' and 'environmental' footprint is larger and affects more than the 'physical' city. This has resulted in a growing body of literature exploring how the cities and the networks they have created have developed into new models of multi-level governance.

A central component of urban climate change action is thus the political and jurisdictional complexity that shapes decision making between the diverse actors and its outcomes. In the EU in particular, cities must navigate the multiple scales and processes of policy making at the national, supra-national and international scale (Kemmerzell, 2018). In addition, they must act on issues that are interdependent, interconnected and often trans-boundary. For example, solid

waste disposal from a city which affects tourism in a national park nearby. Deliberations about such an issue would involve public and private actors on local and national scales.

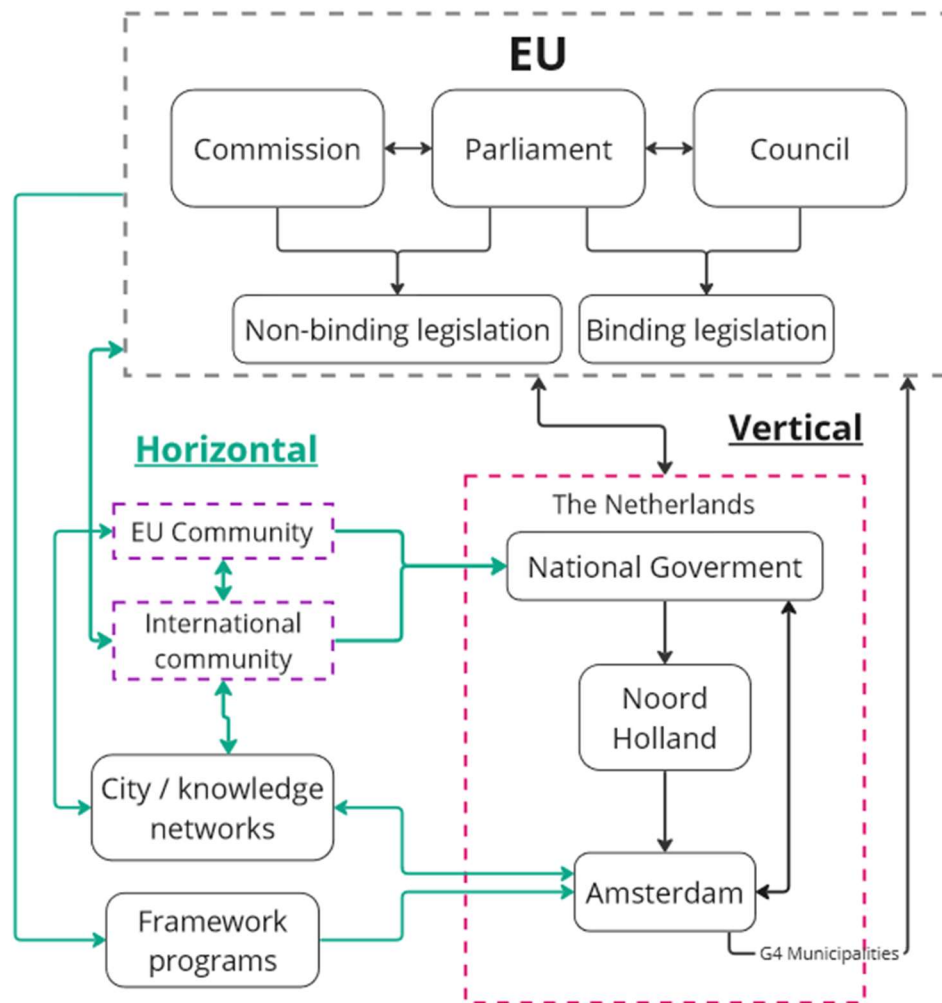


Fig 3. Multi-level governance in the EU

In the Netherlands, municipalities enjoy a significant degree of autonomy and independence from regional and national government structures (Koninkrijksrelaties, 2021, 2023). They are involved in governance both vertically, up through the region of Noord Holland, regional organizations, national Dutch government, and the EU legislation. As well as horizontally, across municipalities in the region and with private and non-governmental actors like the AMS institute and the Climate Neutral Cities Association (Bran et al., 2019). Amsterdam also self-organizes with other actors to discuss issues directly at the national and EU levels. Municipalities have independence to set their own climate action plans, though they are required to develop and enact them by national law. Amsterdam aims to achieve 55% emissions reduction by 2030, and 95% reduction by 2050 (Climate Neutral City Association, n.d.; Gemeente Amsterdam, 2020). If these targets are met, 5% of the budget will remain as residual emissions which will necessitate the use of carbon-dioxide removal technologies (CDR) to reach carbon-neutrality.

1.2.3 Innovation policy:

Innovation policy research is a broad field generally aimed at describing how innovation systems operate and impact the quality and speed of innovation. This field of research explores the relationship between public policy actors and the diffusion of policy innovation, and its effects on the development and adoption of innovative technologies (Cohen & Amorós, 2014). In his broad definition, Lundvall (1992) includes economic structures, its supporting fields, and the interactions between all the elements (Schot & Steinmueller, 2018).

A Policy (or policy instrument) is best understood as a set of guidelines or rules of operation within a legal framework set by a governing entity. They are formed as part of the legislative cycle of an entity where input may be taken from stakeholders and supplemented with research and or larger policy goals (EU Commission, n.d.). Policies can be either legally binding, which oblige requirements onto an activity/product/entity or non-binding which often take the form of recommendations and reports. For example, within the legal system of the European union, a directive is legally binding requiring nations to adopt the policy. Whereas opinions simply state a stance of the EU commission and require no action by other actors implicated in it. Within the limits of member nations, policy can affect almost all sectors including the environment, trade of animals and the use of carbon dioxide removal technologies.

Borrowing from sustainable transition management research the field utilizes socio-institutional approaches and macro-economic theory in addition to structural analysis to understand the interactions of the elements within. Socio-technical systems as it is known in transition study is predominantly interested in how innovation and infrastructure influence transitions (Bergek et al., 2015; Loorbach et al., 2017; Wieczorek & Hekkert, 2012). Three other equally important systems are recognized: Socio-ecological, socio-institutional, and socio-political. These approaches are defined by the conceptual basis from which their perspective stems. Socio-Ecological focuses on concepts derived from ecology such as tipping points and complex interactions. Whereas socio-institutional approaches focus on studying the role of agency and governance in transitions (Loorbach et al., 2017).

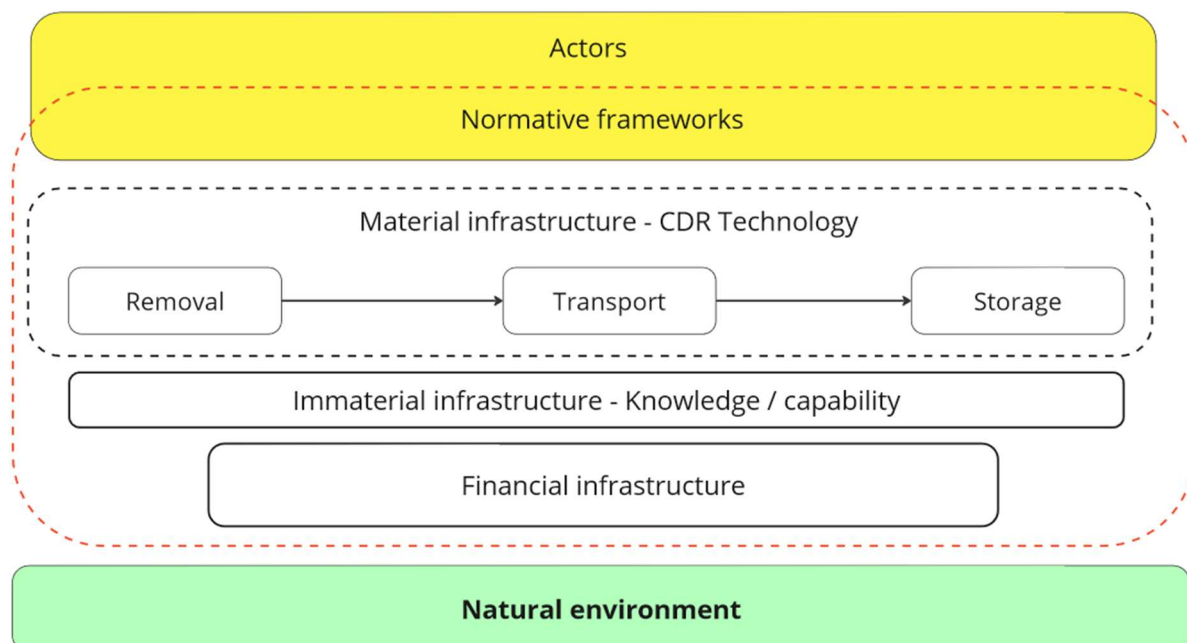


Fig 4. Components of the CDR technological innovation system (Wieczorek & Hekkert, 2012 & own elaboration)

Seen as a whole, systemic innovation conceptualizes the collection of elements, processes and actors as a technological innovation system (TIS) encompassing all the above elements as a unit

(Carlsson & Stankiewicz, 1991). In the case of the EU applied to CDR technology I visualize the above elements of that system. Infrastructure, outlined in black and separated into three components are governed by Normative frameworks (in red). Actors, as networks, individuals or as part of the government influence and define the institutional rules for the system. This all sits above the natural environment which it aims to protect. Any issue with the system which may slow adoption and diffusion are seen as systemic failures or weaknesses. An example of a systemic failure may include poorly articulated demand by incumbents or lock-in problems with existing technology. Klein-Woolthuis defines four general categories for systemic failures: infrastructural, institutional, interaction and capability problems (Klein Woolthuis et al., 2005).

Innovation policy and sustainable technology diffusion categorizes institutional instruments into two categories: supply-side & demand-side. While this approach comes from a neo-liberal perspective of markets it provides a useful framework. The effectiveness of these instruments is generally measured from an economic perspective in how they impact economic growth (Jacobsson & Johnson, 2000). The concepts of supply-side and demand-side are borrowed from macroeconomic theory which defines supply and demand as the two major components of an economic system. Supply being defined as the stock of goods and services available, and demand is the need in the system for those goods and services. Policy instruments are thus designed to either support adoption and innovation by increasing demand, defining new functional requirements, or better articulating the demand of new goods or services (Edler & Georghiou, 2007).

Examples of some supply side and demand side instruments include:

Supply side	Demand side
Innovation hubs / Incubators	Procurement plans
Government R&D grants	Regulation
Loosening of regulation	Incentives (rebates, waiving fee's)
Infrastructure investment	Voluntary certificates / labels
Equity financing	Consumer subsidies
Tax breaks & Grants	Awareness & knowledge campaigns

Table 2. Supply & demand side instruments

Historically, policy innovation and research regarding CDR technologies has been focused on supply-side instruments (Minx et al., 2018; Nemet et al., 2018). The goal being to encourage R&D, experimentation, and technology progress to quickly bridge the gap to large-scale and cost-effective implementation. However, no one policy or instrument can address field completely, and there is need for a suite of policies that are aligned to a common goal (Smits & Kuhlmann, 2004). The field has begun to investigate the effectiveness of demand-side instruments such as certification mechanisms, regulation, and public acceptability. Interest in demand-side instruments is not limited to CDR as this trend can similarly be found in the field of the energy transition with innovative procurement initiatives for demand-side solar and storage (Sharifi et al., 2017). These instruments change socio-political elements in a regime and encourage innovative technology implementation and adoption by changing perception and structural elements (Edler & Georghiou, 2007). This highlights co-benefits and co-costs for the socio-political aspect of CDR adoption and implementation and contributing to its adoption overall.

The mainstream use of this technology is not a given however and researchers have pointed out several policy and systemic failures present in the system. Issues run the gamut from public distrust of the technology (Cox et al., 2020), missing regulation regarding carbon accounting (Tamme & Beck, 2021), a physical lack of near-term storage options for storing carbon (Samaniego et al., 2022) and high cost of operation limiting commercialization (Scott & Geden, 2018; Tamme & Beck, 2021).

1.3: Conceptual framework

To identify how those elements are affecting CDR adoption and implementation I propose a framework for categorizing CDR policy instruments. I adapt frameworks by Ostrom (2005), Cambell & Coreley (2010) and Wieczorek & Hekkert (2012) from innovation policy, and urban policy studies. The aim of the framework is to present a methodology for categorizing policy and examining its goals, mechanisms, and interactions with other policies in the landscape relevant to a specific technology. In this way allowing me to examine how the policy landscape will or does affect the adoption of that technology. In this case I apply it to understand how the multi-level landscape of CDR relevant policy affects Amsterdam's use of CDR to address residual emissions.

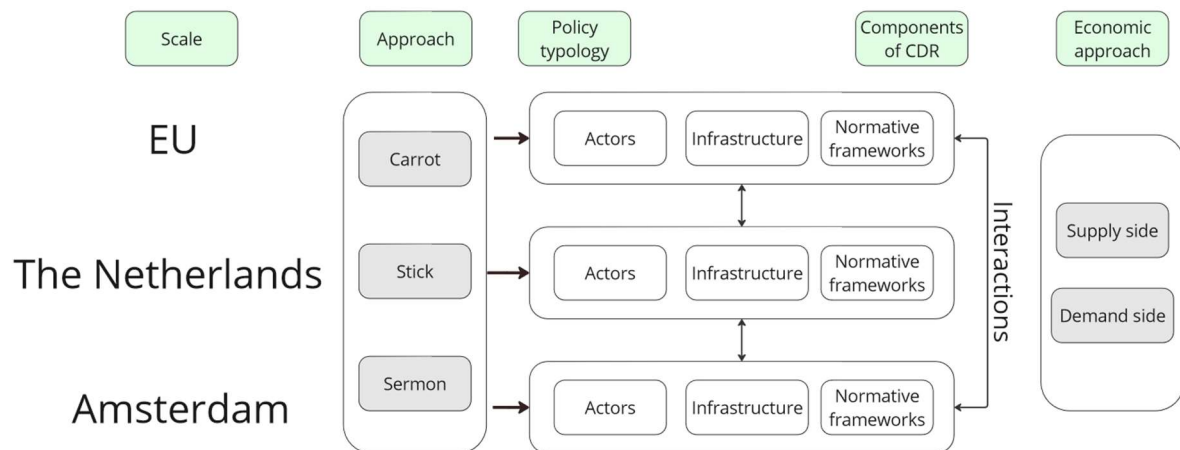


Fig 5. CDR policy/instrument framework

The framework (fig X) identifies 3 layers by which policy can affect a technology (Actors, Normative frameworks, and infrastructure). Actors refer to three scales of human organization: as individuals, working together as organizations (voluntary or legal) and networks of them. While Wieczorek & Hekkert delineate by the actor's role in the economic activity this framework is interested in their relation to other process in the policy landscape, including legislation, and advocacy. Normative frameworks have two categories: informal and formal. Where informal frameworks are established practices, culture and norms, aspects that agreed informally by actors in the system. Formal normative frameworks refer to legally binding policy and legislation. Both informal and formal frameworks can be nested and thus be present at the local, national and intranational scale.

Under infrastructure there are 3 categories. Material, Immaterial and Financial. Material infrastructure refers to the physical components or elements that would remain should all humans suddenly leave (Campbell, 2010; Campbell & Corley, 2015). Carbon dioxide removal technologies require 3 main infrastructural components: A process or method of capturing carbon from a source, transport, and storage. Operating these technologies requires knowledge, capacity, and skills from actors in the network which is defined as immaterial infrastructure. The knowledge may not be contained only in operating firms, but also within government, universities, and non-governmental organizations. Financial infrastructure, defined by Wieczorek and & Hekkert (2012) constitutes the third component, referring to economic support mechanisms including subsidies, grants, tax waivers, etc.



Fig 6. Breakdown of categories

Policies are categorized by origin scale into the Municipal, National, and EU levels. This distinction is interested in at what scale it operates at with the highest scale being the defining category. For example, the CCS directive is transposed into national law, but as the EU is the origin of the policy and it applies to full EU member states that is the category. I will focus only up to the intra-national (EU) scale and ignore how the EU and member states interact with other nations and international organizations.

CDR Policy is then categorized by approach (supply vs demand side) and mechanism (Carrot, stick, sermon). Where supply and demand refer to how the policy aims to create change in the market; through suppliers of a technology or actors who may need the technology. This classification will allow me to understand if and how policies are affecting the market for CDR which depends on a favourable market to reach widespread adoption (Scott & Geden, 2018). Similarly, the carrot, stick, and sermon typology of policy (Bemelmans-Videc et al., 1998) distinguish between regulatory, economic, and communicative instruments. It simplifies a complex discussion of instrument choice and the target actor and continues to contribute to the study of policy more than 20 years after its inception (Pacheco-Vega, 2020).

I suggest 4 layers by which the instruments can create systemic failure in the system (barriers and gaps) by examining components of the system (actors, normative frameworks, infrastructure, interactions) at different scales. These categories correspond to the same elements above, though, I add the interactions component here to involve the dynamics of multi-level governance. In the interactions the framework is interested in which party is initiating the contact and for what reason.

Using the framework, I will define the boundaries of the CDR technologies policy and innovation system spatially, temporally, and conceptually. Spatially, the system is limited to within Amsterdam's municipal borders at the local level, the Netherlands sovereign borders at the national scale, and the interconnected 'Schengen area' of the EU. The Schengen area refers to a legal agreement between the 27 member states of the EU to keep open internal borders. I however do not consider Lichtenstein, Norway, Iceland and Switzerland in the EU scale as they are not EU members and so not required to transpose all laws. I temporally use the public repository of in progress EU policies on the 'legislative train' (European Parliament, 2023a). This means that I consider policies and laws that are either in force today, or scheduled to come into effect and are tracked by the EU's legislative process reporting mechanism. For the Netherlands and Amsterdam, I will consider strategy documents but only refer to in force policy as the relevant legal documents. Conceptually, the policies I consider must be relevant to components above (actors, infrastructure, frameworks) of carbon dioxide removal technology.

Chapter 2: Methodology

To address the research questions this study will engage in qualitative research with a case study design. The municipality of Amsterdam will serve as the case-study selected for several factors. First, Amsterdam was chosen due to the progressive nature of its climate action plan, in addition to its membership of several progressive climate action city networks. This includes the Climate Neutral City Association, Covenant of Mayors and 100 climate-neutral and smart cities by 2030 mission. Thus, it may be in process or already have engaged with the issue of residual emissions in its carbon budget. As a result, Amsterdam represents a forward case regarding urban environmental policy. The case presents an opportunity to investigate the national and EU scale surrounding carbon-neutrality with a focus on actors at the local scale. Amsterdam is also within travelling distance, making it possible to conduct interviews with local officials and other actors face to face.

The case study will investigate Amsterdam in 3 parts; First, I will analyze all of Amsterdam's public facing documents regarding their climate action. This includes the municipalities climate-neutral roadmap, climate reports, circular economy report and technical reports. Additional documents will be incorporated if the municipality points to them directly either in their reports, websites, or official media channels. I will focus primarily on their plan to utilize CCS. Second is to identify, retrieve and analyze the regional, national and EU policy documents regarding CCS. Of additional interest might be policy documents that regulate what municipalities are able to implement or plans that would impact their use of CCS. For example, policy regulating the use of CO₂ as input for industrial purposes. I will use this analysis to outline the multi-level policy environment and identify where gaps or barriers exist or opportunities to align policy are possible. CDR and CCS technology is mostly addressed in industrial policy so there may be opportunities for Amsterdam to align here. Third is to conduct interviews with municipal and non-governmental experts alongside independent analysts connected to the Amsterdam case. I will try to connect specifically with actors who are decision makers or provide input into Amsterdam's plans regarding carbon-dioxide removal. Answers from interviews may point out new policy documents and insights into how relevant actors interact with them.

2.1 Data collection:

2.1.1 Document analysis:

To be able to answer the general research question, this paper analyzes and discusses findings from Amsterdam, National policy, and European legislation as well as non-governmental actors including actions, communications and directives related to carbon neutrality and carbon removal technologies. Other sources will be consulted including relevant municipal documents, city network communications, news articles, websites, datasets, and academic articles.

I will first analyze documents published by the municipality of Amsterdam regarding its climate action plan. A detailed analysis of this document and other policy will reveal what actions the city has indicated it will take to reach this goal. The goal will be to answer how Amsterdam will address residual emissions, and how much the city intends to rely on CDR technology.

Alongside conducting interviews, I perform a content analysis of policy documents in the regional, national and EU policy environment. I will construct an updated and consolidated mapping of the policy framework of this environment. Using keywords derived from the theoretical framework and analysis of interview transcripts to improve search for relevant documents. I use these keywords CDR, CCS, BECCS, DACCS, Net-zero, Policy, legislation, regulation, directive, law, act EU, The Netherlands, European Union. From this list I combine them to address technology, legislative instrument, and scale (ex: CDR Policy EU). These searchers are then applied to the repositories and search engines in table X.

Major resources:

Name of repository	Type	Description
EUR-Lex	Repository	Open repository of EU legislation
EU Science Hub	Science Communication	Science communication portal for EU Commission research, communications, projects.
NLog	Repository	Open repository of Dutch oil and gas legislation
Overheid.nl	Dutch law repository	Repository of Dutch legal laws (wet) and other legal documents.
Geemente Amsterdam	Municipal Portal	Municipal portal to access services, communication and reports
EU Parliament legislative train	Legislative reporting mechanism	EU Parliament reporting portal for all policy in the process of legislation. Reporting is published by rapporteurs.
Google	Search engine	A public search engine. Tailors results according to hidden algorithm.
DuckDuckGo	Search Engine	A public search engine. Does not tailor results to geographic region, cookies, or other personal information.

Table 3. Repositories

From searchers I list policy instruments, activities, decisions, and other material. I then filtered by relevance to my conceptual framework. I located relevant non-governmental or private actor reports and knowledge papers as they become known to me in my research.

2.1.2 Semi-structured interviews:

I plan to conduct qualitative semi-structured interviews with municipal actors, non-governmental partners (e.g., city networks) and independent analysts. Semi-structured interviews are motivated by a structured interview plan but allow the interviewer to deviate should a line of questioning become more valuable (Kvale, 2011). These interviews will cover several topics related to the city's climate action plan, emissions reductions, and needs so that residual emissions can be addressed.

I will identify interviewee's using several elements: the municipalities climate plans, municipality organizational structure, other relevant documents. I will contact identified parties by phone to discuss interest and then outline interview times.

Name	Organization	Title	Aspect	Progress
Marieke Van Doornick	Gemeente Amsterdam	Sustainability and planning council commissoner	Policy / municipal planning	Pending response
Ruth Noorduyn	Regional Energy Strategy (RES)	Noord-holland region contact	Technicality	Pending response
Laura Harkvoort	Gemeente Amsterdam	Planning and sustainability	Municipal planning	Contact Identified
Johan van Zoest	Gemeente Amsterdam	Planning and sustainability chief strategist	Technicality / municipal planning	Contact identified
Net zero-cities	EU commission	NET-zero cities mission	Policy / Technicality	Not yet contacted
Carbon neutral cities association	CNCA	CNCA	Policy / municipal planning	Not yet contacted

Table 4. Interview candidates

I intend to conduct at minimum 4 interviews and a maximum of 8. Interviews will run for a maximum length of one hour. These interviews will be transcribed and examined with assistance from the ATLAS.ti software package. The interviews will be conducted in English where possible.

I identify interviewees based on my conceptual framework to cover different aspects of sub-research question 1. These topics are:

- Technicality (knowledge of the context from a physical science standpoint)
- Policy (knowledge of the policy environment regarding carbon-neutrality)
- Municipal planning (responsible for solutions to the problem within the municipality)

Interviews will be conducted in-person where possible and recorded to be transcribed with interviewees consent. A document detailing interviewee rights and consent agreement will be drafted ahead of interviews (see ANNEX 1). This document will be shared with interviewee's along with a topic list. It will provide interviewee's time to prepare if they feel less confident in English. If they prefer not to or cannot speak English, I will send them a question form with questions translated into their language of confidence. I will use Deepl translator to translate transcripts post interviews.

2.2 Data management:

All data pertaining to interviews including consent forms, recordings and transcripts will be hosted on OneDrive through my university licence. This data will become available to the university upon request or at the end of my thesis.

Interviewee's will be given the option to request for anonymity in the thesis report through the consent form. They will also have the right to retroactively redact parts or the entirety of the interview upon request.

Questions:

Prior to interviews I will cover components of the literature and document analysis regarding CDR and Amsterdam's state climate goals. This will form a basis for the formulation of topics (Kvale, 2011). The full interview and structure will be available in ANNEX 2. As the goal of the interview is to explore Amsterdam's plans, needs and knowledge questions will be posed as exploratory. I will define a short list of open but relevant topics that I intend to cover during the interview. Questions will be posed during the interview so that the interviewee has maximum input on the direction and nature of the conversation. The interview is also conducted in this way as to ease interviewee's so they may express their knowledge freely.

Questions will cover these topics:

1. Views on Amsterdam's emission reduction targets & timeline.
2. Perspective on residual emissions left in the climate budget.
3. Amsterdam's options to address those residual emissions.
 - a. Are they restricted?
 - b. What opportunities does Amsterdam have to improve their options?
4. What is the nature of these barriers?
 - a. Financial?
 - b. Knowledge?
 - c. Regulatory?
5. Views on the policy environment.
 - a. Where is support provided? Is it enough?
 - b. Where is support lacking? How much might be needed?
 - c. Are there any relevant adjacent policies that could be changed to help?

2.3 Data Analysis:

The processing of interview transcripts provides the researcher with solid ground from which to conduct analysis (Kvale, 2011). I intend to transcribe all interviews that I conduct if given permission by interviewees. Interviews will be translated and transcribed with assistance from Deepl translation service. As my goal is to identify policies, documents, other points of interest, needs, and barriers from the results of expert interviews this approach is well suited.

I will also conduct meaning condensation of interview using the coding framework that is constructed. This refers to the process of taking statements and phrases from transcripts and condensing them into single word or short phrase codes. In this way critical elements of interviewee's statements can be identified and contextualized (Giorgi, 1975 as cited by Kvale, 2011). This will aid my analysis by making quick referencing to relevant information easier. The results from this analysis will be used as input to identify policies, initiatives, and other legislative material at the national and intra-national scale. This will then lead into a document analysis of National and EU policy.

Question matrix:

To accurately place each policy instrument in this framework and identify where barriers are present, I will ask these questions of each item:

- What is the purpose of this legislation / policy?
 - Does it directly or indirectly affect CDR technologies?
 - What **sphere** does it affect?
 - Which **approach** does it take?
- At what **scale** are the actors targeted by this policy?
 - What scale is it being deployed from? (Municipal, national, EU)
- Can it affect Amsterdam's treatment of residuals? How?
- Does the policy / legislation do as intend?
 - Does it rely on or is relied upon by other policy/legislation?
 - Does it create a barrier / miss the gap?
 - To what degree of severity is it a barrier? Gap?

Additionally, I develop a diagram to aid in the categorization of policy & legislation which can be found in Appendix B.

Each scale of the governance structure is recognized to have distinct but often overlapping responsibilities to the citizens within its jurisdiction. To this end, each scale is also allocated the rights to develop and enforce legislative tools to implement the necessary actions towards those goals. Those tools can be categorized by legal status as binding and non-binding instruments respectively. This categorization aids in determining the severity of the barriers that they present to a particular technology's development. Though, there is a large degree of nuance in the mechanisms that those tools utilize to enact their goals.

I present qualitative information regarding the instrument and motivate reasoning regarding severity of barrier that it presents. The table will head each policy instrument with digestible information of its contents according to the conceptual framework.

Title of legislation / policy / action (identification #)		Direct or indirect impact on CDR	Date of publishing or enforcement
Legality	Target:	Deploying institution	
Sphere of effect		Specific component being affected	
Approach (carrot, stick, sermon)		Economic approach: supply / demand	

Table 5: Generic instrument content panel

Each policy instrument panel will correspond to an entry that can be found in appendix B with additional contextual information. Entries can be located by the identification scale and number (ex: EU 14, NL 1, Ams 3).

Chapter 3: Results

In this section I present findings from my analysis of Amsterdam's climate action plan as well as national, EU policy and action regarding CDR and relevant sectors. I bring into the discussion interviews conducted with experts in policy landscape and where they believe gaps and barriers exist.

Following the conceptual framework, I will first discuss actors directly and indirectly relevant to the context. I will present the municipality of Amsterdam and its climate neutral roadmap 2050. An overview of CCS operations in Amsterdam that are best placed to achieve those goals follows to answer SRQ 1 and 2. I then outline the Dutch government, EU and networks so the interaction them and policy instruments can be clearly seen.

Finally, I present all relevant policy to infrastructure and normative frameworks in the EU and the Netherlands to answer SRQ3. The policy landscape surrounding carbon-dioxide removal overlaps with a number of other sectors in society including energy, health, and industry. Each policy is detailed alongside perceived barriers and gaps in order to answer SRQ 4 in the discussion.

3.1 Actors

3.1.1 Municipality of Amsterdam

The municipality of Amsterdam (or Gemeente Amsterdam) is the competent authority regarding the built environment, planning, social services amongst others within its municipal borders. The city is obliged by law to be the responsible authority regarding the environment and planning within its borders. While Amsterdam and other municipalities are not required to specifically address the impacts of climate change in their territories, they are obliged to ensure quality of the environment. As a result, many municipalities are or have developed a roadmap for addressing impacts to environmental value. In the case of Amsterdam this takes the form of its climate action plan.

Amsterdam's climate action plan is notable for its ambition to reach climate neutrality by 2050, in line with national and EU targets. The climate action plan was developed by the office of economy and spatial planning and published in 2019. Developed alongside this plan is a circular economy action plan and the template for annual reports regarding progress towards their goals. These reports communicate the current administration's goals, what pathways will be taken and progress along these lines. Stakeholders from all sectors were involved in the development of this plan including civil society, private business, national and European governance.

The city has developed the circular economy roadmap because it acknowledges that material imports to the city have an emission impact outside of its borders. However, inline with international and EU emissions accounting methodologies it splits its total emissions into two parts. The climate-neutral roadmap addresses scope 1&2 emissions tied to energy production for and direct emissions from activities in the city. The circular economy roadmap thus addresses emissions from scope 3 and is the more significant emissions portfolio.

It is not clear if the current administration of Amsterdam, which did not develop the plan, intends to revise it, or if there is a mechanism to revise it at set dates.

Amsterdam Climate Neutral Roadmap 2050		Direct	2020
Non-binding	Target: Org & Individuals	Deploy: Ministry of economic affairs and climate	
Sphere: Infrastructure		Material – Immaterial	
Approach: Sermon - Carrot		Economic approach: Demand	

The ultimate objective of the municipalities climate neutral roadmap is to reach climate-neutrality in 2050. The administration has defined this as the need to tackle the “energy transition in Amsterdam” and develop a circular economy plan. To this effect the roadmap outlines two goals: an interim goal of 55% reduction of GhG emission in 2030, and a final goal of 95% reduction in 2050. These GHG reduction goals are measured against the level of emissions in 1990 which are estimated to have been 3,8199ktCO₂-eq. In 2050 then, the goal is to arrive at 190ktCO₂-eq emitted per year. These targets are aligned with both the national and EU’s targets which aim for the same level of reduction by 2050 (Erbach et al., 2022).

Amsterdam has calculated its emissions pathways to 95% reduction based on four factors. Emissions reduction by the national government, replacement of polluting technologies, the growth of the city, and impact from measures towards climate-neutrality from the city. The city expects significant support from national measures, particularly in greening energy production, standard setting, and regional strategies. The Netherlands Environmental Assessment Agency (PBL) in collaboration with the city projected that the current actions would lead to 48% GHG emissions reduction in 2030.

A stock taking of Amsterdam’s current emissions put the city at somewhere around 5,000ktCO₂-eq per annum. This figure was calculated in 2017 in the most recent effort to calculate the city’s total GHG emissions. Emissions were calculated using the methodology outlined in the *2006 IPCC Guidelines for National Greenhouse Gas Inventories* (Jim Penman et al., 2006). Amsterdam separates these emissions into four sectors; harbor and industry, electricity, mobility, and built environment with emissions calculated at 920kt, 1,960kt, 870kt, and 1250kt respectively. Agriculture within the city’s borders is also considered but those emissions are negligible. Key emission points are from energy production and heat supply, represented in emissions tied to the built environment. This is in large part because Amsterdam considers the Port of Amsterdam emissions, and particularly that of the waste-incineration plant AEB to fall under its purview despite a substantial amount of heat from AEB going to other municipalities (Gemeente Amsterdam, 2020, pg 40). This is a departure from the Global protocol for Community Scale Greenhouse Gas emissions inventories (GPC).

The climate neutral roadmap is split further into 16 pillars, which each represent an aspect of the actions that the city will take to address emissions in the four categories. An additional four pillars are included which outline guiding principles to direct those actions in a just and sustainable manner.

The Built Environment	1	Scaling up Natural Gas phase-out
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	2	Developing Sustainable sources for heat distribution
	3	City wide heat infrastructure
	4	Energy efficient housing
	5	Energy efficient businesses
	6	Energy efficient municipal buildings
	7	Energy-neutral construction
Mobility	8	Limiting polluting traffic
	9	Cleaning up polluting vehicles and vessels
Electricity	10	Maximize roof-top solar
	11	Optimize wind energy in the city
	12	Develop future-proof energy infrastructure
Harbor and Industry	13	Transform harbor into a sustainable battery
	14	Develop green hydrogen economy
	15	Carbon capture, storage, and utilization
	16	Energy efficient industry

Table 6. Pillars and categories of action

Four guiding principles are included to outline how Amsterdam wants to act.

Working together	17	Building the movement in the city
Climate Justice	18	Working towards a fair energy transition
Learning and Experimenting	19	Boosting knowledge development and innovation
Space and Infrastructure	20	Making space for the energy transition and related infrastructure

Table 7. Guiding principles

Actions under each of the 16 pillars are expected to drive emissions down, but a few are expected to have significant reductions. Amsterdam expects that its most impactful actions are to move to energy production from renewable and to encourage the uptake of renewables in solar and wind within the city's borders. To mitigate curtailment, the city is also pursuing storage technologies in the port. Following this is the disconnection of building from the natural gas supply for heating and replace it with renewable systems such as heat pumps. Reducing the presence of internal combustion emissions and improving energy efficiency in buildings also represent significant actions. Combined these actions will aid significantly in reaching the 95% reductions target.

Reduction of scope 1&2 emissions in Amsterdam will run into significant barriers, and lead to reliance on BECCS to address residuals. In the Netherlands, despite the abundance of strong winds coming in from the north sea, only generates 4.2% of its total energy consumption from wind (Statistics Bureau Netherlands, 2023). Total projected renewable energy production in 2030 is at 27-35% and is not set to reach 100% in 2050 under the current pathway (Government of Netherlands, 2019). In all, the work needed to reach 95% emissions reduction is immense, expensive, and moving too slowly. Unless the situation changes, the city could need AEB to capture more than 5% of the residual emissions. This would signal to other cities that reliance on CCS is a valuable pathway to carbon-neutrality.

Amsterdam Circular 2020-2025 Strategy		Indirect	2020
Non-binding	Target: Org & Individuals	Deploy: Gemeente Amsterdam	
Sphere: Infrastructure		Material	
Approach: Carrot		Economic approach: Demand	

In addition to the climate-neutral roadmap 2050, the Municipality of Amsterdam published a strategy report which outlines its plans to address the city's climate impact from material and product consumption and waste. The report's methodology is based on the concept of the city doughnut, an extension of doughnut economics introduced by Kate Raworth in a 2012 Oxfam report. Amsterdam in partnership with C40, Doughnut economics action lab, Circle economy and Biomimicry 3.8 have adapted it to apply to cities. It outlines principles and strategies for creating balance between the needs of the city and the ecological limits of the planet by encouraging circularity in the city-region (Raworth, 2012).

The city's scope 3 emissions are tied to the circular strategy and were measured to be 8,000ktCO₂ in 2017, representing 63% of the city's total CO₂ emissions. These emissions are calculated from life cycle analysis (LCA) of the materials and products that come into the city (1,216 kt), including waste that the city generates. Amsterdam then categorizes the materials and products into 3 value-chains: food and organic waste streams, Consumer goods, and the built environment. Each value chain will be addressed by three ambitions and a number of courses of action dependent on its complexity.

Despite being a critical line in Amsterdam's ambitions to become climate-neutral, its scope 3 emissions are not relevant to its climate-neutral roadmap. This may change in the future as the discussion on the international scale regarding fair-share emissions for nation-states develops. If the accounting frameworks change then Amsterdam will have to increase its ambition to match. It may equally mean further reliance on CCS to capture additional residual emissions, worsening existing barriers in the policy landscape if they exist.

3.1.2 CCS & CCU in the Port of Amsterdam

In order to address this potential shortfall in emissions reductions, Amsterdam has identified a potential pathway for the removal and storage of its residual emissions through CCS. Located in the Port of Amsterdam a number of companies are pursuing CCS and CCU technologies that could sequester roughly ~11% of Amsterdam's emissions from 2017 (SouthPole & Bellona, 2022). Currently however, emissions from the harbor are 18% of cities emissions. This excludes the biogenic CO₂ emissions by AEB (63% of 1,537ktCO₂), which are currently not counted under the IPCC's methodology (Gemeente Amsterdam, 2020). If these emissions return to the atmosphere, they are considered neutral and not additional CO₂ emissions (Jim Penman et al., 2006). If captured for transport and storage (T&S), they would be counted as negative emissions and contribute to a net reduction of Amsterdam's yearly emissions.

A 2021 quick scan report by SouthPole and Bellona commissioned by the municipality of Amsterdam describes the potential capacity of CDR by firms in the city. South Pole is a private consulting organization with expertise in CDR. Bellona is a non-profit organization consulting on sustainable environmental solutions worldwide. The report outlines two major categories, standalone CDR, and Bio-CCS, differentiating options by presence of bio-genic CO₂ in its process.

Few standalone CDR options exist, with very poor potential for carbon capture and storage that is permanent. Only enhanced weathering and DACCS represent potentially permanent options, with CNI (~1MtCO₂ total) and Carboyn (1MtCO₂ per year in 2030) planning pilot programs of each technology respectively. By 2050 however, if DACCS technologies and processes improve significantly, it could represent significant capture potential in the Netherlands. Currently however there has not been any significant research efforts into modeling potential pathways of DACCS in the Netherlands.

The report identifies biomass with capture and storage (BioCCS) as having the greatest potential for CCS in the European Union and specifically in the North Sea commercial area (NSCA). Significant amounts of biogenic CO₂ could be emitted, roughly 916ktCO₂ per year optimistically. At the EU level this is closer to ~200Mt of biogenic CO₂ which could be captured with ambitious deployment across the continent. These figures do not account for additional emissions of biogenic CO₂ from hydrogen production, which Amsterdam and the EU have indicated as a priority in their respective climate action plans. Currently only AEB has plans for capture of its biogenic CO₂ in the NSCA and plans to start capturing CO₂ by 2028 using amine scrubbing technology for capture. The other firms are expected to contribute biogenic CO₂ for sequestration closer to 2050 (SouthPole & Bellona, 2019).

AEB Amsterdam (hereby AEB) is a municipal waste-incineration plant that was partly owned by the city's administration but has become a private business more recently. The plant produces energy for the city in the form of heat by incinerating non-recyclable waste and sludge. In total the plant emits roughly 1.1MtCO₂ per year 37% of which is from fossil origins and 63% from short-cycled biomass. Heat generated by the plant is in part sent to Amsterdam and neighboring municipalities.

In the 2020 climate-neutral roadmap 2050, Amsterdam indicated that AEB was the only CCS operation significant enough to incorporate into the plan. AEB is aiming for the net-capture of 500ktCO₂ per year by 2028 (Gemeente Amsterdam, 2020, pg 148). However, this figure has been cited differently across multiple sources and may continue to change. Once operational, the BECCS operation would reduce the city's emissions by roughly 10% of 2017 levels. In addition, it would reduce AEB's carbon emissions by 85% from 2022 levels. As the only firm within the municipality with concrete plans for capture, it positions AEB as Amsterdam's only pathway to realizing carbon neutrality currently. To this end, Amsterdam has pledged significant support for AEB under pillar 15 of its roadmap as the firm faces an uphill battle to operation.

3.1.3 AEB & BECCS operation chain

The operation of BECCS in the current technology landscape requires the coordination between multiple private and public actors. In addition, the capture and sequestration of the bio-genic CO₂ requires significant infrastructure and management across the Netherlands. Infrastructure for capture (firing plant, capture stack), for transport of biogenic CO₂ (pipeline, trucks), compression, and storage (injection plant, geological formation) are required to store carbon safely and effectively. For AEB, the relevant pieces of infrastructure are visualized in the diagram below including under whose jurisdiction they fall under (fig X).

Currently, there are 2 major ongoing transport and storage projects in the Netherlands; the Porthos project and Aramis project. The Porthos project is a collaboration between Shell, Total energies as private operators and Gasunie and EBN as state owned operators. It will operate 3

storage points in the Dutch continental shelf; P18-4, P18-2, and P18-6 with a total capacity of 14MtCO₂. Included is the development of an extended transport network with options to extend to Westphalia, Germany. Porthos will sequester 2.5MtCO₂ per year through this network with a maximum rate of 5MtCO₂ per year. It has been given the go ahead to begin operations soon following a project contestation by environmentalist groups concerned with nitrogen deposition in Natura 2000 protected area (Kombrink H, 2023). Porthos' 41MtCO₂ capacity has been fully booked by Shell, Total energies, and other industrial operators in the Rotterdam port area and will not provide storage potential for AEB or other capture projects. Plans for other storage points are in progress, in addition to inter-country T&S network development.

Aramis represents the only other sequestration project currently being operated in the Netherlands. It is managed by the same partners as the Porthos project but with differing goals, though synergies are being considered in cases of joint action (The CCUS Hub, 2021). The storage point has a capacity of roughly 400MtCO₂ put it at 10x the capacity of the Porthos project. Technical reports indicate an injection rate of 5MtCO₂ per year. Additionally, Aramis will connect the port of Rotterdam, Antwerp and locations in France and Germany through multiple shipping options (on-shore and off-shore pipeline, shipping, road). Due to its pan-european nature it has been given a Project of Common Interest (PCI) status, making it eligible to co-funding (5.8billion for energy projects) from the Connecting Europe Facility (CEF). The partners project an operations start-date in 2027 though it is likely that a lengthy environmental impact review similar to the Porthos project will be necessary and delay operations. AEB has indicated that it intends to work with the Aramis project to sequester the 500ktCO₂ per year that it captures. Aramis thus represents at present time AEB's best storage options given its proximity.

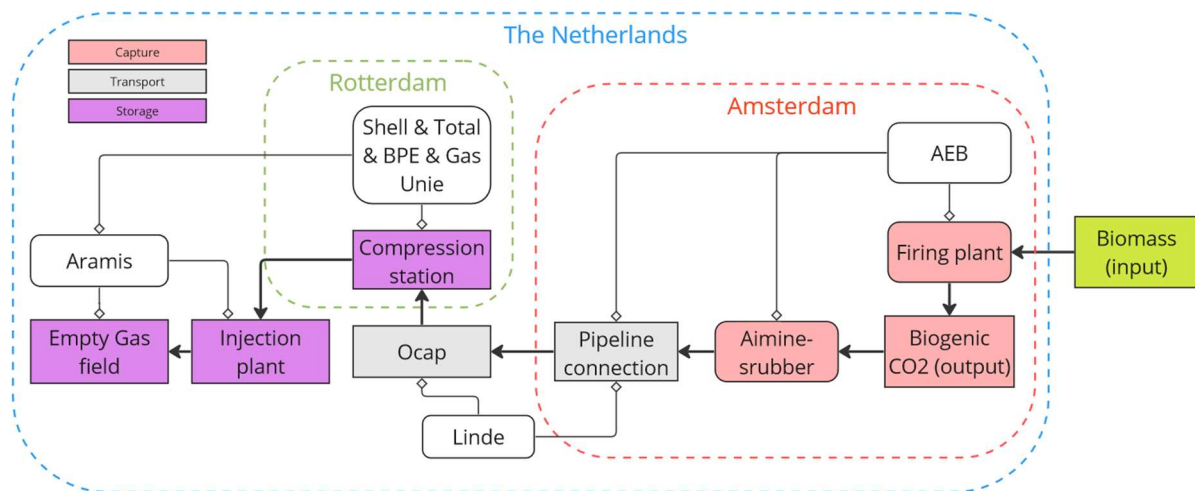


Fig 7. CCS operations from AEB

AEB is dependent on a CO₂ pipeline connecting to these storage points through the OCAP network. The transport network is owned and operated by Linde an energy generation company in the Netherlands which utilizes the pipeline to move CO₂ captured from natural gas and industry to horticulture applications. AEB must connect to OCAP in order to transport its captured CO₂ to compression stations in the port of Rotterdam. The connection, while only 2km in length is currently under consideration though operators at AEB expect it to be implemented without issue (Interview 1, 2022). Thus, the transport and storage infrastructure are currently on track to be operational for AEB and don't represent a significant barrier. It is unlikely that Aramis or the OCAP connection projects fall through, despite continued public distrust in the Netherlands. For

AEB to operate successfully, it will rely on Amsterdam to advocate in its favor in cases where complications arise.

3.1.4 Netherlands

In the Netherlands, climate action is largely managed by two ministries and a handful of agencies. The ministry for Internal affairs and kingdom relations (BZK) is responsible for management of national land and defines laws for municipal responsibilities (The Netherlands, 2021). The Ministry of Economic Affairs and Climate Policy is responsible for setting climate goals and developing mechanisms to lead the economy towards them (The Netherlands, 2021). Agencies differ from ministries in that they are semi-independent on government administration. For example, the environment assesment agency (PBL) is contracted by ministries above it to carry out impact assessments, write position papers, and perform other duties that are more knowledge based. Other responsibilities are delegated to provinces (regional administration) and municipalities (local administration).

In the Netherlands planning responsibilities and rights are outlined by the RVO in the forthcoming Environment and planning law (omgevingswet), expected in 2024. The act outlines what rights and duties authorities have in regard to managing the environment in their jurisdiction. Currently, the laws are enforced by the environment managment law (wet milieubeheer). The powers afforded to municipalities to address CC adaptation/mitigation are the ability to one; voluntarily set 'environmental values', meaning the municipality can further define what 'good air quality' for its jurisdiction. Two, to define env permitting requirements for certain activities and land uses which would require council approval if they fall within the municipality. In this case the council refers to a committee set up by the municipality to advise on application proceedings. Three, to define rules for activities that impact the env by order of council. This council refers to a body in the national government that has powers to make legal decisions clarifying aspects of laws. Four, levy costs on applicants for environmental permits. Though this law defines the municipality as the competent authority within its jurisdiction, when land use questions fall across multiple municipalities decision making powers are ceded to provincial authorities.

Environment and Planning law (Omgevingswet)		Indirect	Expected arrival - 2024
Binding (regulation)	Target: National and sub-national planning authorities	Deploy: Ministry of the Interior and Kingdom relations	
Sphere: Normative frameworks & Actors		Hard (Binding) & Organizations (Government)	
Approach: N/A		Economic approach: N/A	
Environment management law (Wet milieubeheer)		Indirect	In force
Binding (regulation)	Target: National and sub-national planning authorities	Deploy: Ministry of the Interior and Kingdom relations	
Sphere: Normative frameworks & Actors		Hard (Binding) & Organizations (Government)	
Approach: N/A		Economic approach: N/A	

This law decentralizes actions from the state level to municipal by principle of subsidiarity without comparable increasing of immaterial and financial infrastructure. It may lead to an increase in the administrative burden for municipalities, particularly smaller ones that cannot maintain technical expertise in-house. While Amsterdam is large enough to dedicate a large team to environmental sustainability, the law does not provide the city significant powers to support CDR. Decisions regarding transport will cover multiple municipalities and so be the remit of the provincial government. Storage options and permits are overseen by the national council and disallowed on National soil under the mining law. Thus, the municipality only has powers to support CDR if capture plants fall within their municipality. The support being limited to permitting requirements and costs and questions of 'environmental value'.

The Legal system in the Netherlands also plays an important role for Aramis and by extension AEB. If environmental groups put forward objections to the Aramis project, the courts will be required to consider the case to ensure environmental impact assessments are valid. This precedent was set when Porthos, a CCS pilot project by the same actors as Aramis, was contested by environmental groups. This is due in large part to transport pipelines that fall across Natura 2000 protected nature areas off the coast of Rotterdam. While the courts ruled to allow the project, Aramis will run pipeline across the same area and so will likely run into the same legal battles.

Netherlands National Climate Agreement (NCEP)		Direct	28/06/2019
Binding (regulation)	Target: Org & Individuals	Deploy: Ministry of Economic Affairs and Climate Policy	
Sphere: Normative frameworks & Actors		Hard (Binding) & Organizations (Government)	
Approach: Sermon – Stick		Economic approach: N/A	

Pursuant to the Paris agreement and the Governance of the Energy Union and climate, the Netherlands has developed its own National Climate and Energy Plan (NCEP). This document outlines the country's aim to reach a 40% emissions reduction compared to 2005 levels by 2030 and 95% reductions by 2050. The 2030 target has been updated, following the fit for 55% package at the European Union which will then apply to other obligations including the 'no-net debit' rule applied to LULUCF. The climate and energy plan outlines targets and actions regarding four major sectors: the built environment, mobility, industry, and Agriculture including land use. While the EU ETS covers energy-intensive industry, the Dutch government implements a suite of additional policies to encourage action. In particular, are policies regarding the environment and planning act, National carbon levy, and SDE++ fund.

While the Dutch government indicates the necessity of CDR and CCS in modeled reduction pathways to reach 'virtually net-neutral emissions' it has put limitations on their long-term implementation. The governments reasoning is to ensure that investment in CCS does not negatively impact the availability of resources for renewable energy and emission reduction technologies needed for a 'long-term transition' (Gov of the Netherlands, 2019. Pp 93).

3.1.5 The European Union:

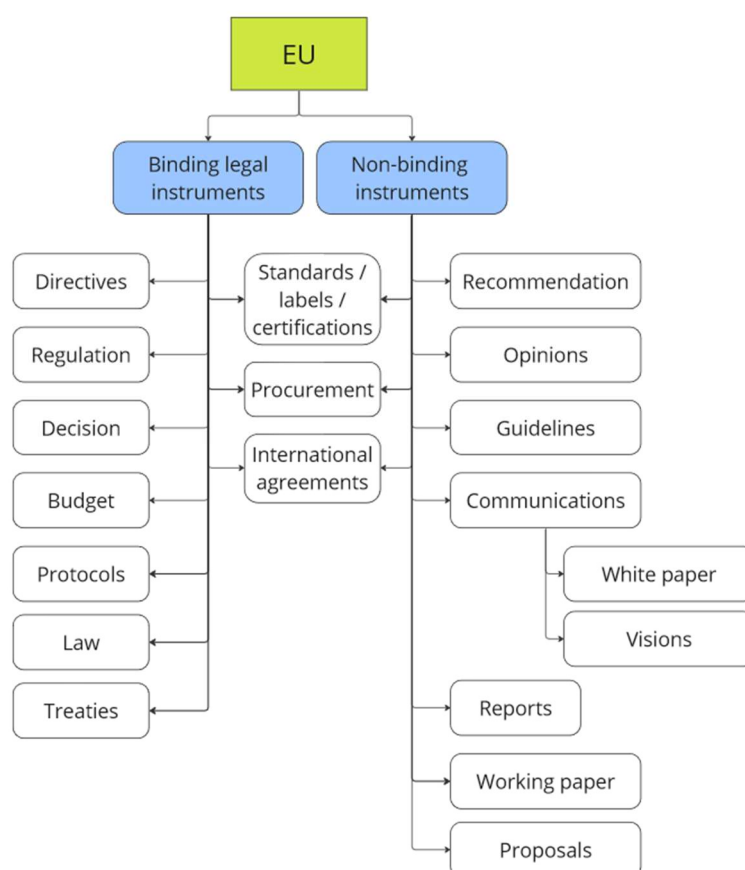


Fig 8. Policy instruments of the EU

In recent years the European union has revised many of its legislative instruments in the direction of sustainability and good climate stewardship following the Paris agreement, with the war in Ukraine accelerating and strengthening this action. The most critical pieces of legislation in this respect are the EU Green Deal, EU Climate Law, and Governance of the Energy Union and Climate and the bloc's 2050 climate neutral strategy to the UNFCCC. Many other directives, regulations and strategies have been and are being developed at the imperative of them.

EU Green Deal (& Investment deal)		Indirect	11/12/2019
Non-Binding (Communication)	Target: EU wide	Deploy: European Commission	
Sphere: Normative frameworks & Actors		Informal (non-binding) & Organizations (Government & networks)	
Approach: Sermon		Economic approach: N/A	

Action in a sector often begins as a discussion by relevant stakeholders in Brussels and is developed into a proposal to regulation or communication by the EU Commission. In 2019 the European Commission presented the European Green deal communication, outlining the existential imperative for the union to address climate change and biodiversity loss (EU Commission, 2019). It reported that if implemented the growth strategy would set it to become the first carbon neutral continent in 2050. Alongside the communication, a set of proposals were developed for the policies needed to reach this goal. These proposals have since been developed

into regulations, plans, directives, finance mechanisms and amendments of critical importance to significant climate action. Many of these instruments are directly relevant to CDR related technologies.

EU Climate Law		Indirect	30/06/2021
Binding (regulation)	Target: Member states	Deploy: European Union	
Sphere: Normative frameworks & Actors		Formal (binding) & Organizations (Government & networks)	
Approach: Sermon – Stick		Economic approach: N/A	

Member states agreed to enshrine this ambition into law in 2021 in what is known as EU Climate law, obliging member states to plan and act to attain climate-neutrality by 2050 (European Parliament, 2021). It creates the imperative for the bloc to pursue technologies that support reaching the target, specifically CDR. The regulation sets out critical avenues of action (mitigation and adaptation to climate change), and mechanisms by which to guide action by member states. It was amended by the ‘fit for 55%’ package of proposals, which also amended other important climate legislation in many sectors. The regulation relies on a sermon approach by creating moral imperative for member states to aim for the 2050 target. No penalizing mechanisms are outlined which reduces its efficacy, though some are present in other legislation implicated in articles 6, 7, and amendments. However, if a member state does not reach interim targets set out then the EUC may issue a recommendation for corrective action. The member state in question is not obliged to follow it or communicate reasons for not doing so. Though it is reasonable to expect that offending member states may lose access to EU funding and its national administration tried in the European court of Justice (ECJ).

Amsterdam and other cities in the union are not subject to the law, though it is implied by transitive property that they will need to act in accordance. Other directives and regulations aligned with EU Climate Law establish mechanisms pursuant to it.

Governance of the Energy Union and Climate		Indirect	20/11/2023
Binding (regulation)	Target: Member states	Deploy: European Union	
Sphere: Normative frameworks & Actors		Formal (Binding) & Organizations (Government)	
Approach: Sermon – Stick		Economic approach: N/A	

The Governance of the Energy Union & Climate regulation seeks to establish governance mechanisms that will implement strategies to meet the targets enshrined in EU Climate Law and international agreements such as the Paris Agreement (Governance of the Energy Union and Climate Action, 2018). It establishes mechanisms for collaboration between member states, financial support for renewable energy development and public participation in national development of climate plans. Critically, it establishes mechanisms requiring the monitoring, accounting, and reporting of GHG emissions and removals in an annual cycle following the UNFCCC and Paris Agreement Secretariat. However, there are no mechanisms for forcing

compliance or addressing non-compliance directly. It is likely that similar to EU Climate Law, the member state may lose some standing in the EU and access to funding initiatives. While not directly affecting CDR and related technologies, the regulation requires member states to develop climate action plans. Inclusion of CDR in these plans would positively signal to sub-national and private actors that investment in the technologies has national backing.

While the Netherlands does include CCS in its National NCEP, it is considered a transitional technology by the current administration. For Amsterdam and AEB this means that they have limited opportunities to receive support from the National government. Thus, the city will need to both find support elsewhere and advocate to the Dutch Government the value of BECCS.

3.1.6 Networks

Policy instruments rarely affect actors directly other than regulation regarding the rights, responsibilities and powers afforded to actors in the governance structure. All policies naturally, however, indirectly affect actors in the relevant fields. Many of the funding schemes and regulations detailed below have a network or knowledge components to support their objectives including Horizon Europe, Connecting Europe Facility and TEN-E. However, there are some initiatives, networks, and activities whose main aim is to support the actors involved.

Other relevant actors include non-profit organizations, knowledge centers such as universities and experimental organizations such as AMS institute. These actors affect the policy landscape indirectly through academic research, reporting, and advocacy. They rely on the existence of networks to carry normative knowledge, opinions and reports to decision makers integrated in the vertical governance hierarchy.

Vertically initiated networks (100 climate neutral cities, Green city accord)		Indirect	n.d
Non-Binding	Target: Organizations (cities)	Deploy: Various	
Sphere: Actors		Organizations (government)	
Approach: Sermon - Carrot		Economic approach: Demand	

The EUC as the EU's implementing body has a number of initiatives aimed at supporting municipalities and regional actors achieve sustainability goals. One such network for cities is the Green City Accord. This initiative by the European commission to make cities greener, cleaner, and healthier (EU Commission, 2023). Membership is voluntary and non-binding, and municipalities are encouraged to monitor and report on priority areas using an its framework. Through this initiative the EUC makes available learnings from other cities, funding, and soft power through publicity. Amsterdam, however, is not a signatory though many Dutch cities including Utrecht are (EU Commission, 2023). It is unlikely that this initiative would offer significant value to Amsterdam, at least in respects to CCS activity, other initiatives are more relevant. Other cities in Europe tackling CCS (in particular Copenhagen) are similarly not members of the accord.

An alternate vertically initiated network is the net-zero cities initiative of the 100 Climate-neutral and smart cities program implemented by the EUC. Unlike the Green City Accord its focus is specifically on driving GHG emissions reduction action by members of the network. It is a project

of the EU Missions program of the Horizon Europe research and innovation scheme. As a result, the program is better funded and can provide administrative support under work package 10 and 14 (Net Zero Cities, n.d.). Through network events and other activities, the network can provide Amsterdam a platform to transfer knowledge with other cities that are also including CDR in climate action plans.

Horizontal networks (c40, G4 municipalities)		Direct	N.d
Binding (regulation)	Target: Organizations (cities)	Deploy: Various	
Sphere: Actors		Organizations (Government)	
Approach: Sermon – Carrot		Economic approach: N/A	

In addition to EU initiated networks Amsterdam is in cooperation with a number city networks and other knowledge sharing organizations working to support municipal action on the issue of climate adaptation and mitigation. In particular, are the networks C40, Covenant of mayors, and Local Governments for Sustainability (ICLEI) which Amsterdam are members of. These networks operate at the international level and include members from around the world of differing scales. In addition, these networks often work with larger governance bodies including the UN and EU, providing a platform for municipalities to push for change at the international level. This can be seen most explicitly at the Ministerial meetings at the Convention of the Parties (COP) 27 and most recently, COP28 (United Nations Environment Programme, 2023). Here mayors from global cities convene at the largest convention for climate action to speak on issues, needs, opportunities and solutions for cities to address the climate for the benefit of people.

Amsterdam uses some of these networks to lobby to the Dutch government, and the EU in brussels. In particular is the cooperative network ‘G4 municipalities’ that makes up the four largest cities in the Netherlands (Amsterdam, Rotterdam, The Hague, Utrecht) in what is known as the ‘Randstad’. Together they represent a significant portion of the country’s economic power and population, which provides this lobbying cooperative some heft (Netherlands Environmental Assessment Agency, 2016). The group has offices in Brussels, giving the cities a direct line to the legislative activity of the EU’s bodies (VNG, 2023). In its climate roadmap 2050, Amsterdam indicated it would use this line to lobby for CDR in favor of the port of Amsterdam. Though, this lobbying power is debatable given the 5,000 registered lobby groups (PBL, 2016 pp.25), and most certainly tightly related to issues of concern for cities.

Closer to home Amsterdam is connected to the Association of Netherlands Municipalities (VNG) and Regional agency for the environment. These groups act as networks for sub-national governmental bodies, facilitating advocacy and communication between themselves and the national government. The VNG in particular works towards advocacy to support changes that would empower municipalities to better act in the interest of residents. These networks are also voluntary for Amsterdam.

Through the interactions and interrelations with the actors in the MLG structure of the EU Amsterdam can make an impact on policy where it is needed. Favorable perception on the issue of climate change and the necessity of BECCS is critical to implementation of policy supporting capture, transport, and storage of CO₂.

3.2 distribution of policy instruments relevant to CDR

In all, this study identifies X policy instruments that directly or indirectly affect CDR relevant spheres (actors, infrastructure, normative frameworks). Additionally, 5 networks were identified to be operating on direct and indirect issues. Using the categorization framework developed I tag each instrument by sphere of effect, target actor, approach (market and incentive), legal status (and mechanism). Given the nature of the EU's structure, many of the regulations are pursuant to the identified directives, which could be counted as a single instrument together. I decided to count them apart given that each has a different mechanism for enacting its policy goal.

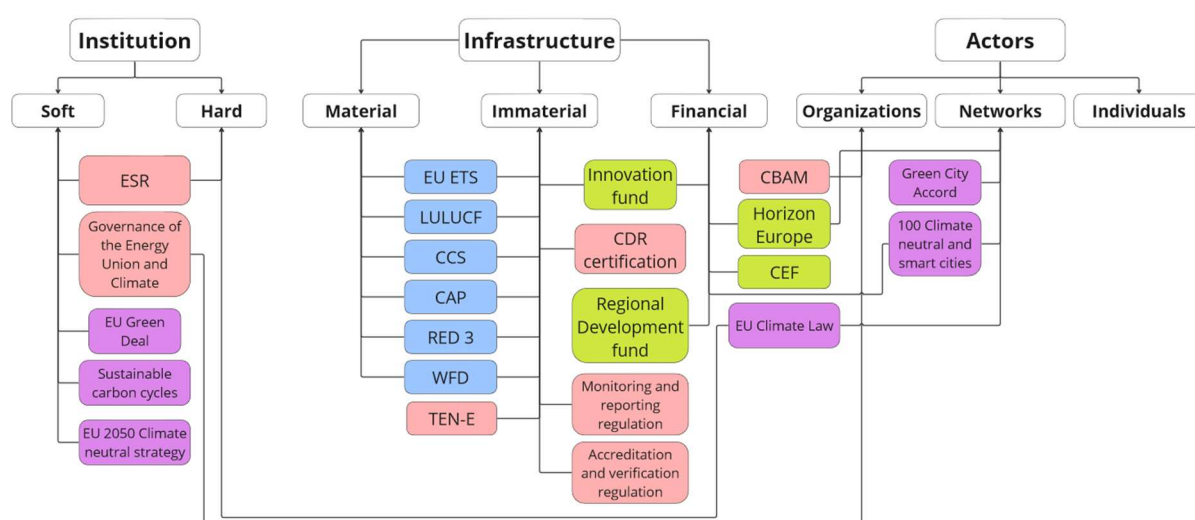
Of the 38 policy instruments identified 23 are deployed by EU bodies:

Instrument type	Number
Directives	6
Regulation	7
Finance/support schemes	5
Strategy / vision	6

Table 8. Distribution of instruments

The sphere by which the instrument effects is differentiated is by mechanism and the content with content being the defining factor. For example, the CCS directive defines requirements for the physical infrastructure of geological storage (content), it does so by requiring member states to transpose the directive into law by establishing a national standard (mechanism). Whereas the Innovation fund is a funding scheme (mechanism) for emissions reduction innovation and research (content).

Of the EU legislation, 9 target material infrastructure often overlaps with immaterial infrastructure which accounts for 12 of the identified instruments. 6 Financial support mechanisms are identified that are open to funding CDR / CCSU; most significant are the innovation and Horizon Europe fund as they represent billions of euros in available funds. regulations are identified with relevance.



Fix 9: Distribution of EU instruments

Many EU legislative instruments affect more than one sphere, in particular Directives which often cover content regarding a material component (agriculture) and immaterial component (practices for increase soil storage in agriculture).

In the Dutch legal environment, a total of 10 legislative instruments were identified. The landscape is heavily skewed towards material infrastructure and financial mechanisms with 5 and 4 instruments respectively. CDR and CCS is eligible for funding under 4 schemes, though the MIA is a tax rebate mechanism. The SDE++ is the most significant funding scheme and AEB has previously applied to it.

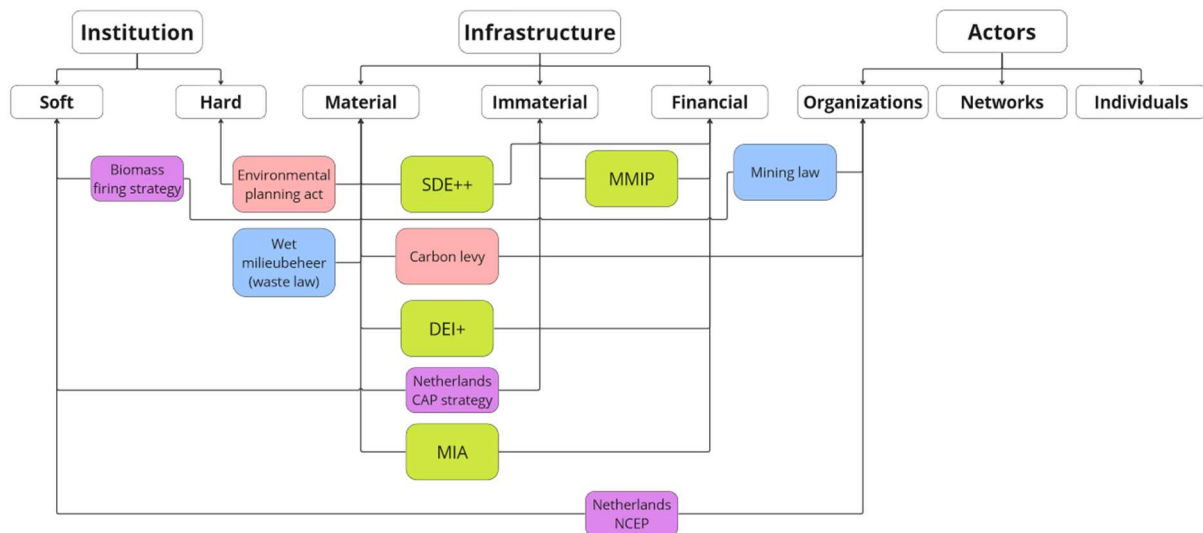


Fig 10: Distribution of instruments in the Netherlands

3.3 Infrastructure

By conceptualizing Amsterdam's CCS needs as the requirements to operate by AEB I categorize infrastructure according to three components: material infrastructure, immaterial infrastructure, and financial infrastructure. The most critical element of the three is material infrastructure, which refers to the physical components of the system including but not limited to amine capture stacks, biomass, and geological formations. I will look closely at this aspect as without these components CSS is a thought experiment. Financial infrastructure refers to financing mechanisms and instruments that are available to support investment in material and immaterial infrastructure. Here I will detail the policies and how specific mechanisms in them may or do create barriers for AEB and Amsterdam directly or indirectly.

Finally, immaterial infrastructure refers to technical knowledge for operation, administration, and networking. Additionally, it can refer to permits, certificates and data collected by various parties. Many of the following policy instruments primarily affect material and financial spheres but include mechanisms that affect immaterial infrastructure. For clarity I organize them under the primary sphere of effect, but secondary spheres are noted in each policy panel.

3.3.1 Material infrastructure

CCS Directive		Direct	24/12/2019
Binding (regulation)	Target: Organizations	Deploy: EU	
Sphere: Infrastructure		Material & Immaterial	
Approach: Sermon		Economic approach: Supply	

Important to the success of AEB's operation is the CCS Directive 2009/31/EC defined by the European Union in 2009 and amended in 2018 (EU CCS Directive, 2018). This directive establishes procedures for the environmentally safe and permanent storage of physical carbon dioxide in geological formations. It applies to all territories of member states including exclusive economic zones and continental shelves pursuant to the United Nations Convention on the Law of the Sea (UNCLOS). It additionally prohibits the storage of physical carbon dioxide in the water column, which removes ocean fertilization as a permissible CDR method.

To ensure safe geological storage the directive sets out mechanisms for member states to establish. These include permitting and monitoring requirements, the requirement for an accredited third party responsible for the inspection of storage sites to be set up, and technical requirements for storage. The permitting body will be responsible for defining those technical engineering and permitting requirements. The Netherlands has charged the Ministry for Economic Affairs and Climate as the permitting body. Additionally, it establishes a mechanism by which storage sites must be detailed to the European Commission which may then issue an opinion. The opinion must be considered by the permitting body of the member state and the EUC before the project can go ahead. There are no penalties for not incorporating the opinions of the EUC, however member states are required to define penalties in national legislations for infringement of the directive.

Since 2009, few CCS storage projects have been carried out in the European Union, though several member states have issued exploration permits. An exploration permit, covered by the Directive under Article 4 & 5, is required to collect necessary information regarding the suitability of the site for storage. Only the Netherlands has issued exploration and storage permits under the Porthos project, with permits expected to be issued for the Aramis project in the future (The CCUS Hub, 2023).

The CCS directive does not contain any potential barriers as itself as it leaves the responsibility of permitting and storage requirements to the decision of independent third parties in member states. However, only minimum requirements for CO₂ streams are defined by the directive and not for geological storage. No definition for 'permanent storage' is given which could be problematic if member states opt for geologically risky options. A definition is however given under the CDR certification framework in Article 2 (CDR Certification Framework, 2022) which members will be required to follow once passed. Commission has noted in its 3rd implementation report that the Netherlands has satisfactorily transposed the directive (European Commission, 2023).

For AEB the policy does not represent a barrier directly as the firm does not operate transport or storage infrastructure. The firm instead will purchase storage space and thus must rely on Aramis to be given permission to operate by the RVO, and a positive assessment by the EU Commission. In 2022 Aramis received a positive opinion from the EU Commission for storage off the coast of the Netherlands (European Commission, 2022).

Mining law (mijnbouwwet)		Direct	2011
Binding (law)	Target: Firms	Deploy: Ministry of Interior and Kingdom relations	
Sphere: Infrastructure		Material & Immaterial (operations & permits)	
Approach: Stick		Economic approach: Supply	

In the transposition of the CCS directive, the Netherlands amended its Mining law (Mijnbouwwet) in 2011 to enshrine the necessary articles regarding capture, transport, and storage of carbon dioxide in geological storage (Ministry of Economic Affairs and Climate Policy, 2019). The directive identifies the State supervision of mines (SSM) as the relevant authority for the permitting of exploration, transport, and storage operations. Permit applications are done in a competitive manner, and permits must be reviewed by the European Commission before being issued. There are no specific technical requirements outlined by the law for storage.

Relevant rulings have been made pursuant to this law on safe operating limits and areas that are off limits to mining operations in 2011. Specifically, two CCS projects in 2011 were shut down, and onshore CCS was indefinitely postponed due to public pressure. This stemmed from the public backlash to a 2007 decision to pursue CCS projects under Groningen (Global CCS institute, 2015). Court orders following the issue ruled only CCS projects off the coast of the Netherlands can be legally considered for permitting. This significantly reduces the total storage capacity of the country, which has led to an interest in connecting the transport network to other European countries.

Legislators have identified that additional legislation will be needed to address gaps such as preconditions for CCS, financial commitments, and liability for damages. It is unclear if these gaps have been addressed as of yet though it is likely that once the CDR certification and verification regulation is adopted that the mining law will be amended to reflect some of the requirements regarding technical operation. Favorable rulings on the legislation for CCS would lead to streamlined implementation of new storage and availability of funds that AEB can access to invest in its BECCS operations.

Net-zero Industry Act		Direct	2023
Binding (law)	Target: Firms & Government	Deploy: EU Parliament & Council	
Sphere: Infrastructure & institution		Material (CCS infrastructure) & Hard (binding targets)	
Approach: Carrot		Economic approach: Supply	

New amendments tabled for the Net-zero Industrial Act set out in April of 2023 mark a significant shift in favor of CCS(U) and CDR technology (European Parliament, 2023b). If passed by the EU Parliament and Council, the act would lead to significant easing of barriers present in the policy landscape for AEB and Amsterdam. The NZIA is integral to the EU's 'Green Deal industrial plan for the net-zero age,' which outlines a European approach to bolster the EU's net-zero industry (European Parliament, 2023b). Most crucially the ITRE endorsed the setting of binding targets for

GHG storage capacity EU-wide, with targets for 50MtCO₂/y in 2030, and 550MtCO₂/y in 2050. Furthermore, amendments will expand the storage obligation to all entities selling crude oil, petroleum products, or natural gas in the Union (Global CCS Institute, 2022).

Amendments to the NZIA and the Carbon Removal Certification Framework (CRCF) were approved by responsible European Parliament Committees. Notably, on October 24, 2023, the Environment, Public Health and Food Safety (ENVI) Committee approved changes to the European Commission's Proposal for Regulation Establishing a Union Certification Framework for Carbon Removals. These amendments address issues like double counting of carbon credits and establish a clear distinction between permanent removals and short-term sequestration activities.

Proposal in the NZIA could significantly support AEB and Amsterdam by accelerating permitting, increasing funding, and stimulating job creation connected to net-zero technologies. It would define strategic net-zero projects by establishing a "one-stop shop" in each country for coordinating permitting of manufacturing projects. It additionally detailed timelines for permitting procedures, giving priority to strategic projects. This ease of permitting aligns with similar actions including the 'presumption of public support' principle under the RED. While not applicable to Aramis anymore, it would ease exploration and storage permitting for other storage capacity in the Netherlands, potentially including the Groningen fields.

Currently, the amendments by ITER to the NZIA has been tabled (EU Parliament, 2023), with a vote expected in November 2023 in plenary. The vote will determine if it is passed as is, rejected or sent back to discussion for changes. Overall, these developments reflect a significant step towards the EU's commitment to climate neutrality and a sustainable energy future, with a strong emphasis on the development and implementation of CCUS technologies. It will create strong signals which may convince the Dutch Government to change its stance on CCS as a 'transitional technology' and lead to significantly bolstered support for AEB and Amsterdam.

3.3.3.1: Renewable energy and Biomass

While the infrastructure necessary to sequester carbon may materialize, AEB also requires a significant amount of biomass for firing in order to generate negative emissions. Concerns over the sustainability of biomass as a renewable energy source have increasingly become the subject of debate within the EU. In response, a number of EU and Dutch legislative instruments have been revised to address those concerns. The direction of those decisions may impact AEB's sourcing of biomass.

Renewable energy Directive (3)		Direct	Est. 2023
Binding (law)	Target: Organizations (government)	Deploying: EU	
Sphere: Normative frameworks & Actors		Hard (binding & non binding) & Networks	
Approach: Carrot		Economic approach: Supply & Demand	

The Renewable Energy Directive 2 (EU 2018/2001) is the legal framework for the development of clean energy across all sectors of the EU economy, supporting cooperation between EU countries

towards this goal (Renewable Energy Directive, 2023). It sets rules and targets to ensure the uptake of renewable energy by member states in all sectors. The directive is currently being revised to be recast with an increase in ambition from 40% renewable energy mix by 2030 to 42.5% with 2.5% indicative top up. It additionally brings transport and maritime under the directive requiring 14% renewable energy use in those sectors by 2030. This revision was driven by the EU Green deal and fit for 55% package, making it more in line with the REpowerEU plan to phase out energy dependence in Russia and accelerate the clean energy transition. The revised RED3 directive has reached provisional agreement and will enter into force later this year (2023).

Within the Directive, member states are encouraged to address both Supply (by investing in renewable energy production supply) and demand (by supporting consumer side renewables and encouraging the requirement for renewables on new or renovated buildings). Regional and local administration are also encouraged in this directive to include RES growth in plans. To support development, the amended RED3 puts into place financing mechanisms allowing member states to provide renewable 'support schemes' to projects within and outside of their borders. Additionally, the directive would put into place a mechanism for accelerating the permitting and project approval under the "public interest presumption principle".

In response to the growing discussion regarding the sustainability of biomass as a fuel source given the uncovered negative impacts that wood pellets have had on old-growth forest in Europe a number of national and EU legislation has been revised. The recast RED3 directive crucially puts into place sustainability criteria for biomass for energy and other biofuels. It directly addresses the use of Biomass for energy (electricity and heat) and supports the tightening of sustainability requirements aligning with the CAP, LULUCF directive, and Biofuels regulation (EU 2019/807).

The environmental criteria set out under the RED 2 (RED2, 2021) is:

1. Source: sustainable Biomass shall not be sourced from high-ILUC sources.
2. GHG savings: supply chains must meet emissions reduction targets.
3. Waste and residue: need to fulfill emissions savings and address soil quality.
4. Reduce risk of unsustainable production.
5. Provision of verifiable data for sustainability for above criteria.

Under the RED3 article (26.2) measures are in place to ensure that biofuel production does not negatively impact the use of land for food or feed. Biomass coming from these areas that displace the original functions are labeled high indirect land-use change (ILUC) risk fuels and are subject to strict requirements. The consumption of these fuels will not go towards renewable energy targets and may not surpass consumed levels in 2019 (which will reduce to 0% in 2030). For low ILUC-risk fuels to go towards targets they must be monitored and certified by member states. The EU commission is empowered to recognize national and international sustainability schemes that certify low ILUC-risk sources.

The directive will also implement the Cascading principle whereby woody biomass for bioenergy will be favored for "highest economic and environmental added value" (EU Parliament, 2023x, pg 8). Such as using only feedstocks with little market competition should be used. In addition, the directive enforces a ban on direct financial support to specific feedstock from forestry for energy (EU Parliament, 2023x, pg 180).

The sustainability requirements under this recast directive will require the Netherlands to adopt more stringent biomass sourcing criteria. While this will ensure that any use of biomass by AEB

is sustainable for the planet and people producing it, the criteria will shrink global stocks of biomass that can be sourced in this way. If stocks do become reduced, AEB may be unable to operate enough BECCS to capture Amsterdam’s residual emissions in the long-term.

LLUCF regulation		Indirect	21.4.2023
Binding (law)	Target: Organizations (Firms)	Deploy: EU	
Sphere: Infrastructure		Material & Immaterial (operations & permits)	
Approach: Stick		Economic approach: Supply	

The Land use, Land Use Change and Forestry (LULUCF) regulation was developed to incorporate GHG emissions and removes from land use into the EU’s climate framework (LULUCF regulation, 2023) (LULUCF Regulation, 2023). It requires that member states must reach GHG emission reduction and removal targets in land uses for the 2021-2030 period. The regulation, however, does not set a target for increasing the LULUCF carbon sink, but rather includes a ‘no net debit’ target for LULUCF (Forests and Agricultural soils). By linking Land use and forestry with the certification of removals regulation the regulation aims to ensure quality removals in the sector.

The regulation alters emissions capture accounting from biomass harvesting so they are accurately reported under the climate framework (IEEP, 2021). Under the LULUCF, bioenergy is ‘zero-rated’ meaning no additional emissions come from its utilization as the emissions are already accounted for in agriculture and forestry. As a result, from its normal operation only 37% of AEB’s emissions are counted towards its emissions budget.

By enforcing regulation on forest management practices, the regulation is expected to reduce woody biomass and agricultural biomass availability for bioenergy (Nabuurs J et al., 2018, pg 6). This is particularly important for AEB as the firm relies on low-grade woody biomass for firing (AEB Amsterdam, 2022). However, possible mechanisms for counteracting the decline have been identified and could be included in future amendments to the regulation. Additionally, it is expected that the regulation may not have a significant impact in limiting biomass in reality as on paper (Nabuurs et al., 2018, pg 7).

Common Agriculture Policy		Indirect	09/09/2021
Binding (law)	Target: Various	Deploy: EU	
Sphere: All (primarily infrastructure)		All	
Approach: Stick & Carrot		Economic approach: Supply & Demand	

The common agriculture policy (CAP) is a 60-year-old piece of legislation which outlines shared goals of European union members regarding agriculture (Common Agriculture Policy, 2023). Having been revised recently, the new CAP aims to foster a sustainable and competitive agricultural sector that can support the livelihoods of farmers and provide healthy and sustainable food for society, along with vibrant rural areas. It provides a framework of regulation, directives, and shared goals for reaching those objectives. The package will provide 387 billion euros in funding towards policy goals for the 2021-27 period. This includes support to promoting biomass use for more eco-friendly energy, as long as they are in accordance with environmental

standards under the cross-compliance system (EU 1306/2013, 2023). Studies on the CAP's effect on biomass availability project growth from all sources for 2030 and 2050 in all scenarios (Soler Abia, 2022). This reflects positively on biomass pricing and availability for AEB, translating to a stronger business model for BECCS operation. If scenarios by Soler play out, AEB may find that biomass sourcing won't be as significant an issue.

Netherlands Biomass policy		Direct	2022
Non-Binding	Target: Firms	Deploy: 2022 Dutch Cabinet	
Sphere: Infrastructure		Immaterial	
Approach: N/A		Economic approach: Supply	

Due to questions about the sustainability of biomass, in particular wood pellets, the Dutch government indicated that it would consider biomass' future place in the economy. After discussions the government communicated that despite the controversy over wood pellets, biomass and other biofuels will continue to play a significant part in the renewable energy strategy of the country. Though it was indicated that tighter sustainability requirements would be put into place in line with changes to the Renewable Energy Directive and LULUCF directive. The decision pointed out that biomass sourcing would only be supported for high-value purposes, and purposes without a sustainable alternative. This includes advanced biofuels, bio-chemicals, and the use of biomass for firing (above 100C), BECCS and CCU. To this end, the Dutch government will base its sustainability criteria and monitoring program on the RED3 directive.

In addition, the Dutch government decided that it would not limit biomass sourcing to the EU, as domestic supplies are expected to be insufficient (USDA & GAIN, 2022) to reach national renewable energy targets. This reflects the growing use of biomass and renewable municipal waste for energy purposes (IEA, 2021).

The tightening regulations surrounding biomass use at the EU and National level are positive changes reducing the negative impact of its production on environmental limits and food safety. However, they will necessarily lead to rising prices of biomass in the EU, and reduction in supply for many uses. As such AEB may face a reduction in support from financial programs and stakeholders including government regarding biomass firing. If this were to coincide with an increase in public pressure against its use on sustainability concerns these changes would represent a significant barrier to operation. The cost and sustainability issues may lead residents of Amsterdam and the municipality itself to view BECCS plants as incompatible with its climate neutral roadmap.

3.3.4 Immaterial infrastructure

Waste Framework Directive		Indirect	2023
Binding (law)	Target: Member states	Deploy: EU	
Sphere: Normative frameworks		Hard (Standards & regulation)	
Approach: Carrot & Stick		Economic approach: Supply	

As both parties across the EU look at measures to encourage a more sustainable and circular economy, policies have been developed to update the management of 'waste'. Under the Waste-

framework directive (WFD) and Dutch national legislation, biogenic-CO₂ is considered waste for handling and management purposes (Waste Framework Directive, 2018). Handling of waste has specific requirements, and the use of these materials has strict regulation. The directive also outlines the end-of-waste (EoW) label which can be applied to certain materials indicating it as a product or secondary raw material. Materials with this status would send a clear signal to other actors that the material is safe and of appropriate quality for reuse. The aim is to increase the reuse and recycling of those materials towards a circular economy. The commission is empowered to set regulations regarding the criteria of what can be considered EoW, though member states are free to make their own decisions for each material. In 2023, the WFD was amended with a specific focus on textiles.

In the Netherlands, the WFD has been transposed under the “Wet Milieubeheer”, however the lack of clear legal ruling at the EU and national scale is deleterious for firms. Currently, unless specific EoW criteria exists, firms are open to develop self-assessment criteria and can claim adherence to that criterion (Mense, M et al., 2023). A ruling on the self-assessment can be sought by relevant authorities for a “rechtsoordeel einde-afval” or EoW ruling, which is not legally binding. Many gaps remain in the regulatory environment that causes more confusion and greenwashing. For Biogenic-CO₂, it means a lack of clarity regarding CO₂ stream quality and other technical requirements for AEB.

As it applies to storage in geological formations, these regulations are not strictly relevant as the biogenic CO₂ does not need to be classified as end-of-waste to be sequestered. However, for CCU the ‘waste’ must meet criteria of the importer after which it can be considered to prove achievement of the status. If the biogenic CO₂ goes towards stable products (at the timescale of millennia) or continuously recycled, then the policies positively contribute to removals.

3.3.5: Financing and investment instruments

For operation, BECCS requires a network of costly technology that has not yet been proven for large-scale deployment. As a result it requires financial support to develop pilot projects that will provide learning and lessons for scaling to where it can begin to drive significant negative emissions under the IPCC’s 1.5C pathway (Tamme & Beck, 2021). The European union and the Netherlands have set aside financial support for various aspects of CDR technology through grants, tax returns and investment programs.

Horizon Europe		Direct	2021
Non-Binding	Target: Various	Deploy: EU Commission	
Sphere: Infrastructure & Actors		Various	
Approach: Stick & Carrot		Economic approach: Supply & Demand	

The Horizon Europe is a research, development, and innovation funding scheme with a significant budget, perhaps one of the largest of its kind (Negative Emissions Platform, 2021). The aim is to support technologies to move from the lab to early-stage demonstration with the goal of large-scale deployment and commercialization. Horizon Europe is part of an ongoing program and the successor to the Horizon 2020 program, covering the period 2021-2027. The fund has earmarked 14.7 billion Euros in funding for carbon dioxide removal technology under the new Cluster 5 work

programme: Climate, energy, and mobility. Seven funding calls have been set out though only 3 of them may be relevant to BECCS. The results from the funding may not become visible for some time but may become relevant for Amsterdam in the future if certain technologies become cost and space efficient. Though, the scheme is not relevant for AEB as the relevant calls only target research and CCS technology price reduction. This is a gap in the scheme, however other funding sources cover it.

While Horizon Europe work package 5 is not currently relevant to Amsterdam, it may be useful for other cities experimenting with more novel forms of CCS. For example, the establishment of distributed small-scale DACCS within cities powered by renewable energy. Such an application, while very far off, could create co-benefits beyond negative emissions for residents. It is also possible that Amsterdam could apply it in collaboration with universities and DACCS firms. In this case the horizon Europe program would be applicable.

Innovation Fund		Direct	15/09/2023
Binding (Budget)	Target: Various	Deploy: EU Commission	
Sphere: Infrastructure		Material	
Approach: Stick & Carrot		Economic approach: Supply	

One major funding source with direct relevance is the EU Emission Trading Scheme's Innovation Fund (IF). The financing of this fund comes from the sales of allowances by member states to firms that are covered by the ETS (Innovation Fund, 2021). The Innovation Fund supports private large and small-scale projects that have the potential to bring major emissions reductions in energy intensive industries, renewables, energy storage, CCS&CCU, net-zero mobility and buildings. Projects being funded must demonstrate a robust business model and capacity to deliver results. It has a potential budget of 40 billion euros over 10 years (2020-2030), under the assumption that carbon dioxide prices are 75 euro per ton. It was revised in 2023 to increase its value alongside the strengthening of the ETS and establishment of the cross-border adjustment mechanism. Additionally, funds allocated under the innovation fund must apply the Do no significant harm principle and must make stronger reference to environmental impacts.

The IF is not state aid and requires that remaining costs are covered in other ways including other support programs like the Just Transition fund, and Projects of Common Interest (PCI). The Innovation fund was developed to fit with Horizon Europe (targeting early proof of concepts and pilots) and the CEF (targeting technology roll out). In this way CDR firms in Amsterdam could, if eligible, receive support through their technology's development and operation.

A potential issue with the innovation funds system is that its funds are tied to the price of carbon in the ETS which has historically fluctuated wildly. Changes to the system which will come into force in 2027 will tighten regulation on the amount allowances being removed each year. This will create more stability in the price change of CO₂ and additionally bring stability to the Innovation fund. The competitive bidding process may cause firms in Amsterdam some issues if they cannot successfully demonstrate their eligibility according to the necessary criteria. Additionally, application and reporting in accordance with schemes of this nature is administratively taxing, particularly if done in tandem with other funding.

TEN-E		Direct	28/04/2022
Binding	Target: Firms	Deploy: 2022 Dutch Cabinet	
Sphere: Infrastructure		Material	
Approach: Carrot		Economic approach: Supply	
Connecting Europe Facility		Direct	2021
Binding (Budget)	Target: Firms	Deploy: 2022 Dutch Cabinet	
Sphere: Infrastructure		Financial	
Approach: Carrot		Economic approach: Supply	

To expand on an earlier point; The TEN-E regulation provides funding to projects that are identified as Projects of Common Interest which have significant impact on EU energy systems and its sustainability (European Parliament, 2023c; TEN-E Regulation, 2022). Established in 2013 with rounds every 2 years, the 5th PCI list was submitted in April 2022 with 6 CO₂ network projects identified. Funding will come from the Connecting Europe Framework specifically for energy systems towards decarbonization goals under the Europe Green Deal. The TEN-E regulation explicitly mentions CO₂ transport and storage infrastructure projects for the “permanent geological storage of Carbon Dioxide” (Annex 2.3). The CEF provides 5.84 billion euros in funding towards supporting the transition to clean energy. As noted earlier, Aramis has already been selected as a PCI and will receive funding from the CEF to develop its T&S infrastructure between Belgium, The Netherlands with potential to connect to France and Germany. While not directly beneficial to Amsterdam, it does provide significant assurance that the Aramis project will be completed and be operational before AEB’s capture stack.

SDE++		Direct	2023
Binding (Budget)	Target: Firms	Deploy: 2022 Dutch Cabinet	
Sphere: Infrastructure		Material	
Approach: Carrot		Economic approach: Supply	

Within the Netherlands, the Dutch government has come to the decision that CCS should be viewed as a transitional technology, providing negative emission till the point that national emissions are “virtually zero”(Government of Netherlands, 2019). To support CDR technologies to provide this transition the SDE++ and DEI+ state subsidies are available. The SDE++ subsidy with 13 billion euros of funding is designed to provide a 15-year contract for difference scheme where the ‘uncommercial’ costs of operation (where the firm would pay ETS allowances) are covered. It is of particular importance to early development of technologies, in particular as operations invest in new processes for decarbonization. Renewable energy and CO₂ reduction techniques eligible include: wind energy (on and offshore), solar and hydro-power. Projects using biomass, biogas and geothermal energy are not eligible for funding under this scheme.

To ensure that CCS remains a transitional technology several limits are placed on how the fund will contribute. Specifically, a cap, horizon and sieve are applied to applications from CCS projects. To receive funding the project must demonstrate that there are no demonstrably cost-effective alternatives, and only a cap of 7.2 MtCO₂ will be funded. All funding for CCS will have a horizon, ending in 2035 unless the project contributes towards negative emissions.

In 2021, AEB with the aid of Amsterdam’s municipal environment team applied for a subsidy from SDE++ to capture 440KtCO₂ for storage. The application was eventually rejected, but the firm is

allowed to apply in future proposal calls. Without the covered costs, the firm is not able to obtain a permit to operate its capture stacks. Additionally, it will not be able to construct CO2 capture for its biomass power plant.

Environmental investment Allowance		Direct	n.d
Non-binding	Target: Firms	Deploy: Netherlands Enterprise agency	
Sphere: Infrastructure		Material	
Approach: Carrot		Economic approach: Supply & Demand	

The environmental investment allowance allows firms and individuals to deduct sustainability investments from taxes (Netherlands Enterprise Agency, 2023). Up to 45% of the cost can be deducted based on criteria including firm size, risk, and innovativeness of purchase. The MIA is state aid, and there is a cap on state aid for all actors. If AEB is not receiving state aid from other sources, the investment in the capture points could be covered for 45% of the cost. This is a significant cost saving mechanism for Net-zero technologies for the Netherlands.

3.4 Normative frameworks

Normative frameworks as defined by my conceptual framework refer to the legal and non-legal rules, regulations, beliefs, and norms agreed or contested by actors in the system that define the sector. In the EU and the Netherlands, several mechanisms are designed specifically to managed how rules for the sector are defined. Most important to CCS and CDR are the EU's emissions trading scheme (ETS) and CDR certification framework.

EU ETS		Direct	2023
Binding (law)	Target: Firms	Deploy: EU	
Sphere: Normative frameworks & Infrastructure		Formal (binding) & Material (output)	
Approach: Stick		Economic approach: Supply	

The EU ETS has been hailed as the most visible and comprehensive environmental policy in the EU (Jänicke & Quitzow, 2017). The EU ETS is a regulatory market establishing a system for greenhouse gas emission allowance trading within the union. It enshrines the 'polluter pays principle' into operational procedures for the union's greatest sources of GHG emissions. The regulation requires installations to pay for GHG emissions above an agreed benchmark, with mechanisms to ensure costs are not unduly forked onto consumers. The ETS is the world's largest cap-and-trade scheme for GHG emissions and has been expanded and tightened several times since its inception in 2005.

The ETS covers all GHG emissions relevant to climate warming including NOx, Carbon dioxide and Methane. Many of the most polluting industries are covered including heat & energy production, heavy metal production, and the chemical industry (EU ETS, 2023). In 2023, the EU ETS was expanded to create a separate emission trading scheme covering other industries including

buildings, road transport, aviation, maritime transport, and crucially waste-incineration. The ETS2 will slowly roll out coverage from 2025, requiring waste-incineration plants to purchase allowances from 2027 for emissions in 2026.

Firms and sectors covered under the ETS and ETS2 must measure and report GHG emissions annually covered under the monitoring and reporting regulation (Monitoring and Reporting Regulation, 2020). The reports must then be verified by accredited national bodies in each member state, and firms then must relinquish allowances equal to emissions in the preceding year. To ensure sufficient coverage, firms may purchase EU allowances (EUAs) by auction or from the secondary market, traded to them by more efficient firms. To reduce undue burden the ETS provides a number of free allowances calculated according to benchmarks set in 2007-2008 of the 10% most GHG efficient operators in the sector (Life ETX, 2022). Thus, firms will only pay for allowances if they emit more than the top 10%, though this has historically not been the case. If firms do not pay for their allowances, they are subject to 100 euros fine per ton emitted. While mechanisms will adjust the figure based on market rates for EUA's the figure is drastically under the costs of damages calculated at 185 euros by recent studies on the social cost of carbon emissions (Rennert et al., 2022).

To support continued emissions reduction the ETS dictates that half of all revenues from the sales allowances by national authorities should go towards environmental action. This provides additional funding that member states can utilize in addition to the Innovation fund. This stipulation is not binding however, and laggards in climate action often do not comply. Additionally, during phase 3 (2013-2020), while 75% of revenues were spent on 'climate action', it was found that many countries either misreported, or intentionally overplayed the amount of ETS revenue utilized (Life ETX, 2022).

Crucially, and in accordance with its intentions to drive emissions reductions, CO2 emissions from biomass (biogenic CO2) do not count towards AEB's (and by extension Amsterdam) emission budgets (European Commission, 2022). For biogenic CO2 to be 'zero-rated' in this way must first pass sustainability and emissions-saving criteria following requirements set under article RED3 and LULUCF (Directive 2018/2001; (European Parliament & European Council, 2023). This means that as AEB fires with 63% biomass, its emissions are already reduced significantly, and likely place the firm in the top 10% performing municipal waste-incinerators. However, negative emissions, regardless of source (BECCS, DACCS, OCCS) are not currently considered under RED3 or the EU ETS and ETS2 directives. For installations with fossil-based CO2 emissions captured emissions are counted as 0, but for negative emissions there are currently no incentives from EU legislation under M4 article 3a (EU Commission, 2009).

CDR certification framework		Direct	Not in-force
Binding (regulation)	Target: Firms	Deploy: EU Commission	
Sphere: Normative frameworks		Formal (binding)	
Approach: N/A		Economic approach: Supply	

While discussions regarding carbon removals are taking place, international knowledge institutions including the IPCC and UNFCCC have not established recognized methodologies that set a precedence for carbon removal processes. While the IPCC has published recognized methodologies for GHG emissions and sinks it does not cover removals such BECCS and DACCS.

In lieu of established international standards, the EU commission has proposed a voluntary Union certification framework (Pereira & Drimie, 2016).

To bring negative emissions under European legislation the Carbon Dioxide Removal certification proposal to regulation (CDR Certification Framework, 2022) intends to cover the quantification and verification of carbon removals for industrial activities. The methodologies it will use for verification of GHG emission avoidance are outlined in Annex C of the innovation fund . This regulation is critical for the robust monitoring and reporting of removals, creating binding regulation addressing technical details crucial to effective removals including specifications of 'permanent storage' and 'additionality'. The regulation will cover removals potentials from known methods outlined in Fig X, establishing methodologies tailored each. Clarity and consistency in verification will reduce risks perceived by actors financing carbon removal technologies, plugging the 'market failure'.

The regulatory framework will set out an overall certification scheme based on the Q.U.A.L.I.T.Y criteria for carbon removals. The criteria will consider Quantification, Additionality with reference to baselines, Long-term storage, and sustainabilITy. Quantification of removals will be calculated according to this formula:

$$\text{Net carbon removal benefit} = \text{CR}_{\text{baseline}} - \text{CR}_{\text{total}} - \text{GHG}_{\text{increase}} > 0$$

Read as the sum of the removal activity minus the baseline and emissions from the activity. Additionality is defined as having two components; that removals go beyond statutory requirements and that they occur due to the incentive effect of the certification. The definition given for Long-term storage, seemingly interchangeable with 'permanent carbon storage' is storage for 'several centuries' by BECCS and DACCS (Proposal to regulation 2022/0349 COD, pg 10). Finally, carbon removal activities must have neutral impact or generate co-benefits for the environment. The regulation will require that a nationally accredited independent body verify removals.

It is difficult to outline what barriers or gaps the regulation might create once adopted as its content is not yet in force. In particular the methodologies for establishing Q.U.A.L.I.T.Y of removals are still being developed by the expert group supporting the EUC (CDR Certification Framework, 2022, p. 6). However, this policy instrument aims to create transparency and clarity for investment by creating a standard. In this way it would bridge a knowledge gap for investors.

The methodologies cited in the CDR certification framework under the Innovation fund Annex C cover emission avoidance activities for various industries. Updated in 2022, amendments were added to include net carbon removals (section 1.1.2.1). For projects to be identified as contributing net negative emissions they must exclude credits for non-principal products, and time operation. Direct-air capture projects will be given a blanket relative emissions avoidance score of 200%. Stipulations of section 3.1 place "CO₂ capture from biomass power installation" under calculation rules for section 4: Renewable energy. It makes the added stipulation that "use of biogenic resources is calculated with a **zero emission-factor** from the point source CO₂ emission that is captured" and that there is no distinction between fossil and biogenic CO₂ for calculation of a negative emission term (hereby 'credit').

Net project emissions for CCS projects are calculated as the sum of the emissions term and the credit term. Where the credit term is calculated as:

GHG emission credit	=	Sinks	-	Sources	
$CC_{credit,y}$	=	$\sum_{y=1}^n (CC_{storage,y} + CC_{use,y})$	-	$\sum_{y=1}^n (CC_{capture,y} + CC_{injection,y} + CC_{EHR,y} + CC_{pipeline,y} + CC_{transport,y})$	[3.1]

Table 9. Credit term calculation (EU commission, 2022)

Where the emission credit (CC credit for year y) is equal to the sum of total removals and sources of emissions in the CCS process, for example from road transport. Projects that represent only one component (capture or injection) or that are less 5,000km travel in total can ignore this. The emission credit is then included in the project scenario to calculate absolute and relative GHG emission avoidance. These calculations will be used in accordance with the monitoring and reporting regulation pursuant to the ETS. As noted previously, operators are not obligated to surrender allowances for CO₂ that are captured and certified. While the use of clear methodologies that account for negative emissions is valuable, the ETS does not yet incentive it beyond waiving the need to purchase EUA's equal to the captured emissions.

This is a significant missed opportunity to incentivize negative emissions by the framework. While the cost of CCS per ton of carbon is dependent on the method, they are all exorbitantly expensive at current rates. Firms implementing DACCS for example need to charge between 100 and 300 euros per ton of CO₂ removed, which only the largest of firms may be able shoulder as operating costs (IPCC, 2022; Williams, 2022). To at once incentivize an increase in supply and demand, mechanisms in the CDR framework and ETS should subsidize negative emissions. A change like this would improve the business case of AEB's BECCS operation and confer greater value to support the action by Amsterdam.

National Carbon Levy		Direct	2021
Binding (law)	Target: Firms	Deploy: Ministry of Economic Affairs and Climate Policy	
Sphere: Infrastructure		Material (output)	
Approach: Stick		Economic approach: Supply	

In 2021, the Netherlands introduced a national carbon tax (or 'levy') targeting activities covered by the ETS, in particular heavy industry (European Commission, 2021b). The aim of the instrument is to incentivize efficient GHG operation in the industrial and waste incineration sectors by setting a progressively rising tax on carbon emissions from fossil sources. A cost of 30 euros per ton of CO₂ will be applied starting in 2021 and reach 125-150 euros in 2030, taxed on top of the allowances they must pay for to the ETS. Only firms that fall below the benchmark set by the ETS are obligated to pay minus a set amount based on a predefined reduction path. As discussed earlier, this benchmark is set by the performance of the top 10% of operations in the sector. Firms that perform less efficiently than this will pay the difference in EUAs. Thus, companies in the Netherlands are incentivized to invest in emissions reduction so as to avoid paying the higher

costs of the ETS allowances overall (currently 79 euros as of November 1st, 2023). A recalibration of the system will be requested in 2025 to adjust ETS benchmarks. Significant effort will be expended by the government to ensure that businesses can avoid the levy through investment in measures to reduce emissions, speeding up permitting processes and providing subsidies in a timely manner.

The carbon tax currently does not cover waste incineration firms, as they are not yet covered by the EU ETS. The setting of their reference scenario emissions will take place over the year 2025-2026, after which they will be required to follow regulations set under the ETS and this levy. The establishment of this carbon levy will not likely affect AEB as the firm already operates within the 10% most efficient operations.

Chapter 4: Discussion

In this discussion I assume that Amsterdam will fail to successfully realize all emissions reductions defined in the climate roadmap 2050. Though it is possible that Amsterdam will reduce emissions to the point that no CDR is necessary. However, in this scenario Amsterdam could still benefit from CCS(U) activities in the port.

In reading of the relevant policy, three of four systemic failures can be seen in the policy landscape. Failures in infrastructural, institutional and capability problems exist across the system. The severity of the barriers is largely dependent on how necessary and how quickly Amsterdam needs to reach carbon-neutrality. In this paper I hold that the IPCC modeled pathway to 1.5C with no or limited over-shoot is necessary (IPCC, 2022, p. 26). Under these pathways the Netherlands must sequester 0-1070GtCO₂ through CDR from BECCS and DACCS cumulative from 2020 – 2100 (Fekete, 2022) to account for historic emissions.

Amsterdam's plan to reduce emissions by 95% is ambitious, but current analysis of national emission pathways indicates that greater action is needed to reach it. Analysis by various models for the Netherlands and the EU indicate that insufficient action is being taken towards a 1.5C world (Climate Action Tracker, 2023; IPCC, 2022, p. 202). The Netherlands, and by extension Amsterdam, has not modeled its emissions reduction pathway in regard to historic emissions. Total emissions reduction would need to reach 94% by 2030, or emissions will exceed the Paris Agreement 1.5C world carbon budget by 1070MtCO₂. Additionally, current CCS projects the Netherlands with the capacity of 414MtCO₂-eq in fall far short of being able to capture it. In light of this, and with respect to Amsterdam's capacity to act, barriers regarding physical infrastructure, biomass cost and availability, and gaps in financial support and carbon accounting must be solved.

In the European union and the Netherlands, there are a wide range of policies, laws, activities, and other instruments that cover activities related to carbon dioxide removal. Here I will discuss the implications of each instrument set out in the results to answer my research questions.

4.1: Barriers to addressing residual emissions

4.1.1 Amsterdam's relationship to CDR & CCS:

Sub-research question one asks, "What are Amsterdam's plans to addressing residual emissions in its carbon-dioxide emissions budget"?

Amsterdam's relationship to CDR & CCS is direct but largely opportunistic. The municipality plans to address its residual emissions by relying on AEB's BECCS technology. While AEB used to be a municipal company it is now a private organization after the municipality sold it to private operators. The privatization of public works in the Netherlands including water and energy procurement is commonplace, and cities less and less, own the operations of these services (). These services now operate on economic principles such that they can provide services at the lowest cost to their customers. Regardless municipalities are still these services' largest stakeholder and so must operate within planning and policy goals of the city as a result. As such the municipality is concerned primarily with the viability of the company for financial reasons and

for waste management requirements. In a discussion with the business manager for AEB it became clear that AEB's plans are very 'convenient for the city' (Interview 1, 2022). The environmental and sustainability team at the municipality see this plan by AEB 'contributing to policy goals' in the city, namely reaching net-zero emissions (Interviewee 1, 2022). While the city was not party to the plan's development, they now aim to support it.

In the Port of Amsterdam, South Pole and Bellona's report identified a number of companies implementing CDR & CCU with varying potential to sequester carbon permanently. Amsterdam's climate report and roadmap 2050 only makes mention of AEB, omitting the other companies operating primarily CCU activities. Though the South Pole report is positive that a few firms may be able to contribute towards emissions reductions in the city (SouthPole & Bellona, 2022). Though the firms would need to scale up significantly, Particularly Advanced Mentanol Amsterdam (part of Gidara Energy) and Cargill. By portioning some output of biogenic CO₂ towards Aramis it could translate to additional negative emissions for the city as a whole. However, this potential is only 'promising' after 2030 and presumably why they are absent from the Circular economy roadmap currently. As such AEB is the only feasible pathway for Amsterdam to address its residual emissions.

According to AEB the plan will contribute roughly to an emissions reduction of ~10% or 500ktCO₂. Though this figure is expected to change, depending largely on administrative decisions between AEB and the Dutch national government. These decisions depend on operating capacity of AEB, subsidy and permitting decisions by the Netherlands Environmental Agency (NEA), and the progress of Aramis. Which in turn depends on the wider policy environment and maturity of the sector. As such, AEB's plans to capture and store biogenic CO₂ is an extension of Amsterdam's ability to address its residual emissions. Vanishingly little can be done by Amsterdam directly.

The Dutch government has made clear that it considers CCS to be a transitional technology and has no intentions to support projects post 2035 (Government of Netherlands, 2019). Regardless of support from the EU through the innovation fund, CEF or Horizon Europe, the move would likely lead to a downturn in interest for CCS in the country. If Amsterdam, Rotterdam, or other municipalities at point still expect to require CCS then it would present a significant barrier.

4.2 Policy landscape

In answering research question 3 the orientation of policy towards industrial actors becomes clear. Of the 38 policy and legislative instruments analyzed, 28 are oriented towards supply and 12 have mechanisms to stimulate demand. This result comes from the fact that the major actors in CCS are private industrial firms in energy production, and the goal of many of the policy instruments is to reduce emissions in that sector. Given the industrial nature of the sector and the requirements to operate CCS, this distribution of market instruments is not surprising. Additionally, many instruments from the EU cover both supply and demand given their scope. The Renewable Energy Directive stimulates both grid level and distributed renewable energy production.

Overall, the EU, Dutch government and Amsterdam choose to incite action primarily through 'carrot' policies, providing economic or administrative benefits to actors. Several large funding schemes exist which favor net-zero technologies in addition to other mechanisms that funnel

investment funds towards those projects. Other instruments then reduce administrative burden or permitting requirements to lower the bar to action where applicable. The distribution is 18, 13, and 7 for carrots, sticks and sermons respectively. The sticks primarily show up in connection to obligations for climate action including emissions reduction (ETS) and sustainability (RED3).

As CCS is still a relatively novel technology, policy directly related is predominantly support focus, targeting pilot projects and knowledge development. This is apparent in the financial and investment programs which are aimed at developing material infrastructure either to pilot technology or enable future action. Where barriers exist, it seems that actors in the EU are aware of them or at the very least working on solutions. The same cannot be said for the Netherlands currently.

4.2.1 Infrastructure:

For Amsterdam to address its residual emissions a number of infrastructure projects (physical construction) must see successful development in the coming decades. This includes developments within its municipal boundaries and beyond in the port of Rotterdam and offshore in the North Sea. This includes the construction of new pipelines, storage facilities and capture facilities. Spatial planning in this regard is not a competency of the EU, and relevant legislation and policy is directed by national governments.

Close to home, Amsterdam depends on AEB to build its capture stack, planned for completion in 2028 (Interview 1, 2022). The capture system will utilize an amine-scrubber process which is a proven but costly technology and has been widely applied (Kearns et al., 2021; Panja et al., 2022). AEB will need to find support to finance the investment, and the permits to operate it. Nothing in conversation with AEB or Bellona suggests that engineering is a significant challenge for the organization (Interview 1 & 2., 2022, 2023).

For the captured Biogenic CO₂ to be stored, it needs to be transported from Amsterdam to a compression plant in the port of Rotterdam. To do this AEB intends to connect its plant to the OCAP system which carries high-quality CO₂ to horticulture applications in the region. AEB is confident in this arrangement (Interview 1, 2022), but it is not clear if it has been contractually arranged yet. To make the connection 2km of pipeline must be laid within the municipality of Amsterdam which necessitates an environmental permit to be issued. The municipality of Amsterdam is the competent authority to issue the permit as it falls within its boundaries and will arguably not cause 'significant environmental damage' (Koninkrijksrelaties, 2021). This is not a significant issue for Amsterdam or AEB as both parties are in close contact with one another in the development of the BECCS operation.

Beyond Amsterdam, major infrastructure elements will need to be developed for the Aramis CCS project to become operational. The current timeline expects operations to begin in 2028, though operators acknowledge that it is likely that that date is postponed due to public resistance and appeals to permits (Aramis, 2023a). Still to be developed are interconnections between the port of Rotterdam and industrial partners, the main trunkline from the port of Rotterdam to the storage site (Aramis, 2023b), and storage licenses. To do this Aramis requires environmental permits from the RVO, with the BZK in cooperation to conduct. This process to conduct environmental impact assessments (EIA) is underway and permits are expected to be issued in 2024. However, there is a good chance that like the Porthos Project, environmentalist groups will contest the permit and project, delaying it (Reuters, 2023b).

The decisions regarding Aramis are beyond Amsterdam's remit but for the city and the Netherlands to continue to utilize CCS as a technology to achieve its fair share emissions reductions significantly more infrastructure is required. Porthos and Aramis will only have 414MtCO₂ capacity between them, though explorative studies indicate between 2700 – 3200MtCO₂ of storage excluding Groningen. The largely untapped field only represents an estimated 9,000MtCO₂ of storage, though utilizing it will be a severely unpopular move (Reuters, 2023a, 2023b). Even in combination, storage capacity in the Netherlands cannot capture its total fair emissions budget of 1070MtCO₂-eq. Requiring capacity in other parts of the EU to become open to storage of Dutch CO₂. This will necessitate an expansion in CO₂ transportation, potentially through Aramis and rail lines.

4.2.2 Biomass:

For AEB to operate its BECCS operation it will require the continued supply of Biomass which may increasingly become an issue as the resource becomes contested. The current and future regulatory landscape affecting biomass as a renewable energy source is tightening and will restrict AEB and other firms in the port of Amsterdam that utilize it. In conjunction will be a continued rise in use by the EU as a renewable energy source.

Strategic vision setting in Amsterdam and the Netherlands set biomass to be a continued necessary resource for the foreseeable future. Additionally, the EU sees biomass, and other biofuels to be a significant contributor to the renewable energy stocks for the union (European Parliament & European Council, 2023). However, in response to continued concerns regarding the sustainability of biomass, significant sustainability criteria have become progressively tighter in the union.

The municipality of Amsterdam draws a significant portion of its heat and energy from natural gas and biomass firing in part from AEB. However, due to sustainability concerns it has stated that it looks to end its use of biomass for electricity and heat (Gemeente Amsterdam, 2020, p. 149). For situations where it must, the city will apply strict biomass sustainability sourcing criteria. Presumably based on national certification frameworks based on criteria in the RED2.

Several policy instruments are aligned with this trend of tightening sustainability criteria, namely the RED2 with the LULUCF, CAP and now the NZIA applying the criteria. The Recast RED3 further tightens the criteria, and additionally expands its coverage to smaller energy and electricity plants that use bioenergy almost certainly covering AEB. It additionally tightens requirements for the sourcing of biomass from forestry, defining any biomass taken from an area that was of high value in 2008 to be High ILUC risk material. Specifically advising against the use of direct financial support towards certain woody biomass. In addition to implementing measures to ensure the application of the cascading use principle (European Parliament & European Council, 2023, p. 10). The recast of RED 3 has resulted in a wave of studies attempting to understand its implications for biomass availability for the EU.

The large collection of reports and academic studies is indicative of concerns for the availability of biomass for bioenergy for future renewable energy targets in the EU and worldwide (PBL, 2020; Schueler et al., 2013; Fuels Europe, n.d; EU Commission, 2018x; Thran et al., 2010; Long et al., 2013; Concawe, 2019). Conclusions report a wide range of numbers based each on different viewpoints on relevant scenarios. While most reports agree that total stocks of biomass worldwide will increase, the amount of sustainable biomass is contested. Schueler et al, (2013)

estimate that 10% of biomass in global stocks are 'free of sustainability concerns' based on the RED1 criteria. The same analysis today would likely return a more limited number. A more generous analysis by IRENA overlapped with IEA data reports a potential of 5,700 to 7,000Mtoe (Mega tonne) of sustainable dry biomass for bioenergy worldwide in 2050 (IEA, 2017; IRENA, 2016). This number is very optimistic and is contrasted by a study which estimates a total maximum of 2,150Mtoe/y worldwide in 2050 (Searle & Malins, 2015). A report by EU commission using domestic data up to 2018 projects domestic supply at 170-252Mtoe/y for 2050.

The national government, aware of the reduced biomass availability in the future has a national biomass strategy to increase domestic stocks (Ministry of Economic Affairs and Climate Policy, 2016; Netherlands Enterprise Agency, 2014). However, the amount of biomass that could be expected from domestic supply under this scheme will not be able to satisfy national consumption (PBL, 2020). As such the Netherlands has shifted to a strategy of importing supplies from the EU and other countries, particularly the US. Imports from the U.S. totalled at 223 million USD in 2021(USDA, 2022).

Depending on which scenario is considered AEB may not be able to acquire the biomass it needs to operate and produce Biogenic CO₂. It may in this case have to rely entirely on municipal organic waste which it already uses (AEB Amsterdam, 2022). Unless the Netherlands and NoordHolland's RES continue to support municipal incinerators, this may end biomass use at AEB. If the strategy of imports continues and does not conflict with RED3 criteria, AEB may be covered for some time. However, if availability of sustainable biomass for import diminishes it may call into question the viability of AEB's operation. All of which is speculation and requires more time and study once RED3 and NZIA have been passed to better understand future trends.

4.2.3 Financial infrastructure:

A major barrier to Amsterdam's plan is acquiring financial support for the investment in the amine-scrubber capture point at AEB. Such an operation is often very costly and depends on factors including pollution control design, the scale of the plant and pressure of the gas leaving the incinerator (Kearns et al., 2021). Though, as AEB intends to construct the capture stack and system anew, they can avoid retrofit costs which are often much larger (OSTI, 2002). Additionally, it is also costly to operate, requiring among other resources additional electricity generation to run the system (Panja et al., 2022). Without a subsidy, AEB cannot afford the capex and opex costs.

In the Netherlands and EU 8 funding schemes are available to fund emissions reduction and renewable energy projects. There are many more, but the wording of their eligibility criteria excludes such projects as AEB is planning. Together the funding portfolio covers research, pilot, demonstration, and deployment projects. The support under the Horizon Europe, regional development fund, CEF, and MIP are meant primarily for the support of CDR relevant but indirect components. Covered are but not limited to research, transport infrastructure, and sustainable energy systems. Of the identified funding schemes AEB's capture stack is only eligible to be funded by the EU's innovation fund. In its first and second call 4 of 7 eligible CCS projects in France, Belgium, Finland, and Sweden were awarded funding by the scheme (Global CCS Institute, 2022). In its 2023 third call, the innovation fund will specifically target renewable hydrogen production projects (European Commission, n.d.)(EU Commission, 2023) with an eye towards industries now covered by the revision of the ETS directive. Importantly the IF will stay technology

neutral in funding calls so AEB's plans are still eligible. It is unclear if AEB has made previous applications to the Innovation fund and was not mentioned in the interview.

In the Netherlands, the SDE++ and DEI+ funds are most relevant. However, under the SDE++ scheme biomass projects are not eligible for funding (EU Commission, n.d). Funding towards CCS is only eligible in contexts where there are no other sustainable alternatives to emissions reduction (Netherlands Enterprise Agency, 2022). This criterion implies that only projects aimed at capturing fossil CO₂ are eligible. This is likely why Amsterdam and AEB's application to the scheme in 2022 was rejected (Gemeente Amsterdam, 2022). The firms will need to locate alternative sources of funding or hope that the tabled NZIA regulation will change eligibility criteria in its favor.

4.2.4 Normative frameworks:

The methodology for GHG accounting currently used in EU and Dutch legislations works in AEB favor, but does it not effectively incentivize negative emissions. Current perspectives of the value of CCS are predicated on its use in heavy industry to reach carbon-neutrality. However, BECCS can create net-negative emissions in total project emissions and contribute to overall climate action by the union. However, this is not reflected in the in-force legislative environment.

How AEB's project emissions budget is counted is most crucial in regard to its future inclusion under the ETS 2. The expansion of the ETS to cover aviation, maritime transport and waste incineration will capture a significant portion of EU's emissions and lead to more effective emissions avoidance. Under the directive, AEB will be required to purchase an EUA for every ton of CO₂ that it emits above a benchmark set by the EU Commission. Set according to the top 10% of emission efficient plants, every ton above will cost, currently, 79.65 euros as of the 26th of November, 2023 (Ember, 2023). In 2026, when AEB will come under the ETS2 system and be obliged to purchase EUA in 2027, the price per ton is forecasted at 107.97 euros in 2026. Prices are expected to fluctuate however as energy security continues to be an open question for the union (Vitelli, 2023). AEB may not need to purchase any allowances, given that 63% of its total emissions per year are biomass, placing it in the 10% of installations.

Under current ETS methodology for GHG emissions accounting (2023) biogenic emissions count equal to the emissions of fossil fuels. Amendments to the ETS will align accounting with the LULUCF directive so that biogenic emissions are zero-rated (ETS Directive, 2023, pg 63). Zero-rated refers to the concept that 0 additional tons of CO₂ are added to the atmosphere from combustion. The carbon came is captured by plant matter from atmosphere and released shortly after, leading to no net change in total global stocks after a year. Once in-force the CDR certification framework will then apply the 'zero-rated' concept to installations and activities managed under the ETS.

The methodology and formulas contained in the Innovation funds Annex C are set to be enshrined under the CDR certification framework pursuant to the ETS. The formula for Biogenic CO₂ production from bioenergy and heat is the sum of emissions from all sources and the capture credits from the project. For AEB this means that its project emissions per year will be:

Total Project emissions (Year) = Fossil emissions (37%) + process emissions - captured emissions (~500ktCO₂) + 0 (uncaptured biogenic CO₂)

When the captured emissions are stored in Aramis, AEB's operation will have net-negative emissions or >100% emissions avoidance for the year. Though that depends on the carbon intensity of its fossil emissions for the same period. I was unable to identify total installation emissions from its websites and interviews. However, the firm reported 37% of total emissions are from fossil sources and 67% from biomass. An effective >100% emissions avoidance value means that AEB would not pay any additional costs for EUA's, presumably as it would one of the more efficient waste-incinerators in the union. As a result, AEB avoids payments to the Dutch national carbon tax for the same reason. Though a significant benefit for AEB as it is already far ahead of the curve, the investment in its capture stack does not create additional benefits for the firm. Under the CDR framework or ETS, no benefits are awarded to projects with greater than 100% emissions avoidance. Doing so under the present policy landscape is a superfluous achievement and significant cost for AEB beyond demonstrating that it is possible.

4.3 Significance of the Net-zero Industry act for CCS:

In April 2023 the EU Commission adopted a proposal to regulation named the Net-Zero Industry Act (NZIA). The aim of which is to dramatically accelerate the uptake of net-zero technology and injection capacity for CCS in the union. The regulation includes the EU-wide binding target of 50MtCO₂ annual storage capacity by 2030, to scale up to 550MtCO₂ by 2050. This regulation, if adopted by the European Parliament and Council as is, represents a significant shift in stance by the union towards CCS. Additional support from the regulation and relevant mechanisms could make it significantly easier for AEB to find funding and permits for current and future BECCS plans.

The NZIA will encourage member states to make significant changes to their NCEP's, stances on CCS and funding. The Netherlands will likely reverse its stance regarding the temporary nature of CCS(U) as part of its national climate action plan. This could lead to an expansion of the SDE++ subsidy, possibly making AEB eligible to receive the subsidy it needs. For Amsterdam, it signals that the union is in support of its climate action plan, perhaps as a result of advocacy from the municipality.

Despite the value of this regulation for industry, the potential risks as described by environmentalist groups will grow similarly (Cassier, 2023). The NZIA will oblige oil and gas companies to act as front runners in the union utilizing their industry expertise. If these actors are then permitted to capture their fossil emissions, they will simultaneously avoid ETS costs, and receive significant support to do so by developing CCS capacity. Additionally, the demand for available biomass that passes RED3 sustainability criteria could significantly increase, creating incentive for actors to cheat sustainability criteria in environmentally sensitive land. Without strict checks and balances, the potential for social harm is not negligible. The potential for damage is greatest if alternatives to biomass for high-temperature net-zero industrial operation do not materialize.

4.4 Options to address barriers

Amsterdam's main tools for addressing barriers to AEB's CCS operation are administrative support, and advocacy at the national and EU scale with relevant actors. Additionally, Amsterdam could initiate education and resident engagement projects, but is unlikely to do so for CCS without additional support. Beyond this, as the municipality is only an authority for local spatial planning and management it does not have the capacity to address barrier to CCS more broadly.

A revision to the environmental management law (Omgevingswet) that is expected to come into force in 2024 and expand its responsibilities and powers slightly. Primarily in regard to setting 'environmental values', though the city already has a climate roadmap and adaptation plans. While additional support and resources will come with the expanded responsibility, it is not yet clear on what scale and mechanisms they will be available. In respect to permitting, Amsterdam could decide to prioritize AEB's permit by streamlining the process and reducing permitting costs. This is at most what the city can do directly for AEB.

This comes as part of an increase from national and international actors for increased action at the local level on climate change. Given that 72% of global emissions are tied to cities, there is a significant part for urban areas to play. However, cities rarely have the capacity to act on the growing responsibility placed on them. Only the largest and richest municipalities can afford to have this capacity in-house, such as Barcelona, New York, and Amsterdam. Thus, national and international bodies need to build out more mechanisms so urban centers can systematically interact with the process in constructive ways.

Indirectly Amsterdam is able to do so as it is party to a number of city networks and lobby groups that give it a platform to speak directly with national and EU level leaders. With this position, Amsterdam could advocate for increased ambition regarding CCS, AEB's eligibility for funding, and amendments to existing legislation. This is Amsterdam's most influential line of action for it to address its residual emissions which it is already in the habit of doing. Mentioned in its 2050 climate roadmap, Amsterdam is using its membership in the G4 municipalities and offices in Brussels to speak directly to EU commissioners and Parliamentarians. The association of municipalities in the Netherlands gives the city the power to do so at home as well.

Additionally, Amsterdam's membership to C40, Net-zero cities, and Global covenant of Mayors for Climate and Energy (GCMCE) allow it to work with other cities around the world. These networks are quickly becoming very important to multi-level governance discussions on the climate (Global Covenant of Mayors for Climate and Energy, 2022). Both C40 and the Global covenant of mayors have become significant players at the Conference of the Parties (COP) which brings to global attention the needs and effectiveness of climate action at the local level (C40 Cities, 2022). Through these collaborations, and with sufficient interest by the municipality, Amsterdam could advocate for CCS. The results of which would potentially lead to more countries adopting ambitious targets for CCS.

4.4.1 A global market for BECCS & DACCS

If worldwide capacity for BECCS and DACCS increases significantly in the following decades, it is possible that decentralized CDR will become available in the international market. The development of a global market for CDR offsets would require learning from the past and robust internationally recognized CDR monitoring, verification, and storage frameworks and mature

technologies. Such an international system would optimistically be possible by 2050, though significantly more capacity is needed (Casey, 2023). Analysis by the IPCC's WR6 consistently indicates the need for 1GTCO₂/y by 2030 (IPCC, 2022).

In this context it is possible that Amsterdam or the Netherlands would be able to purchase carbon sequestration offsets to cover parts of its emissions budget it cannot do domestically. The risk is that countries and large firms will then use this market to avoid the work of emissions reductions. This is likely to happen given the current issue with reporting and validating emissions avoidance offsets (Climate Social Science Network, 2022). Lessons learnt by the international community would need to be applied to avoid monitoring, credibility, and cost issues.

4.5 Contributions to the literature

In this paper I investigate how Amsterdam plans to address its residual emissions, as stated in its climate roadmap 2050. I then perform a content analysis of relevant legislation, policy, regulation, and relevant multi-governance structures that make up the policy landscape for CCS in the EU. The results of the study identify barriers in biomass availability, a lack of necessary infrastructure (physical, knowledge, and financial), and a missed opportunity to incentivize CCS in the carbon accounting framework. This paper thus contributes to the discussion on urban centers' potential contributions to CCS and climate-neutrality more generally. Currently the field is largely focused on green infrastructure's capacity for carbon capture in cities (Chen, 2015; Huovila et al., 2022; Shadman et al., 2022).

To aid this work I develop a framework for laying out the policy landscape to identify where barriers and gaps are present in the multi-level governance structure for a technology. The framework categorizes policy by approach, mechanism, target, scale, allowing the user to compare it to the policies stated and implied purpose and uncover barriers and gaps. By connecting the landscape then to relevant actors in the multi-level governance structure, solutions can then be identified.

This framework could be applied to other technologies including sustainable building materials, distributed energy technologies and heat pumps. Additionally, landscapes could be compared to one another to identify missed opportunities in overlapping goals and interests. As well as identify where unintended barriers are created for other technologies. Future research will be needed to determine the value of this framework and its value for integrated policy analysis. Additionally, future research could examine the framework's application to non-technological items such planning practices.

4.6 Limitations of research

To conduct a part of this study I intended to conduct interviews with relevant stakeholders and actors at each scale of governance. However, during the course writing this report only 2 actors agreed to be interviewed, both connected to the local scale. Without an interview or discussion with the municipality of Amsterdam it is difficult to draw conclusions regarding the validity of the analysis of the results. In addition, I do not have a clear understanding beyond written reports

the views of the Netherlands national government, European Commission, and other organizations. While this did not require a significant change in methodology for the thesis, I acknowledge that I am potentially missing important information.

Additionally, this paper was written over the course of an extended period of time than was first intended. As a result, a number of significant changes to the legislative environment are captured that would otherwise have been missed. In particular, are changes to the ETS, CDR certification framework, and NZIA.

Finally, my analysis of Dutch national policy was dependent on translated reports. Not all available translations were official, and many could not be located. In these circumstances, DeepL translator was used to translate the legislative documents. The translations cannot be substituted for the legal documents, and I acknowledge that important details may have been mistranslated by the software.

While the stage for municipal involvement in CDR technology is likely to accelerate significantly as a result of expected amendments to the NZIA the result of this paper has some limitations on explanatory power for other cities in the EU. Amsterdam is an outlier given its ambitious climate-neutral roadmap and a specific relationship with the municipal waste-incinerator AEB. The exclusive status is changing rapidly however, and lessons learnt from how Amsterdam addresses its residual emissions could stand as a template to other EU cities.

Chapter 5: Conclusion

The urgency of achieving carbon neutrality cannot be overstated. The daunting challenges of climate change demand immediate and innovative responses, particularly in urban environments. In this paper I show that cities like Amsterdam at the forefront of ambitious climate action can integrate Carbon Dioxide Removal (CDR) and Carbon Capture and Storage (CCS) technologies. Despite facing numerous obstacles such as the need for substantial infrastructure development, biomass availability, financial constraints, and complex carbon accounting frameworks, Amsterdam's proactive approach offers valuable insights into the policy environment.

Amsterdam's efforts to adopt CDR and CCS technologies, is not a solitary effort but a part of a larger multifaceted approach involving various policy instruments and legislative measures at both the EU and national levels. The city's direct but largely opportunistic relationship with CDR & CCS, particularly through the AEB's implementation of carbon capture technology, highlights the nuanced interplay between municipal planning, private sector participation, and overarching policy frameworks. Moreover, the rapidly evolving policy landscape surrounding CCS with instruments like the Net-Zero Industry Act and ETS provides a backdrop against which cities like Amsterdam navigate their climate action strategies.

In closing, the pathway towards carbon neutrality, as exemplified by Amsterdam, is rife with challenges but also rich in opportunities. This paper's exploration of Amsterdam's strategy in integrating CDR and CCS into its urban framework offers a valuable case study that can be applied to the larger global effort to combat climate change. As cities continue to evolve and innovate in their climate action plans, Amsterdam stands as an example of the transformative potential of localized yet globally impactful environmental initiatives.

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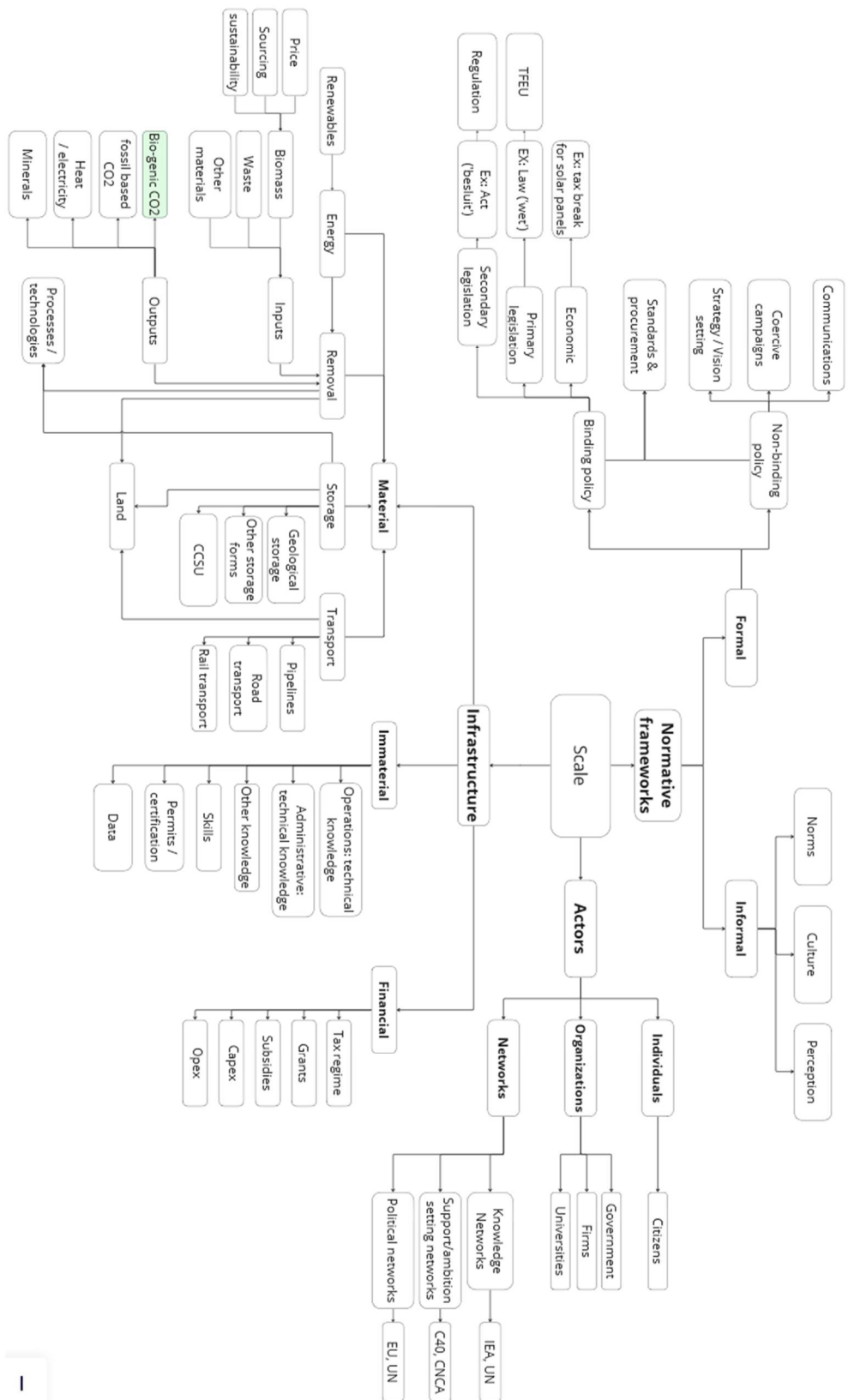
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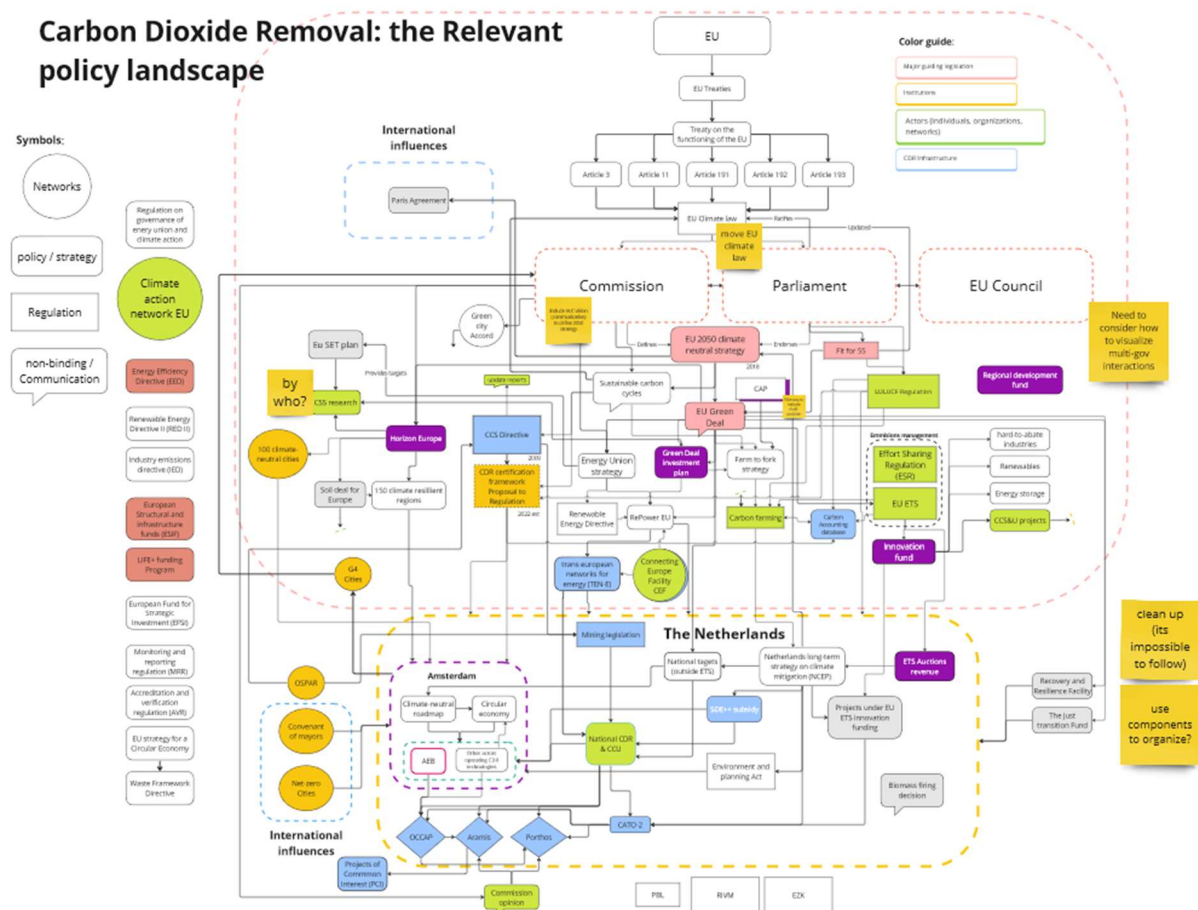
ANNEX A: Additional Figures



CDR/CCS technology sector breakdown

This table is a reference chart of the components of the CDR/CCS technology innovation landscape. This table was used in addition to the question listed in the methodology to categorize legislation and policy regarding CCS/CDR.

Full landscape:



APPENDIX B: Interview transcripts

Interview 1: Micha Hes of AEB Amsterdam, 2022

Before the device comes on, I'd like to ask you maybe a few questions in terms of, um, getting an idea of the context of uh.

The situation.

So.

I think you should understand that I'm doing this as part of my master's thesis here at the university. I'm looking specifically into carbon dioxide removal, and it's specifically in its relations to Amsterdam.

And the context, the policy.

The assignment, is it is it your own or were you given an assignment by someone?

No, this is a this is a personal thesis project I'm working with in collaboration with my professors at the Environmental Policy Chair Group in Wageningen university. But this is a personal, personal thesis work.

Have you already spoken to someone from the city of Amsterdam? Not yet, unfortunately, I have not been able to talk with them.

Um, as of yet, although I'm trying to organize that.

Because there has just been a report written assigned by the city of Amsterdam on carbon dioxide removal potential within Amsterdam's borders. It is being conducted by a consultancy firm called south pole and you may want to get a hold of that report.

And the person responsible for that is Hannah de Valk from the department of sustainability, especially on carbon capture usage and storage.

[omitted audio to protect Hanna de Valk's personal contacts]

On different levels, because we are the largest industrial emitter in the Amsterdam region and one of the largest producers of short cycle carbon. So, within your framework of research that might be of interest.

So, let's start at the beginning. Can you perhaps describe who you are and what you do at the AEB?

Yeah, my name is Michael Hes, I work as a strategic business development manager and strategic manager for AEB. I'm responsible for the CO2 program. And one of the key things I do at work is responsibility for the CO2 capture plant and the storage of the CO2 with AEB.

And what does the, what does the AEB do?

AEB is a city owned company that has been for the last 105 years responsible for the digestion for household waste within the city of Amsterdam. Started as a company within the city and now we are corporatized. We own 6 production lines capable of digesting 1.4 million tonnes of waste every year, and we supply 900 petajoules of heat and electricity. You can find that on our website of course.

We also have a post separation plant for household waste and also a large amount of secondary raw materials that we extract and retrieve from the waste. And besides that we also have a Biomass powerplant. And we are a company that basically working on a clean and sustainable removal of waste and production of sustainable heat and power.

Alright, fantastic. And of course the AEB sits in the municipal borders of Amsterdam. And you work closely with Amsterdam, and how does AEB factor into Amsterdam's climate plans? How much?

Yeah we have a plan of capturing roughly half a Mg tonne of CO₂ which represents roughly 10% of the CO₂ emissions of the city.

And you plan to capture this by is there a date that you plan to do this?

Yes, we want to start capturing by 2028. With amine scrubbing technology capturing plant and then by transporting the CO₂ via the OCAP pipeline starting.

Almost 2km south of us, so we need to make that connection which is the least of our worries. Then transport via OCAP to the portal pipeline still being built in Rotterdam and from there to the 2 Maasvlachte where it will be connected to and will be pressurized until 80 bars to go Aramis trunk line to the storage of the depleted gas fields of total energies and Shell.

The only thing that is currently available is the OCAP pipeline between Amsterdam and Rotterdam with the small gap of 2km. And we are planning to build and that's what we're preparing for next year and we want to start building in 2025.

Okay so by this timeline you should hope to start capturing half a Mgt Co₂ and storing in the offshore oil field.
Gas field.

Yes, excuse me, gas field.

Depleted gas field to be precise.

And is any of this gas going to be used, as I understand to pull out more oil or gas?

No, no no no. Absolutely not. There is no talk of enhanced oil recover as is known, this is sequestering CO₂ period.

Okay.

And this is not only our CO₂, this is the CO₂ of more companies, um a total amount of 5 million per annum.

So you mean to say that there will be, you will take on a contract to capture carbon for other industrial partners?

No we will capture for ourselves. There are other companies that will capture and store their own CO₂. Please be aware that our CO₂ has a small fraction of fossil CO₂, only 37%. The rest is short cycled biogenic CO₂ and what we will do is that 63 percent will actually be met with the fossil emission that we still emit. But that's if the EU commission will allow us then we are able to administratively swap the biogenic part with the fossil part if necessary.

So, and before we do the OCAP will use our biogenic Store CO₂ to swap the fossil CO₂ that they already use for the horticulture industry.

To be administratively agreed, be short cycled.

Ok?

So a part of the CO₂ that OCAP is using at the moment for the greenhouses stems from fossil plants, namely Shell Pernis. And we are able to partly or fully administratively swap it into a biogenic CO₂ because of the sequestration of our CO₂.

Yeah. And this, uh, this will all sort of go into the goal of carbon neutral.

Um, in the our goal is not to carbon neutral. We are able in the end if we could, if we can swap. So if negative emissions are allowed to be swapped with fossil emissions by the European Commission to theoretically abate or reduce our, our CO₂ emissions by 85%.

OK.

So what we roughly have is 5500 and 60,000 tons of CO₂, fossil CO₂ per year. And with the 500,000 tonnes of CO₂. Sequester it, then we can have 85% reduction.

That's more than what we do so far, but at the end of the day, It's more than 10% of the total emissions by the city of Amsterdam.

OK. So to to what capacity, you've already said that 500 kilotons of CO₂, is that the goal in 2028? Are there plans or are there intentions to increase this, this capacity further?

No, no, this is this is maximum, this is also what we are permitted to do. Uh the **subsidy** that that comes with it only allows 8000 hours per year to capture. But since we are able that's more of an administrative thing that, so we only get 8000 hours per year subsidized. We have a flow ton per hour that we can actually work with. So, all together we have a maximum subsidy and that's what we will capture.

If there is another capture blend, we don't have the room for that. But uh, we are also starting on working on a CO₂ capturing on our biomass power plant, but this will only start the earliest 2032 I guess. And this is only biogenic of course because it's a biomass power plant. We can either store that CO₂ or supply it's for the chemical industry mainly I assume for fuel, jet fuel production.

Hmm.

For combination, of course with hydrogen. So hence the 2032. Hopefully there a stream large enough to actually be able to work on that

OK. So. You mentioned that you, um you have you only permitted to, um, capture 8000 hours of CO₂. By which subsidies and by which legislations sort of limit this this number? For AEB.

There a subsidy that if you want to do more you need to get another subsidy but you also have a permit and a capacity of a plant. The permit and the capacity are jointly developed in the sense that we are kept with what we can capture otherwise we need another permit and of course there's a limitation to the size of a plant if we want to build a bigger plant. Then the whole business case that we already have will become very feeble, so we needed to fix the scope of the project which is now quite small. Half a megaton so if we would like to do more.

There's another aspect. So it's space, it's permitting, it's business case, but it's also.

Will this this increase be able to. Are there plans to sort of take on more of that budget of Amsterdam?

Is Amsterdam in the idea of that it might increase. This amount that I might have capture increase or are they sticking to that that target.

It's our it's our project although we are owned by the City of Amsterdam. It's our governance. The only thing we need to do is because of the scale of this project the investment and the contracts we that that lies.

Behind it, we need to get an OK from, from, from the shareholder from the city, but we are this is our project and the fact that without that, doing this project we would lose our license to operate.

That is for us more important than the 10% reduction for the city of Amsterdam. It's convenient that it's aligned with our, with our goals of course. But we are a company and the city of Amsterdam only has a common purpose in that sense. So, the city of Amsterdam is not really in a in a sense pushing this goal of 10%. Not at all. It's a convenient thing for them, because we need to have a business case. If there's not a business case, we can say "yes we want to do it, but please add some money with that wish".

And so that's it's we are on our own when it comes to being a company and to make money and the continuity of the company is of the highest interest for us.

And that's, that's also what the shareholder wants. But the shareholder wants shareholder value. And the shareholder value for a city is not money but is contributing to policy goals. And the policy goals that we contribute to is a clean environment. Namely, no waste in the streets, or the most sufficient and efficient

digestion of waste. And the cleanest. That's what they want. So to reduce no polluting of air, soil or water. That's what we do.

Ok, and does the city of the municipality of Amsterdam put any resources towards this?

Zero none, zip 0. And it's important to understand this. Our project is our business. And the shareholder supports it because it's aligned to the more that the policy goals that they that they are trying to achieve.

OK. And then I think as a final question is there.

There's one more thing there. So it's there's one department looking only at the financial stability of AB as a shareholder and there's the other department that is only interested in the sustainability. So the policy oriented goals and yeah for that and for the other part of the city it's only being a shareholder and shareholder value and the fact that we are able to do this at a good healthy way. In, in financial terms, not in, not in policy terms. So, bear in mind there's a, there's a contradiction within the city of Amsterdam itself, our formal shareholder doesn't give a flying **** so to speak about what we do as long as we do it with a profit.

I'm surprised that sort of answers my question in terms of is the, well actually no. I guess the question then is what could Amsterdam do? Is there anything that the city or the municipality could do to sort of improve the ability for Common Dioxide Removal at AEB?

No, no, no. Not in scope one. They can do it at scope two level, but that means that the stream would need to contribute to our neighbours. There's a scope 2 interest and also a reduction of CO2 at a global level for Amsterdam but it's not administratively going to be helping us at all. But will be helpful for the city of Amsterdam there, you might say well there might be some investments in infrastructure that are not.

On top of AEB, there's port of Amsterdam, as another city owned company, and there's the city itself. So by the port of Amsterdam, it might be the vehicle for the city to actually do some investments. The city itself won't do any investments or contributions whatsoever.

And I need to run. So you have more questions.

No, I think that that should cover a lot of the information that I was looking for the last thing. One last thing is that, whichever way this ETS and certification framework goes, which would be the most what would be the most beneficial pathway for AEB and for common dioxide removal as a whole?

That negative emissions should be accounted for within ETS.

Alright. Well that's very clear. Thank you very much. Thank you time.

I hope that you can sign the document that I sent in the e-mail very recently just so they'll do that on Monday. Is that OK?

Yes, that's alright. So that I can we can interview as part of my thesis.

OK, OK, good luck with that. Hopefully I can read your thesis once it's ready. And please, uh, make a make a contact with Hanna default. She will help you immensely.

I guess I will. Thank you very much for that.

Cheers. Bye, Nicholas.

Interview 2: Samantha Tanzer of Bellona Europe, 2023

nk to recording: https://unitednations-my.sharepoint.com/:v:/g/personal/nikolas_lanjouw_un_org/EeEY1BZF4AtEsRrB1xV38V8Br11rb7MQIM5pQv3MOwEbW

Request permission from Nikolas Lanjouw if you need access.

Start of Transcript:

Nikolas Lanjouw (Intern)

Yeah, just do sort of give of let's let's sort of start at the beginning, Umm, Amsterdam is planning to use some amount or it's it's in 10, it's a calculating umm the use of Cdr in its carbon budget and its carbon accounting.

You.

Well, let's sort of first start with the that we bought.

Can you summarize what the goal of the report was?

UM, between and and how you were collaboration with South Pole sort of figured into that?

Samantha Tanzer

So this was a project with uh socially before we started at Belona South Pole, approached Belona to see if we wanted to be a part of this and they've been approached by the the city of Amsterdam.

I believe it's been I put.

I don't actually know how that tendering process works, but anyway, so the Amsterdam had been doing a couple of different sort of quick scan things.

They have one not just for Cdr, but they also did one on CCU as well.

About what are the different options for reducing the carbon footprint of the city?

And so the goal was that was to really focus on seeing what options existed for different types of removal.

0:1:19.840 --> 0:1:20.30

Nikolas Lanjouw (Intern)

Umm.

0:1:16.60 --> 0:1:28.900

Samantha Tanzer

One first outlines of what is removal, what are the different types of removable and specifically for options for removals within the city of Amsterdam, and sort of that North Sea Industrial area.

0:1:29.510 --> 0:1:29.670

Nikolas Lanjouw (Intern)

Yeah.

0:1:29.630 --> 0:1:33.30

Samantha Tanzer

Ohh so I wasn't looking at say offset.

You know things like that.

Those are different questions.

You know, this is always an issue with cities because cities are always supported by the hinterland, right?

Like, if you're gonna say you don't grow all of your food in the city, you don't have all the resources.

You always have that support ohm from uh places, so you have also that with carbon, you know emissions, you have the emissions that happen in the city limits, which is what the umm, what Amsterdam was looking in terms of.

How do we be carbon neutral with intercity limits, but you also sort of not consider the port, but also important is the idea of the emissions from consumption from all the things that get imported into the city.

But that was sort of outside the scope ohm and so a couple of different things that we were looking at in this report or you know we were looking at sort of Cdr as from bioenergy with carbon capture and storage ohm direct air capture and storage, biochar enhanced weathering, a forestation and storage and soils.

And these were all very different, both in terms of permanent storing things geologically is very permanent mineralization and enhanced weathering can be very permanent, but it's a very slow process.

Trees and soils.

0:3:3.710 --> 0:3:3.940

Nikolas Lanjouw (Intern)

Mm-hmm.

0:2:50.680 --> 0:3:4.130

Samantha Tanzer

You know good and inexpensive way to move carbon, but they are not very stable and biochar is kind of its own special case of having really difficult to calculate, really difficult to say how long it's going to last ohm.

And so we did this partly by source sync matching for you know, OK, what are the sources of biogenic CO2?

What are the views of those stakeholders?

Which a lot of interviews with actual people, you know, the actual people in these companies, you know, for storing it up.

The primary source of Biogen Xunuo to an Amsterdam being the waste and generation, we also talked to the sewer people.

We talked to farm organizations, we talked to different industrial actors and I'm blanking on the name of the company. But I remember it's in the report about who wants to switch their plan to biomethane all and there was another food processor who used food waste.

Umm for ah for energy generation.

So we did that as well as then looking to say OK enhanced weathering biochar, those are things that need ground cover, especially enhanced weathering.

So then we were looking at, OK, what are the areas in the greater Amsterdam region that you could theoretically cover with these materials?

And then with soil and forestry, we were looking at kind of existing forestry initiatives, but again a lot of comes down to you know, so the question of area, how much land is available versus how many emissions and also looking at things like technologic technology readiness, when do we expect these things to be available?

0:4:26.850 --> 0:4:27.50

Nikolas Lanjouw (Intern)

Mm-hmm.

0:4:28.290 --> 0:4:33.720

Samantha Tanzer

And I mean, the main thing is like a lot of these are not very well developed technologies and artsy trees have been around for a long time.

But there's also limits to how much you can plant them.

CCS is gaining traction, but they're sort of bottlenecks also on storage, you have Porthos beam built, so it goes to them and there is that pipeline there.

0:4:43.60 --> 0:4:46.500

Nikolas Lanjouw (Intern)

Umm, right? Yeah.

0:4:46.470 --> 0:4:48.600

Samantha Tanzer

But Porthos is already there.

It's already full up.

0:4:50.110 --> 0:4:50.310

Nikolas Lanjouw (Intern)

Yeah.

0:4:49.590 --> 0:4:58.970

Samantha Tanzer

It's already been committed by industries there and some other thinking arms and sort of the North Sea, but those are, you know, we're looking at 20302035, I think earlier.

0:4:57.850 --> 0:4:59.920

Nikolas Lanjouw (Intern)

Yeah. OK.

0:4:58.980 --> 0:5:24.150

Samantha Tanzer

So near term there is very little potential for Cdr in the Amsterdam region and you know in the question is whether or not the removals can balance out the emissions depends on how many emissions and how much effort Amsterdam makes to reducing their emissions, which is kind of always the primary should always probably be the primary focus because removals is much more resource intensive.

0:5:29.650 --> 0:5:29.810

Nikolas Lanjouw (Intern)

Yeah.

0:5:24.160 --> 0:5:29.820

Samantha Tanzer

It takes a lot more energy to take us out of the air than to stop it from going in the 1st place.

So really, you know, move his only gonna be relevant if Amsterdam can really get its emissions down in the first place.

0:5:36.480 --> 0:5:37.130

Nikolas Lanjouw (Intern)

Yeah.

Umm great.

So in you you were mentioning sort of, uh, the sort of different scopes.

Uh, and how you sort of define the system of Amsterdam's emission, can you can you sort of speak to umm what that system looks like?

Uh, you.

You said you considered, you know, emissions for on the city's emissions from waste that comes into the city and activities.

So the different scopes of.

So how does that that landscape look?

0:6:9.540 --> 0:6:9.830

Samantha Tanzer

Yeah.

So so this project was and you know what they're looking for?

The absolute they were interested in, you know, how is this compared?

Well, mostly looking, you know, not so much admission side, but as they were potential.

But I think the Amsterdam was concerned about their direct admission, so the emissions that happened in the Greater Amsterdam region from people burning fossil fuels at home and power plants and cars, that sort of thing.

So you're very direct footprint, but I'm thing especially when we're talking about Cdr is very important to understand that the important number is not how much do you two you take out of the atmosphere and permanently store, it's how much you take it on a permanently store minus all the greenhouse gases you've emitted in that process.

0:6:51.780 --> 0:6:51.980

Nikolas Lanjouw (Intern)

Yeah.

0:6:52.0 --> 0:6:58.710

Samantha Tanzer

Ohh and so we so need this life cycle assessment framework or sort of cradle to grave is the only one that can give you that number.

But this is made complicated by the time aspect.

So for example, with enhanced weathering, the uptake of carbon dioxide from rocks exposed to the element is actually very slow.

We're talking about a process that happens over decades, but all of the emissions associated with, with the grinding and spreading of those rocks happen up front.

Stupid.

So I think that's a very important consideration with life cycle accounting.

0:7:31.330 --> 0:7:31.530

Nikolas Lanjouw (Intern)

Yeah.

0:7:26.390 --> 0:7:40.220

Samantha Tanzer

That time often gets compressed, and we say all the impacts of 100 years or 1000 years we put in one number, but it's very important to consider not just with the boundaries is, but what's actually sort of happening over time and when things are occurring.

Yeah.

But I don't think Amsterdam is really looking at their consumption emissions, which to be fair, different actors have different reasons for looking.

0:7:52.820 --> 0:7:53.20

Nikolas Lanjouw (Intern)

Mm-hmm.

0:7:51.90 --> 0:7:55.850

Samantha Tanzer

You know, as a city, it has most control over things that are happening in that city.

0:7:55.200 --> 0:7:56.210

Nikolas Lanjouw (Intern)

Directly. Yeah.

0:7:56.220 --> 0:7:59.110

Samantha Tanzer

Yeah, and it has some influence over the indirect emissions. Obviously.

You know, they have procurement plans, they can say, OK, we're gonna build a building and it has to meet these requirements or those sorts of things.

0:8:5.640 --> 0:8:5.820
Nikolas Lanjouw (Intern)
Yeah.

0:8:5.560 --> 0:8:9.270
Samantha Tanzer
But in terms of what's gonna have direct control, it's gonna be what's in that territory.
So I think that's why the focus there, but you have to make sure that if you have offset, I don't say offset, I don't like that word.
But if you're trying to balance your residual emissions that you're balancing it not with gross removals but with net removals.

0:8:22.90 --> 0:8:23.210
Nikolas Lanjouw (Intern)
Yeah, yeah.

The time dependent the time component of planting it, or a four station is maybe 100 years, but then the the the permanence of that storage is very it's it's unsure.
And you have to take that into account.
How likely is it to leak back out into the system?
And so the major components in in the, the research that you have done, the major sort of elements or industries that can sort of take in or do CD in general, will I B and one or two others, right?

0:9:9.450 --> 0:9:10.340
Samantha Tanzer
Yeah, the.

0:9:6.250 --> 0:9:15.760
Nikolas Lanjouw (Intern)
Diabete was sort of recognized as the most, having the the highest capacity for for biology and like UH CS5.
Right.

0:9:16.70 --> 0:9:29.40
Samantha Tanzer
Yeah, by by virtue of the fact that they are the largest emitter of biogenic O2 currently in the area, that could change, especially as some of you know, the fossil images in that region decide to switch to biogenic feedstocks.
But the moment?
Yeah, it's it's the most.
It's the one with the most Parisian xunuo, too, so therefore they're the ones are the most potential for biosci.

0:9:39.350 --> 0:9:48.931
Nikolas Lanjouw (Intern)
But but they also intending to to you in the report it says that uh, I base most of that will go to.
To greenhouses, but.

0:9:52.510 --> 0:9:56.540
Samantha Tanzer
Yes, that's the current plan, because they're just, they're following the money, right?

0:9:57.250 --> 0:9:57.510
Nikolas Lanjouw (Intern)
Right.

0:9:57.570 --> 0:10:2.520
Samantha Tanzer
They capture it, but they also they they need to make it so that they break even.
And currently there isn't a market.
There is nobody buying, you know, buying energy and CS removal credits or anything like that.
If this was someone who was, then they would sell it to them.
But currently the most economic thing for them to do is to sell it to ocap and the greenhouses.

0:10:21.100 --> 0:10:21.610
Nikolas Lanjouw (Intern)

Yeah.

Well, the ocap the the the line and the ocap is not even complete between them and the the main line and the, but they've also, they've also indicated I spoke with Michael Hass at the at the I Bay and they he indicated that they are also planning to to push some of this field to to the hookup pipeline for it to be sequestered offshore and he he indicated that there was about 8000 hours worth of they they were permitted to do it about eight 8000 hours worth of capturing and this sort of came to.

0:11:3.980 --> 0:11:5.980

Samantha Tanzer

They those hours, that's almost a year, right.

0:11:7.90 --> 0:11:7.710

Nikolas Lanjouw (Intern)

I yeah.

0:11:7.190 --> 0:11:11.0

Samantha Tanzer

You're 3000 something hours, so like that's but out of our time span.

0:11:9.420 --> 0:11:26.650

Nikolas Lanjouw (Intern)

So yeah, about they didn't give, he didn't give an indication of like what over what time span they would capture these 8000 hours, but he gave a range of about 500 kilotons as the total capturing that they have permitted by.

But for the company and that they might in the report, it says that this might start around 2028, but they they were, they were thinking closer to 2032.

To what degree?

Like 505 hundred kilotons is in the in the the grand scheme of things.

Not a lot, especially in terms of sort of Amsterdam's historic emissions, but it points to this number quite often, and the number in your report as well.

In the nine, about 9% tent of its emissions being sequestered.

0:12:9.200 --> 0:12:9.340

Samantha Tanzer

Yes.

0:12:10.350 --> 0:12:21.420

Nikolas Lanjouw (Intern)

It's not a very high number and the this is a 2050 when they they they planned to hit net zero is that right?

0:12:22.640 --> 0:12:23.150

Samantha Tanzer

Yeah.

So I mean the thing the thing.

Oh yeah, the number isn't very large and this is kind of the issue with removers in general.

People think, oh, there's gonna be lots of removals because they want to keep emitting things they want to get to net zero.

But the reality of the matter is, is that you know, a lot of removal technologies, especially they involve actually proper secure storage like Dacs and backs are very, very nascent and will not be available at any scale.

0:12:47.620 --> 0:12:47.780

Nikolas Lanjouw (Intern)

Yeah.

0:12:47.900 --> 0:12:59.230

Samantha Tanzer

And the question then becomes, how do people actually stick to their guns and say, OK, we don't have removals, so we're going to actually make harder decisions about what to reduce and abate etcetera.

0:13:1.810 --> 0:13:1.970

Nikolas Lanjouw (Intern)

Yeah.

0:12:59.240 --> 0:13:5.870

Samantha Tanzer

Or do people say, well, we tried, but that, you know, like kind of remains to be seen.

0:13:5.880 --> 0:13:9.700

Samantha Tanzer

But I think you know it's the amount that's going to be available.

So is now starting to, especially with the competition for storage resources that you're seeing from fossil companies, especially in the Netherlands, they're starting to be more of an acceptance of CS.

But of course the issue with CS of course you know is that I think this isn't the first time people have tried to build CCS integral to Dom.

I think it's the third project.

One of them is called Rd.

0:13:32.440 --> 0:13:32.690

Nikolas Lanjouw (Intern)

Mm-hmm.

0:13:32.440 --> 0:13:50.270

Samantha Tanzer

I think one of them called backbone or something, but there have been a few different iterations of it and this one certainly so forgotten the furthest and seems there's actually going to go ahead, but we lost a lot of opportunity for technological learning by not myself sitting on our hands for the past.

What, 30 years? Uh.

0:13:53.110 --> 0:13:53.460

Nikolas Lanjouw (Intern)

Do you?

Do you see the learning curves for for backs and ducks being fast enough in order to make any appreciable bold dent in them, visit to emissions?

Or do you?

Do you see them being too, too far out to to realistically be part of the solution?

0:14:13.770 --> 0:14:20.870

Samantha Tanzer

I mean so learning curves are based on, you know, essentially the doubling of capacity, right?

It's not necessarily specific time, so the question is it basically how much money are we willing to throw at the problem?

I think there's definitely some low.

They think as I really dislike the idea of running down to zero just because something is small solution doesn't mean we don't need it.

0:14:36.20 --> 0:14:36.240

Nikolas Lanjouw (Intern)

Yeah.

0:14:35.270 --> 0:14:53.820

Samantha Tanzer

We just need to be realistic about it, but it needs to be part of this portfolio, especially with things like in backs, you really have a spectrum of you know you have things like that for more uncomfortable with, you know, personally like you know, thinking trees and burning them for electricity generation because trees have a very long rotation period. You're talking a carbon.

0:14:57.440 --> 0:14:57.620

Nikolas Lanjouw (Intern)

Yeah.

0:14:54.900 --> 0:15:1.670

Samantha Tanzer

Those actually move decades ago, and then you use it to produce a resource that we have other ways of producing.

But there are things like paper and pulp mills.

There is, you know, waste to energy, assuming that you've minimized how much waste you're producing.

Is bio gas.

There are other products and processes that produce biogenic CU2, not just for, you know, non energy reasons, but also they've been doing this like we have the plenty of biogenic resource like sources of CO2 that currently exist.

0:15:27.830 --> 0:15:28.80

Nikolas Lanjouw (Intern)

Umm.

0:15:27.390 --> 0:15:33.640

Samantha Tanzer

And I'm personally other opinions that you know we we should be phasing out unabated greenhouse gas emissions.

0:15:35.540 --> 0:15:35.780

Nikolas Lanjouw (Intern)

Yeah.

0:15:33.750 --> 0:15:43.230

Samantha Tanzer

Period doesn't know if the biogenic or fossil or whatever, but that kind of needs to be what the focus and the question is, is there enough willpower, and people are gonna throw the money at that.

Ohh I still hope so.

0:15:48.550 --> 0:15:48.910

Nikolas Lanjouw (Intern)

Mm-hmm.

0:15:46.260 --> 0:15:56.170

Samantha Tanzer

I mean, there's a reason I still work in this field, but you know, it's it's definitely a hope, not a sort of statement of confidence of.

0:15:57.280 --> 0:15:59.300

Nikolas Lanjouw (Intern)

OK. Yeah.

Yeah.

0:15:57.720 --> 0:16:21.640

Samantha Tanzer

So I think it's really hard to say what, but the one thing is to know is that you know technological it is transferable like so you know the the scale enough of DAC in the US with these two large projects that were just announced that will have uh, you know this one feedback here and also there's a lot of work going in bioenergy and CS that starting to grow separately in those both will have uh sort of feedback loops.

So like, it's not they're not independent technologies.

0:16:25.210 --> 0:16:25.430

Nikolas Lanjouw (Intern)

Yeah.

0:16:24.960 --> 0:16:31.680

Samantha Tanzer

Or if it's some of the hardest parts of it, are not actually, they be captured technologies or the storage technologies.

These are all things that exist.

It's the putting it together.

0:16:36.690 --> 0:16:36.890

Nikolas Lanjouw (Intern)

Yeah.

0:16:34.420 --> 0:16:41.430

Samantha Tanzer

It's most of the system learning, especially with pipelines, because pipelines you have the eastern, also social acceptability.

0:16:44.160 --> 0:16:44.340

Nikolas Lanjouw (Intern)

Yeah.

0:16:41.600 --> 0:16:44.900

Samantha Tanzer

People don't like pipelines running on their land, stuff like that.

So you have other issues, not just about the technology itself, which can, you know hinder or speed ohh development.

0:16:53.960 --> 0:16:55.600

Nikolas Lanjouw (Intern)

Yeah. OK.

0:16:55.610 --> 0:16:56.680

Nikolas Lanjouw (Intern)

Yeah, that's.

I mean certainly when you start to talk about pipelines, it's it's clear that the currently at least given Aramis and Porthos, the the capacity for slow, which is already almost full.

And although the port sort of indicates that there is potential for umm storage over off the show of einthoven, when that area for up to a gigaton of CO₂, that seems to be some some ways out and so there needs to be a way to move that.

That's, you know, to to other places.

Do you?

Do you see there being enough work towards an actual pipeline infrastructure within the Netherlands and also within the EU such that you know an appreciable amount of that biogenics CO₂ can be stored?

Right.

0:17:53.940 --> 0:18:11.90

Samantha Tanzer

All within the Netherlands, I think it's more likely just I feel like, you know, besides, you know, the the question mark around the nitrogen issue, which has slowed things down and love sectors, the the Netherlands, I these services are more from my personal experience of living here for the past eight years.

Then you know sort of a professional analysis perspective is that you know, the Netherlands is much more honest about their infrastructure.

0:18:17.660 --> 0:18:18.150

Samantha Tanzer

They accept.

0:18:18.160 --> 0:18:20.150

Samantha Tanzer

That's a part of life, they maintain it.

Part of this might be related to the history of dams and polders and water, so, like so, I think you know the Dutch people are more comfortable with the idea.

Obviously they're very uncomfortable with the idea of putting gas under the ground.

Unsurprisingly, given you know the trauma of the fracking and honing in but, and I think there's more of that, you know, less idealist perspective of saying, oh, no, we can't use fossil fuels.

I think there's a bit more acceptance of fossil fuels will continue to exist and therefore we need CCS.

0:18:49.800 --> 0:18:50.0

Nikolas Lanjouw (Intern)

Yeah.

0:18:51.810 --> 0:19:21.590

Samantha Tanzer

Also, the biomass, and I think if it varies a lot country to country for example, I was talking earlier in the spring at a meeting with some people who are doing research on carbon capture and storage in the list with the Lusitanian Basin in, we have a store that's been a Portugal, but in Liberia and there was actually much more selective people were really wary of offshore storage because of some bad experiences in the past.

There were more supportive of on storage because they felt they could trust people less likely for things to go wrong.

0:19:29.350 --> 0:19:29.590

Nikolas Lanjouw (Intern)

Right.

0:19:29.800 --> 0:19:31.30

Samantha Tanzer

So it is going to vary.

It's also should be noted that while pipelines, if you're transporting things over land, is definitely going to be the most efficient in sort of the near term.

0:19:40.130 --> 0:19:45.20

Samantha Tanzer

Ohh, you know, we're talking about 2/2 from shore boats.

Boat transport CO₂ is an option, at least for now.

0:19:49.940 --> 0:19:50.120

Nikolas Lanjouw (Intern)

Yeah.

0:19:49.510 --> 0:19:57.460

Samantha Tanzer

Truck transport is more of a headache if another than just from the sheer volume of CO2 that you're dealing with.

0:20:2.560 --> 0:20:3.320

Nikolas Lanjouw (Intern)

Do you see the the regulation and the policy being in place to support the multimodal and the the the level of movement that is going to be required?

0:20:18.80 --> 0:20:20.590

Samantha Tanzer

Uh, so in different types of policies.

0:20:23.760 --> 0:20:23.950

Nikolas Lanjouw (Intern)

Yeah.

0:20:20.600 --> 0:20:30.250

Samantha Tanzer

So the there's the CS directive from 2009, which is a very good policy in terms of requirements versus, you know, say characterization and transfer of liability and etcetera.

So basically making sure that secure geologic storage actually is secured.

Geologic storage and if anything happens to it, is the injectors that are on the hook, not the government ohm.

So that's in place.

There is the red, the Renewable Energy Directive, which has stuff about, you know, bioenergy sustainability.

But that's not always great in terms of the criteria still fairly low, in my opinion, in terms of their combination either in backs or indirect error capture.

There's basically no regulation on that at this point.

The CRC app is under development at the EU level, but that will be a voluntary framework.

0:21:12.40 --> 0:21:12.260

Nikolas Lanjouw (Intern)

Yeah.

0:21:12.400 --> 0:21:37.120

Samantha Tanzer

Ohm and you know and now you also have the London protocol which, umm, makes it challenging for you know, so it shipping to to sort of across boundaries like in store it under the sea because you can't dump it but you can make bilateral agreements and that's you know what countries are looking at especially in Scandinavia for example Denmark. I think also looking at on short storage as well.

So that policy, the policy of regulation and quantification is under development and I think for direct route specifically and you know for they totally technical logical removals.

More generally, I think that's gonna be less of a headache than regulating nature based removals because you don't have to deal with the uncertainty of the storage ohm.

But in terms of regulation, in the sense of giving, getting money to these processes or enforcing that companies can't have unabated emissions, they feel like that's sort of less on track.

You do have the ETS is going to have 0 allowances in 2039, but it's kind of unclear exactly what that will mean all.

So I think that that's kind of you know like the the regulation necessary so that you know things don't go wrong is getting better, but the regulation to actually make things happen, the policies rather to make things happen and still lagging.

0:22:40.10 --> 0:22:40.740

Nikolas Lanjouw (Intern)

Right.

Yeah, the the.

This sort of going through a process of being reformed to to reduce the free allowances that are being given out to industries and yeah, as you said, it will sort of attempt to hit zero in 2039.

Do you think it's fast enough?

OK, I like the honesty.

0:22:59.840 --> 0:23:5.910

Samantha Tanzer

No, I mean I understand you, if you're optimized, I mean that's the thing, right?

This is the thing to understand.

This is one thing I really noticed.

You know, working with politicians over the past year.

I know got a lot of them.

Get a bad rap, but at least you know in the Brussels there that I've enjoyed most of them care a lot.

0:23:16.880 --> 0:23:17.170

Nikolas Lanjouw (Intern)

Umm.

0:23:16.170 --> 0:23:17.240

Samantha Tanzer

And they're really trying.

They really want to learn.

They really do want to do a good job, but they're balancing many, many different interests, you know, like I think of and a lot of things think of, you know, climate crisis as the most important thing.

And in many ways it is because it affects so many other aspects of our lives.

0:23:32.120 --> 0:23:32.340

Nikolas Lanjouw (Intern)

Yeah.

0:23:32.50 --> 0:23:35.410

Samantha Tanzer

Ah, but The thing is, those other aspects of our lives are still going out.

There are still people in poverty.

0:23:41.920 --> 0:23:42.100

Nikolas Lanjouw (Intern)

Yeah.

Yeah.

0:23:36.830 --> 0:23:49.200

Samantha Tanzer

There is still the issue of migration of unrest, of economic development of, you know, the comforts and conveniences that people useful and would be really unhappy if they go away of defense, so of health.

So these are all things which you know they can't stop paying attention to.

0:23:54.840 --> 0:23:55.60

Nikolas Lanjouw (Intern)

Yeah.

0:23:55.50 --> 0:23:56.940

Samantha Tanzer

You know, would it be possible?

Like, did you sort of saying almost like what we do with the pandemic?

You know that kind of lockdown rationing situation which will thus use your resources and buy us more time.

0:24:7.330 --> 0:24:7.550

Nikolas Lanjouw (Intern)

Yeah.

0:24:7.240 --> 0:24:9.750

Samantha Tanzer

Obviously that is technically possible.

We we experienced it to some extent ourselves.

0:24:16.720 --> 0:24:16.900

Nikolas Lanjouw (Intern)

Yeah.

0:24:12.720 --> 0:24:22.110

Samantha Tanzer

I see that as much as people did during the world wars, but it's very requires a huge amount of social cohesion and dedication.

Common dedication to a goal that isn't there right now in society, and I think we saw that with the pandemic at the beginning.

You know, people once the lock and people were very supportive and government umm.

0:24:37.690 --> 0:24:37.890

Nikolas Lanjouw (Intern)
Yeah.

0:24:34.830 --> 0:24:45.700

Samantha Tanzer

So your popularity levels were much higher, but then over time the longer that went on, you know people are wrote it that erotics people didn't like having to lose their formal life and freedoms that they were used to

So you know, it's hard for people to make hard choices, and this is one of the issues with democracy.

It's like there's the famous, possibly misquoted thing.

You know, democracy is the worst form of government, except for all the others, you know.

0:24:58.420 --> 0:24:59.650

Nikolas Lanjouw (Intern)
Yeah, right.

0:24:59.750 --> 0:25:1.590

Samantha Tanzer

But it's that's the thing, you know?

0:25:9.100 --> 0:25:9.260

Nikolas Lanjouw (Intern)
Yeah.

0:25:1.630 --> 0:25:9.910

Samantha Tanzer

But it is democracy and you have to respect the will of people, other people's opinions and values, which are not the same as ours, and these things aren't fungible.

You can't say even just like as simple as flying, right?

0:25:13.540 --> 0:25:29.290

Samantha Tanzer

So I'm going back to the US next week for the first time in five years to visit some friends, and it's going to be about what a ton of carbon dioxide equivalent emissions to get there and back should I feel like for can you tell me how? What's the number between?

0:25:33.60 --> 0:25:33.270

Nikolas Lanjouw (Intern)
Right.

0:25:30.720 --> 0:25:34.580

Samantha Tanzer

How often can I see my loved ones versus carbon?

These are numbers that are not computable.

0:25:36.980 --> 0:25:37.190

Nikolas Lanjouw (Intern)
Yeah.

0:25:36.850 --> 0:25:39.320

Samantha Tanzer

How much is the extra year of someone's life worth?

Worth you know you you we put numbers on these when you have social cost of carbon quality disability adjusted quality of life years or whatever, but they're inherently unfun table.

0:25:54.70 --> 0:26:10.550

Nikolas Lanjouw (Intern)

It's certainly a very sticky mobile question about how how you sort of manage all these, this terms and still work towards the the the solution of no net zero but actual zero.

0:26:12.210 --> 0:26:12.410

Samantha Tanzer

Yeah.

0:26:10.670 --> 0:26:21.920

Nikolas Lanjouw (Intern)

And you know, this is net 0 not being not being quite enough with the whole falling of the voluntary market and there

seems to be some future.

So future reports that will be that will be coming that are pretty that seem pretty rough about backs and dacks, but the IPCC, still including decks and decks and it's most.

0:26:38.970 --> 0:26:39.370

Samantha Tanzer

I mean the.

0:26:38.780 --> 0:26:44.500

Nikolas Lanjouw (Intern)

We call the in the in the in the pathways that that seem most plausible.

But OK, so bringing so to bring this back to the sort of the Amsterdam context, so go ahead.

Yeah.

0:26:48.280 --> 0:27:4.740

Samantha Tanzer

I I just so I just want to just say one thing on that before we ever which is you know yes Becks and decks are not good ideas but we are not at the point where we have the luxury of only paying attention to good ideas and that's kind of the problem we were really in a big hole here.

We need to do everything, even things that we're not sure that are going to work.

0:27:13.590 --> 0:27:13.790

Nikolas Lanjouw (Intern)

Yeah.

0:27:8.960 --> 0:27:15.240

Samantha Tanzer

If we want to have a chance of continued to maintain the world and safe operating environment for humanity.

0:27:16.270 --> 0:27:16.590

Nikolas Lanjouw (Intern)

Yeah.

0:27:16.950 --> 0:27:29.120

Nikolas Lanjouw (Intern)

So you're sort of of the opinion that we should do whatever like we should follow every every line, regardless if we think it might turn out to be the line that is most best suited.

You know, there's this whole sort of discussion as if we put so much effort into bags and Dax, we're taking away from umm reduction efforts. And do you?

0:27:39.780 --> 0:27:40.290

Samantha Tanzer

I mean but.

0:27:42.350 --> 0:27:50.100

Samantha Tanzer

Alright, so that is like I think that's a false dichotomy, because I think there's a lot of effort in the world that's going to neither reduction efforts or backs and Dax effort.

So I think, yeah, you really have to worry about avoiding moral hazard and people thinking that just because you can avoid you could doing removals.

0:27:59.930 --> 0:28:0.110

Nikolas Lanjouw (Intern)

Yeah.

0:27:57.970 --> 0:28:4.220

Samantha Tanzer

You don't have to do reductions, but my thing is, no, you have to do both and this is kind of the one of the big issues of net zero, right?

Is that net zero hides the numbers and either side of the equation like you can have net zero where you have 40 giga tons of emissions and 40 giga tons of removals.

0:28:17.710 --> 0:28:17.870

Nikolas Lanjouw (Intern)

Yeah.

0:28:14.440 --> 0:28:19.790

Samantha Tanzer

You're gonna have net zero where you have 5 giga tons of emissions and five gigatons of removals, and I didn't want to point.

Always stress is to make that I think you need to look at that.

You need to look at the less resource intensive option which is going to be focused on remove reductions, but you're still going to need a scale of removals.

That is inconceivably large.

0:28:40.30 --> 0:28:40.210

Nikolas Lanjouw (Intern)

Yeah.

0:28:34.20 --> 0:28:40.320

Samantha Tanzer

With the scale of removals that's going on today and scaling it up, those multiple orders of magnitude will require a lot of work.

0:28:41.600 --> 0:28:58.600

Nikolas Lanjouw (Intern)

And in a lot of the so the modeling they, you know, backs and decks are sort of there because there's sort of the discussion of, OK, even if we do hit zero in terms of emissions, that's a historic, very long, historic emissions.

You know, there's there's a lot of CO2 in the the atmosphere that will have in order to reach a point that was pre pre industrial age you do do you see sort of that being a major component of why becks on deck stick around and?

0:29:14.280 --> 0:29:20.280

Samantha Tanzer

Yeah, I mean like that's kind of the you know, so I like the IPCC framing of it, the kind of three rolls of carbon removals. You know the near term is about speeding.

How fast we get to net zero by sort of being on top of reduction is not in replacement to them.

And then AT-90 like you need to balance out the emissions, you need to make sure.

0:29:35.950 --> 0:29:36.90

Nikolas Lanjouw (Intern)

Yeah.

0:29:32.840 --> 0:29:44.310

Samantha Tanzer

And then beyond that, it's about going net negative to try to minimize the amount of feedback effects you're always going to have some like if you build an ice cap, you're not going to get that back just because you're reducing your historic emissions.

But certainly you can, you know, ameliorate some of the other effects.

Least that's the hope.

0:29:50.950 --> 0:29:52.30

Nikolas Lanjouw (Intern)

Yeah, yeah.

0:29:48.470 --> 0:29:54.630

Samantha Tanzer

We actually, there's still a lot of open questions in this kind of atmospheric or a system response modeling.

0:29:56.820 --> 0:29:57.260

Nikolas Lanjouw (Intern)

Yeah.

OK, so so let's try to sort of bring this back to to Amsterdam, since this is the, at least for for me the context in Amsterdam's system, it's attempting to to reduce as much its emissions. And umm, it's climate action Plan it helps to to reduce it by 90% by UH 9 by 25.

See and then use the the cop and storage in order to reach net zero.

This is of course, you know, there are a lot of holes in this and we we don't know what this will be in the future.

Umm, but in the context of Amsterdam.

Are there's still barriers?

In the pilot, in the legislation in the regulation to to even make that possible, because in the report you, so to speak about the this this labeling of waste though having end of waste umm and label and well if it has end of ways and it no longer has to sort of follow certain regulations and can sort of be moved around and then stored in certain ways do you do you still see that being a problem instead and.

0:31:15.60 --> 0:31:23.630

Samantha Tanzer

Yeah, and that that's still an issue because it does make it difficult for these materials to then be used or reused in these ways.

And then, of course, things are enhanced world and biotrol.

You have the fact that you know they they there is no regulation saying all these are foreign substance.

Can they be put on land, et cetera?

So that is 1 issue.

The other issue is that they're really on end.

Literally any level national EU global or whatever, there's nowhere that has sort of even a good definition of removals.

Or how to account for them?

Especially because removals require lifecycle accounting, but territories use annual accounting.

0:31:51.350 --> 0:31:51.570

Nikolas Lanjouw (Intern)

Right.

0:31:51.490 --> 0:31:54.80

Samantha Tanzer

So how do you put those things together

It's still very much an open methodological question.

0:31:57.120 --> 0:31:57.380

Nikolas Lanjouw (Intern)

Umm.

Ohm, I think in Amsterdam you also have the, you know, a Dutch city is also beholden to what are the policies of the national government.

0:32:6.260 --> 0:32:6.540

Nikolas Lanjouw (Intern)

Right.

0:32:4.570 --> 0:32:20.110

Samantha Tanzer

You know, they can make decisions to some extent, but they can, for example, I well to my knowledge ohm after them does not have the authority to say completely ban fossil vehicles in its municipality.

Maybe it does, maybe I'm wrong, but I think you know.

0:32:21.760 --> 0:32:25.630

Nikolas Lanjouw (Intern)

It is hoping to I believe in exact climate action plan.

It says it wants to have no no ice vehicles in the city limits by 2030. Uh.

0:32:32.950 --> 0:32:33.630

Samantha Tanzer

Are they allowed to?

Do that so maybe they can do that.

0:32:36.790 --> 0:32:37.450

Nikolas Lanjouw (Intern)

Who knows? Yeah.

0:32:36.750 --> 0:32:38.340

Samantha Tanzer

It is, yeah.

So that those are the kind of things like system has to see.

What are its policy levels for those reductions?

0:32:45.10 --> 0:32:45.210

Nikolas Lanjouw (Intern)

Yeah.

0:32:44.610 --> 0:32:56.640

Samantha Tanzer

And of course, you also didn't have to look at houses affect elsewhere to this mean you say, OK, you don't have anyone

driving in the city limits, but there's everyone just parked right outside the city, and there's still a lot of commuters. So that's the kind of thing.

0:32:57.570 --> 0:33:4.680

Samantha Tanzer

What are the indirect impacts of these policies and do they lead to increases in greenhouse gas emissions elsewhere?

0:33:8.400 --> 0:33:8.750

Nikolas Lanjouw (Intern)

Yeah.

So you you were saying that the, uh, the accounting framework is really, really crucial.

0:33:16.950 --> 0:33:17.320

Samantha Tanzer

Yeah.

Because and accounting framework those two things.

0:33:15.30 --> 0:33:19.610

Nikolas Lanjouw (Intern)

There was a logical and and there's a gap in there.

And can you can you sort of speak to why the gap is what what the the gap is and why it's such a problem?

0:33:29.710 --> 0:33:32.510

Samantha Tanzer

So we want to report on this?

Umm, not quite an incident, but I'll send it to you about trend boundary accounting of CO2.

That talks all the differences between life cycle accounting and territorial accounting, ohm and part of the main issue sort of carbon accounting in general is like carbon accounting has two purposes, one of which is, you know track or at least model physical flows of carbon emissions.

0:33:54.920 --> 0:34:8.260

Samantha Tanzer

And if removals as they sort of happen over time, but the other one is the assignment of responsibility for the general assumption that if you admitted this carbon, it is very responsibility to do something about.

0:34:17.830 --> 0:34:18.30

Nikolas Lanjouw (Intern)

Yeah.

Yeah.

0:34:10.830 --> 0:34:27.760

Samantha Tanzer

So and then you know, so This is why we track things like reductions and avoidance, even those effectively 0 like it's not an admission, it's not even something you did not do, but we should practically these numbers because it's a tracking how people are fulfilling their responsibility over time.

And I think people often conflate these two things, and it often leads to situations which people add together.

0:34:44.90 --> 0:34:44.270

Nikolas Lanjouw (Intern)

Yeah.

Numbers just because they're measured in tons of CO2, not realizing that you know tons of CO2 reduce is not the same as avoided or store the geologically or stored in a tree or biogenic or fossil, and realize that these things are not fungible.

I think it's very, very important.

I would even say, you know, I don't think emissions are removal.

Should be fungible.

0:35:7.410 --> 0:35:7.630

Nikolas Lanjouw (Intern)

Yeah.

Yeah.

0:34:58.290 --> 0:35:25.740

Samantha Tanzer

I think there number should always be reported separately because they tell you information about umm, you know, resource intensity and so I think that to me one of the issues with carbon accounting that's highly over aggregated 2,

there's the fact that again with you really want removals to be long term and permanent because you know that you want them to do to say OK well counteracting the climate effect of releasing a permanent greenhouse gas. But if you have territorial accounting, you're just tracking flows, not stocks.

0:35:31.270 --> 0:35:31.470

Nikolas Lanjouw (Intern)
Yeah.

0:35:30.540 --> 0:35:42.550

Samantha Tanzer

You're tracking how much was emitted, or we moved into searching year, and so do you say, OK, this was the amount that was literally extracted from the atmosphere and then we just track releases we releases or whatever.

0:35:46.810 --> 0:35:47.30

Nikolas Lanjouw (Intern)
Yeah.

0:35:42.830 --> 0:35:52.620

Samantha Tanzer

I burns up for us down to track those when they happen, which is sort of the current model, but then that leads to the thing of people saying, well, I use this removal.

So I'm claiming that I've done my duty to balance out these emissions that I had, which is problematic.

0:36:0.310 --> 0:36:2.610

Samantha Tanzer

If that then proofs in the next few years.

0:36:3.460 --> 0:36:5.460

Nikolas Lanjouw (Intern)
Yeah. Suddenly.

0:37:2.990 --> 0:37:14.70

Nikolas Lanjouw (Intern)

The seems to be sort of be a a hole in sort of the the guides and sort of the methodology for, especially for cities.

And this is is doubly so for for developing developing countries and cities not in the global also because a lot of the frameworks and a lot of the information at the national scale for carbon accounting just doesn't exist at all.

0:37:33.930 --> 0:37:34.60

Samantha Tanzer

Yes.

0:37:33.380 --> 0:37:45.900

Nikolas Lanjouw (Intern)

And a lot of the tools that are used in order to do carbon accounting, even I mean at any scale, you know this lifecycle analysis, there's just no passing, there's just no knowledge for that.

And in developing countries, so it's it's a, it's a really big issue that I'm seeing at this level.

0:37:56.410 --> 0:37:57.980

Samantha Tanzer

Where are you right now actually?

[REDACTED] Redacted for privacy

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

0:38:29.870 --> 0:38:34.160

Nikolas Lanjouw (Intern)

I think I have maybe one more question and then I'll let you go.

The Horizon Europe and sort of the the it's excuse me?

No, the ETS and you know the funds that will come from that, that are supposedly supposed to come from that.

Do you see them being impactful or, do least see them having the the value that Belgium that you know Belgium says that they will?

0:38:59.510 --> 0:39:8.940

Samantha Tanzer

I mean, I think they could like currently I think it's only what only 10% of the budget and ETS has to go to essentially environmental remediation.

But you know, especially with decreasing the allowances, if you say, OK, you know this this is the pendance that you're paying essentially for emitting all of that money has to go to, you know, various projects at different atmospheric being.

We should know even other types of environmental resilience, et cetera, I think that could be incredibly impactful.

That could be a lot of money, but the question is there's always the temptation to use it for something else.

0:39:33.650 --> 0:39:33.830

Nikolas Lanjouw (Intern)

Yeah.

0:39:33.630 --> 0:39:39.440

Samantha Tanzer

Ohh kind of to the point a bit about this, but also just so the the difficulty of carbonate county.

I think it's also important we focus so much on carbon accounting.

I think because it's really important to have those measurements.

So it'll tell me shark progress, etcetera, but there's a lot of stuff that can be done without that.

You kind of was OK.

You don't have to know exactly how big of an effect it's going to have, but you know, sort of more behavioral target.

0:40:34.170 --> 0:40:34.370

Nikolas Lanjouw (Intern)

Yeah.

0:40:27.770 --> 0:40:42.80

Samantha Tanzer

Is that we can start to change behaviors towards a lower impact society that are less administrative, administratively burdensome, but then demanding everyone know exactly how much they're emitting at any time.

0:40:43.30 --> 0:40:48.20

Nikolas Lanjouw (Intern)

Umm yeah, that is certainly.

I mean, that's a big thing.
But OK, thanks.
Thanks very much, Samantha.
I actually have to go to another meeting.
This has been really great and.

0:40:58.860 --> 0:40:59.830

Samantha Tanzer
Glad to help.
I remember being a master student.

0:41:1.950 --> 0:41:2.110

Nikolas Lanjouw (Intern)
Yeah.

0:41:1.520 --> 0:41:3.970

Samantha Tanzer
So you know, I'm always happy to.
You know, I feel questions.

0:41:7.150 --> 0:41:7.780

Nikolas Lanjouw (Intern)
Thanks very much.
Yeah, it's UM, getting to sort of the difficult parts of especially the the policy and the policy ramifications for the local levels or little bit complicated.
But I'm happy.

0:41:23.460 --> 0:41:23.590

Samantha Tanzer
Yeah.

0:41:21.300 --> 0:41:25.780

Nikolas Lanjouw (Intern)
I'm glad that you had the time to to speak with me.

0:41:25.580 --> 0:41:26.250

Samantha Tanzer
And I'm glad.
Glad you're doing the work.
It's important.
Well, have fun.

0:41:29.170 --> 0:41:29.370

Nikolas Lanjouw (Intern)
Yeah.

0:41:29.160 --> 0:41:30.750

Samantha Tanzer
Uh, enjoy the rest of your summer.

0:41:31.740 --> 0:41:32.590

Nikolas Lanjouw (Intern)
Yes, thank you.
Well, I'm technically working.
But uh, you too.
You enjoy the rest of your summer and or however much Dutch summer is still left.

0:41:42.520 --> 0:41:43.780

Samantha Tanzer
Yeah, that is nice here.

0:41:43.790 --> 0:41:46.740

Samantha Tanzer
Now so you know, I all do have a great afternoon.

0:41:46.600 --> 0:41:47.230

Nikolas Lanjouw (Intern)

OK.

You too.

0:41:48.720 --> 0:41:49.20

Samantha Tanzer

Bye bye.

0:41:48.500 --> 0:41:49.50

Nikolas Lanjouw (Intern)

Thanks Samantha.

Appendix C: Policy matrix

C.1: EU Policy

ID	Policy Instrument	Latest in-force	Relevance	Primary purpose	Approach type	Market approach	Legal status (binding / non-binding)	Deployed scale (institution)	Deploying institution	Target actor	Sphere of effect (specific)		Barriers: Real and potential
EU 1	EU ETS Directive	2023 Reasoning	Direct	To establish a system for greenhouse gas emission allowance trading within the union. Creates mechanisms for the trading, reduction (of lower time), protections against leakage, the regulation of revenues from allowances, a list of covered activities, and the provision of economic and technical support for various actors affected by the tax. It creates a long list of mechanisms to enact the policy package including regulations for monitoring and reporting, verification and accreditation, certification and verification of CDR projects. The ETS interacts with the RED, LULUCF, CCS directive, CBAM, Innovation fund, and Governance of the Energy union and climate. The ETS operates on a regulatory market and secondary market. An expansion in the coverage of the ETS2 will cover Aviation, Maritime, road and rail transport among other smaller industry installations.	Stick	Supply	Directive (binding)	National government	EU Parliament & Council	Various (networks, organizations, individuals)	Infrastructure & Normative frameworks	Hard (law & standards) (Material - output - fossil based CO2)	The newly agreed upon mechanisms will not come into force until 2027, municipal waste incineration plants will be included under the ETS from 2026 and must purchase credits. This means that AEB will have to shoulder additional costs for 2 years before it is able to reduce them as its capture stacks come online in 2028. Recalling that this is only true if it's emission are higher than a benchmark that the EUC will set at some point before 2026. Depending on the costs of CO2 this will require those plants to swap to firing biomass that can be captured, stored or used. Meaning that they will not have to purchase allowances for the amount captured and stored, though the decision still represents an increase in cost for the firms. If the allowance system does not recognize the negative contributions of CDR processes; it would miss potential to incentivize CDR. If revenue from the sale of allowances does not go to CDR or related industry/infrastructure. If the cost of allowances does not go up fast enough, or the number of free allowances is not addressed it will reduce the disincentive in the cost of allowances.
					The EU ETS taxes carbon emissions, which makes operating high emission activities costly. The directive incentivizes emissions reductions through several mechanisms including cost to pollute, tradable allowances for additional income, support through the innovation fund. Compliance is enforced through a fine at 100 euros per ton CO2 or greater depending on the market cost of an EUA.	The ETS creates regulation regarding the price of emitting carbon through the credit. It predominantly affects firms, and suppliers of goods and services.							
	CCS Directive	24/12/2018 Reasoning	Direct	To establish procedures for the geological storage of carbon dioxide.	Sermon	Supply	Directive (binding)	National government	EU Parliament & Council	Organizations (firms)	Infrastructure (Material & Immaterial)	Storage & permit/certification	This directive defines physical science requirements for the safe, permanent and effective storage of Carbon dioxide in geological formations. The directive could create a gap if it is too loose in regards to 'safe geological storage' as the required 'opinion' from the commission on proposed storage points is non-binding. This could lead to unsafe storage if memberstate transposition is lax.
	CDR Certification framework proposal to regulation	N/A	Direct	To establish a framework for the certification of carbon dioxide removals to ensure QUALITY. It will establish requirements for removals to be recognized. It establishes that CDR activity is verified by a certification body. It will cover all emissions under the ETS and additionally those not currently covered (though this may change when the ETS 2 comes into force).	None	Supply	Directive (binding) Not yet in-force	Intranational (EU) & National government	EU Parliament & Council	Organizations (firms)	Infrastructure (immaterial)	Permit / certification	There are a large degree of methodologies for the removal of carbon from the atmosphere with differing results, if the directive erroneously determines methods to be valid or invalid this would be a significant issue for firms operating those processes and dependants. CDR is currently very heterogeneous which poses challenges for certification, potentially positioning the directive and slowing progress by innovators. If certification is too strict it may constrain innovation of CDR, though it is not likely to affect BECCS and DACCS as much as other less proven methods. In it's current form, the directive will create a union system for deployment of 'voluntary' certification programs in member states. The EUC will monitor the certification programs to ensure adherence to the proposal. However voluntary programs have been shown to be less effective, enable greenwashing and other issues.
	Trans European network for energy (TEN-E) Regulation		Direct	To align efforts to better support the modernisation of Europe's cross-border energy infrastructure towards the objectives of the EU Green deal. It outlines 5 criteria for determining eligibility for support through the connecting europe facility (CEF). TEN-E explicitly covers CO2 transport and storage infrastructure projects.	Carrot	Supply	Regulation (binding)	Intranational (EU)	EU	Various (Networks, organizations, individuals)	Infrastructure (Material)	Transport & Storage	Certain projects of common interest (PCIs) can receive EU financing from the connecting Europe Facility (CEF), though this will only come into effect on December 2027. This project does not create any barriers for CDR as is. Though CEF may not provide sufficient incentives for CO2 transport and storage projects. Purposes has been identified as such a project and will receive funding from the CEF.
	LULUCF Regulation		Indirect	legislative regulation for incorporating greenhouse gas emissions and removals from Land Use, Land Use Change and Forestry (LULUCF) under its 2030 Climate and Energy Framework. The LULUCF regulation aim to incentivise EU Member States to decrease greenhouse gas emissions and increase removals in the LULUCF sector. The regulation, however, does not set a target for increasing the LULUCF carbon sink, but rather includes a 'no net debit' target for LULUCF (forests and Agricultural soils). Linking with the certification of removals regulation will ensure quality removals.	Stick	Supply	Regulation (binding)	National government	EU Parliament & Council	Organizations (agriculture firms)	Infrastructure (immaterial)	Operations: Technical	The LULUCF regulation is not likely to create gaps with CDR technologies in Amsterdam. Though if in the process of this regulation, it affects biomass availability and price that may negatively affect biogenic CO2 availability for storage by firms in amsterdam. A WUR report expects that the regulation will constrain biomass availability for bio-energy from forestry and agriculture.
	Horizon Europe	2021	Direct	Horizon Europe is a research and innovation fund available until 2027. It has a budget of 95.5 billion euros. It's goal is to tackle climate change, support achieving the SDG's and boost EU competitiveness and growth.	Carrot	Supply & Demand	Economic (Non-binding)	International (mostly EU)	EU Comission	Various (networks, organisations)	Infrastructure (Material, immaterial)	Various	Funding programs like this often come with strict reporting and compliance requirements to ensure money is being spent on what actors promised. This might put additional undue pressure on Amsterdam or actors in amsterdam receiving the funds if the capacity to manage the administrative. Though the project outlines that it will provide support for this it remains to be seen if that will follow through. While there may be many barriers to receiving the support for amsterdam and CCS projects in the netherlands, they have already jumped most of the hurdles it seems.
	Effort Sharing Regulation		Indirect	The Effort Sharing Regulation (ESR), as adopted in 2018, sets national targets for emission reductions from road transport, heating of buildings, agriculture, small industrial installations and waste management. These sectors cover 57% of EU GHG emissions.	Sermon	Potentially both	Regulation (Binding)	National government	EU Parliament & Council	Organizations (Government)	Institution (Hard)	Binding and non-binding policy	The ESR may cause an emissions bubble and lead to lower ambition from memberstates, reducing action. Without compliance measures the ESR will have significantly lower impact, causing a gap in what is needed and what is being achieved. Net removals between LULUCF and sectors under the ESR are not equivalent and not covered under any monitoring & reporting mechanism currently. However these issues do not represent a barrier or gap in policy for Amsterdam, which already has ambitious climate action targets. It could however lead to reduced action by other memberstates reducing action on TEN-E PCI projects.
	Innovation fund		Direct	A funding programme for the innovation of low-carbon and negative carbon technologies. Has a budget of 40 billion euros over 10 years (till 2030), if carbon prices are 75 euro a tonne. The funding will come from revenues of ETS auctions and sales and is primarily for industry and energy use. It was revised in 2023 to increase its value (as the ETS was strengthened). It was also given some teeth in the Do no significant harm principle.	Carrot	Supply	Budget (Binding)	Intranational (EU)	EU Comission	Organizations (firms)	Infrastructure	Material & Immaterial	A major potential gap in the innovation funds system is that its funds are tied to the price of carbon in the ETS which has historically fluctuated widely. Changes to the system which will come into force in 2027 to tighten regulation on allowances being removed each year. Though the size of the innovation fund is now 530 million ETS allowances.
	Regional development fund		Indirect	The European Regional Development Fund (ERDF) is designed to strengthen economic, social and territorial cohesion in the European Union. It aims to do this by correcting imbalances between regions enabling investments in a smarter, greener, more connected and more social Europe that is closer to its citizens. 30% of overall finance must go to climate objectives.	Carrot	Both	Budget (Binding)	National & regional government	EU Comission	Organization (Government)	Infrastructure	Material & Immaterial	This fund is not directly targeted at CCS or CCU but regional government could use some of that money to support development in it if the dutch government agreed. If not, it does not represent a major gap in financing as there is significant funding available in other programs. However, for small and medium enterprises it represents significant funding and could lead to the continuation or innovation of CDR technologies outside BECCS and DACCS.
	Common Agriculture Policy (Directive)	09/09/2021	Indirect	The common agriculture policy is a 60 year old piece of legislations which outlines shared goals of European union members regarding agriculture. The new CAP aims to foster a sustainable and competitive agricultural sector that can support the livelihoods of farmers and provide healthy and sustainable food for society, along with vibrant rural areas. It provides a framework of regulation, directives, and shared goals for reaching those objectives. The CAP provides 387 billion euros for the 2023-27 period.	Various	Both	Binding & non-binding	Intranational (EU)	EU Parliament & Council	Various (networks, organizations, individuals)	All (primarily Infrastructure: material - capture - inputs - biomass)		The revised CAP 2023-27 could impact CDR operations in Amsterdam by affecting the availability of biomass as an input for energy and heating. Though the revised policy promotes the use of biomass for more eco-friendly energy which would positively affect the operations of BECCS, that remains to be seen. It's interaction with other policies that require more stringent sustainability certification might create a gap in supply and use.
	carbon cycles		Direct	The communication 'Sustainable carbon cycles' is a report by the European commission outlining both the importance of carbon to the EU and the negative effects felt worldwide of its use. The communication addresses how the agriculture, industrial and large economy as a whole can contribute to reduction of emissions and capture/storage as carbon neutrality is reached to further reduce CO2 stores in the atmosphere. It outlines a certification scheme that would support this goal (the CDR certification proposal to regulation).	Sermon	Neither	Non-Binding (Communication)	EU	EU Comission	Networks & organizations	Normative frameworks & actors	Soft (norms) & Organizations (government)	No potential barriers, it advocates heavily for CCS and CCU and has impacted several legislative instruments in favour of CDR technologies.
	Green City Accord		Indirect	The Green city accord is an initiative by the European commission to make cities greener, cleaner and healthier. Membership is voluntary, municipalities are required to monitor and report on priority areas using an indicator framework supplied by the initiative. Amsterdam is not a signatory.	Carrot & Sermon	Supply	Non-Binding (Network)	Municipalities	EU Comission	Organizations (Government - Municipalities)	Various (Institutions & Actors & Infrastructure)		The monitoring reports add administrative burden, but actions are being taken to reduce this. Amsterdam is however not a signatory and so not subject to any of it. As such there is no barrier to their plans.
	EU 2050 climate neutral strategy		Indirect	Europe's 2050 climate neutral strategy is the culmination of vision setting, discussions and and strategizing. It started as a vision set out by the EUC, which was incorporated into an endorsement by the EU Parliament and Council resulting in the EU's long-term strategy to the UNFCCC (4/2020). It outlines how the European aims to address its GHG emissions to reach climate-neutrality in 2050.	Sermon	N/A	Non-Binding	European Union	EU Comission	Network & Organization	Normative frameworks & actors	t (strategy) & actors (network - organization)	Though ambitious the EU's climate action plan, the NDC's of memberstates do not amount to what is needed for a <1.5C world agreed to under the Paris agreement. The EUC, Parliament and Council make the argument that CCS will still allow the union to make that target, after 2050. Additionally arguments regarding whether the EU's 2050 climate-neutral strategy makes ambitious enough point to its fair share emissions budget for a 1.5C world. This line of reasoning is based on the "common but differentiated responsibilities and respective capabilities, in light of different national circumstances" (Paris agreement, Article 4.3). Despite this, the EU acknowledges the need to utilize CCS and includes it in the strategy which will support Amsterdam's plans.
	EU Climate law	30/06/2021	Indirect	To set into law the EU 2050 climate neutral strategy outlined by the European Green Deal. It includes the increased ambition targets of the 'fit for 55%' communication. The EU climate law obliges memberstates to adhere to the climate-neutral 2050 target and creates precedence for the Governance of the Energy union and climate regulation.	Sermon - Stick	N/A	Binding (regulation)	European Union	EU Parliament & Council	Network & Organization	Normative frameworks & actors	t (strategy) & actors (network - organization)	EU Climate Law puts into law the ambitious targets of the European union to addressing climate change, regardless of discussions around fairness; it is a significant achievement by the union. However, it includes a major gap in compliance mechanisms and as such reduce its impact considerably. It's affect on Amsterdams climate action plan are not straightforward but essentially it has strengthened action by the netherlands which has set ambitious goals based on the the regulation. This target then transposed into its NCEP & NDC's subsequently has coercive and legal power encouraging Amsterdam and other cities in the netherlands to follow suit. As a result Amsterdams action plan leaves only 5% of its carbon budget as residual emissions and not more.

Governance of Energy Union and Climate Action Regulation	20/11/2023	Indirect	To ensure the EU reaches its 2030 climate and energy targets, the Regulation on the Governance of the Energy Union sets common rules for planning, reporting and monitoring of emissions and sinks. The Regulation also ensures that EU planning and reporting are synchronised with the ambition cycles under the Paris Agreement. Memberstates must monitor, and report GHG emissions in accordance with the IPCC's emissions accounting framework. It additionally creates a EU renewable energy financing mechanism. The EEA is defined as a implementing body, supporting the EUC under the regulation.	Stick	N/A	Binding (regulation)	Intranational (EU)	EU Parliament & Council	Organizations (Government)	Normative frameworks & Actors	oft (strategy) & Organizations (Government)	This regulation does not affect Amsterdam directly nor its plans to use CCS/CCU. However, the actions of Amsterdam are reflected in these reports, giving incentive for the Dutch government to support Amsterdam's climate action plan. It also creates precedence for the netherlands to give Amsterdam responsibility for its carbon budget through the environment and planning law.
Monitoring and reporting regulation		Direct	Defines the monitoring and reporting plan of annual emissions and the contents of the emissions report pursuant of the ETS directive. Contents include monitoring plans, annual emission reports, verification reports, and improvement reports. Operators must surrender annually the number of allowances corresponding to their emissions in the preceding year. Penalties are defined for failing to do so. A MRR is required to receive a permit to operate.	Stick	N/A	Binding (regulation)	Firm (local)	EU Comission	Organizations (Firms)	Infrastructure	Immaterial	The MMR and AVR are compliance mechanisms to ensure that the number of allowances for each tonne of CO2 emitted are counted and received from operators. While the EUC makes significant effort to reduce the administrative burden of monitoring, reporting and verification they could still represent a barrier to operation for small operators such as AEB when it begins operating under the ETS in 2026. However, the firm is required to have a robust business model and has already taken steps to ensure it is able to operate in 2026.
Accreditation and verification regulation		Direct	Defines the verification rules for the emissions report. Accredited bodies in the Netherlands are then required to verify reports every spring in line with the compliance cycle of the ETS. The verifier in the Netherlands is the NeA (Dutch environmental Agency) and is accredited by the Dutch accreditation Council (RvA).	Stick	N/A	Binding (regulation)	Firm (local)	EU Comission	Organizations (government)	Infrastructure	Immaterial	The AVR requires the verifier to have understanding of the ETS regulation and procedures, have the skills and accreditation to perform the verification and the background knowledge to contextualize the emissions reports.
Renewable Energy Directive	07/06/2022	Direct	The Renewable Energy Directive is the legal framework for the development of clean energy across all sectors of the EU economy, supporting cooperation between EU countries towards this goal. It sets rules and targets to ensure the uptake of renewable energy by member states in all sectors. The directive is currently being revised with an increase in ambition from 40% renewable energy mix by 2030 to 42.5% with 2.5% incentive top up. This revision is more in line with the REpowerEU plan to phase out energy dependence on russia and accelerate the clean energy transition. The revised directive has been voted on and adopted and will enter into force later this year (2023). This directive directly addresses the use of Biomass for energy (electricity and heat), and does support tightening of sustainability requirements aligning with the CAP and LULUCF directive.	Carrot	Both	Binding (Directive)	National & Local	EU Parliament & Council	Organizations (Government)	Institution & Actors	inding and non-binding policy) & Actors	The directive is relevant to Amsterdam's plans on a number of sectors; First it supports the continued uptake of renewable energy and encourages the increase in support for this by member states. This will positively affect the rate of renewable adoption in Amsterdam and the port, including the use of biomass, biofuels and bioliqids for energy. Second, and restricting the first point, it strengthens the sustainability requirements of those fuel types in order to be counted at a renewable source and contribute to meeting the RES targets (Article 6). The commission is however committed to monitoring the commodity prices of biomass and its effects on food security (Article 30). It additionally enshrines a cascading use of biomass requiring use in high-value cases first. The revised directive is not yet deployed and will need to be transposed into national law first to impact the Netherlands. The national government however has already indicated their intention to tighten biomass sustainability requirements for sourcing. Eu
CBAM regulation		Indirect	The Cross Border Adjustment Mechanism is aimed at capturing those emissions that are created from the importation of products of European firms that are sourced from outside the EU. This addresses some of the Scope 2 emissions of the Union and the methodology is developed to be consistent with the 'production based' emissions accounting of the UNFCCC. It will operate similar to the ETS in that CBAM allowances must be purchased for each ton of CO2 released by the imported goods.	Stick	Supply	Binding (regulation)	Intranational	EU Comission	Organizations (Firms)	Infrastructure	Financial	The CBAM will not negatively affect AEB as biomass is not covered by the CBAM.
Waste Framework Directive	2023		The Waste Framework Directive enforces regulations regarding the identification, handling and disposal of waste. It was developed to enshrine principles of cascading waste management and incentivize recycling. The directive additionally sets out binding targets for member states to reach targets for the reuse of material.	Carrot & Stick	Supply	Binding (Directive)	Intranational	EU Parliament & Council	Organizations (Government & firms)	Institution	Hard (Standards)	End-of-waste status for biogenic CO2 would ease handling and use requirements in the EU. It does not negatively affect AEB necessarily, the status of the material is largely determined by the producer, so AEB can make the decision themselves.
EU Green Deal	2019		The EU Green Deal was a communication and proposal package from the EUC to the Parliament and Council of the EU. It outlined the existential imperative to act on the climate issue in the EU and outlined pathways to climate-neutrality.	Sermon	N/A	Non-Binding (Communication)	Intranational	EU Comission	Actors		Various	
Connecting Europe Facility	2014	Direct	The Connecting Europe Facility operates to encourage, support and improve projects with pan-european reach. It is a key funding institution to achieve policy goals of the European Green Deal. The facility specifically funds, infrastructure, transport and digital projects. Funding is available through the facility for the period 2021-2027. The regulation outlines that the facilities purpose "is to accelerate investment in the field of trans-european networks and to leverage funding from both the public and the private sectors, while increasing legal certainty and respecting the principle of technological neutrality." The CEF has three components, energy, transport & digital.	Carrot	Supply	Binding (if funded)	Intranational	EU Comission	Organizations (firms)	Infrastructure	Financial	The CEF could potentially support AEB by supporting T&S projects that AEB would rely on.
Horizon Europe		Indirect	The Horizon Europe is a research, development and innovation funding scheme with a significant budget, perhaps one of the largest of its kind (Negative emissions Platform, 2021). The aim is to support technologies to move from the lab to early-stage demonstration with the goal of large-scale deployment and commercialization. The fund has earmarked 14.7billion Euros in funding for carbon dioxide removal technology under the new Cluster 5 work programme. Climate, energy and mobility. Under the EU Missions projects Horizon Europe aims to drive actors to take up ambitious goals relevant to major issues (climate, health, biodiversity, etc). Of particular relevance is the 100 climate neutral and smart cities mission.	Carrot	Supply	Binding	Intranational	EU Comission	Organizations (various)	Infrastructure	Material & Immaterial	Horizon Europe could support any component of the CCS chain connected to AEB that would support its goals and amsterdam by extension. The only barrier would be if those components cannot find funding through it.
Net-zero Industry Act		Direct	The aim of the NZIA is to increase the competitiveness and resilience of Europe's net-zero technology industrial base. The act was set out in response to the US Inflation reduction act which has dramatically increased investment in green technology in the US. In addition to other green technologies, the NZIA will set out binding regulation for the development of net-zero technologies in memberstates to address its historic emissions. This includes renewable energy, energy storage, a revision to the EU's electricity grid,sustainable biofuel and crucially CCS and usage technologies. Critically it outlines an EU-level objective for the annual injection an storage capacity for 50MCO2 in 2030, with aims to reach annual storage of 550MCO2 in 2050. Member states will be required to perform exploration and publish data regarding potential domestic storage sites. In addition, report on a yearly basis progress made on CCS projects in their territory. The regulation will include measures mechanisms to accelerate capacity development including research, funding, and the possibility of a Regulatory sandbox to test innovative net-zero technologies.	Carrot & Stick	Supply & Demand	Binding	Intranational	EU	Organizations (various)	Infrastructure	Material & Immaterial	The NZIA represents a significant stance change from the EU towards CCS. It would likely be a significant benefit to AEB. Gaps and barriers are difficult to determine at this point as the act is not yet in-force.

C.2: Dutch Policy

Policy Instrument	Latest in-force document	Direct or indirect?	Primary purpose	Approach	Market approach	Legal status	Deployed scale (institution)	Deploying institution	Target actor	Sector	cont'd	Potential barriers:
National energy and climate plan (NCEP)		Indirect	The National energy and climate plan (NCEP) is part of the monitoring and reporting procedure under the european unions Governance of the energy union and climate action regulation. The netherlands has communicated in this document it's strategy for reaching climate-neutrality in-line with the EU's Climate Law. This document sets out the policies and measures to fulfil the climate plan under the Climate Act. A significant part of this strategy is addressing emission in buildings, mobility and industry. CCS is explicitly mentioned as a potentially substantial part of the strategy.	Carrot (in relation to CO ₂)	Both	Binding	National	Ministry of economic affairs and climate	Organizations & individuals	All		There are many potential barriers in the NCEP that might negatively impact the activity of CO ₂ firms in Amsterdam, though there are also many mechanisms for support that are included. I detail specific potential pitfalls later when discussing the individual mechanisms.
Environment and Planning Act		Indirect	To outline the duties and powers of the municipalities, provincial gov, ministers and state of the environment within the Netherlands. The handling and management of waste is also regulated by this legislation. The revision of the act is meant to reduce complexity in the legislative environment and provide more power to municipalities to permit sustainable energy projects. It will draw together several decree's of the dutch government on environmental planning.	N/A	N/A	Non-binding (Primary)	National (Dutch Gov)	Ministry of the Interior and Kingdom relations	Municipalities, provinces, National ('state).	Infrastructure	Immaterial (Administrative: technical)	The act outlines what rights and duties municipalities have in regards to managing the environment in their jurisdiction. The only powers afforded to municipalities to address CC adaptation/mitigation are the ability to 1) to voluntarily set ambitious 'values' for the env 2) Define env permitting requirements for certain activities and landuses (requires committee approval) if they fall within the municipality 3)Define rules for activities that impact the env (by order of council) 4)env costs on applicants of environmental permits. However, it falls to provincial gov to manage land use questions that fall across multiple municipalities. This law decentralizes more actions from the state level to municipal by principle of subsidiarity without comparable increasing of immaterial and financial infrastructure.
Mining Act (Mijnbouwwet)	2011		This act outlines the rules applicable to the exploration and extraction of minerals (including hydrocarbons) and the development of mining-related activities. It establishes that minerals are the property of the state and can only be explored, produced, or stored with a license. It additionally covers the capture, transport and storage of carbon pursuant to the CCS directive.	Stick	Supply				Operators of mining, excavation and other industries operating under dutch sovereign soil. (Firms)	Infrastructure	Immaterial (Operations: technical) & Material (R,S,T)	The act outlines rules and requirements to operate activities including drilling, extraction and storage of material in the earth. Firms are required to obtain environmental permits, and a storage permit. There are also ruling on safe operating limits, and what areas are off limits to any mining activities (biocorridor fold). This Act creates limits to CO ₂ storage in NL due to public pressure. This act affects capture, transport and CO ₂ storage and so CCS technologies as it defines storage as 'storage of substances' >100m deep. Legislators have identified that additional legislation will be needed to address gaps such as preconditions for CCS, financial commitments and liability for damages (van der Wielen, n.d.). It is unclear if these gaps have been addressed). The ministry of economic affairs and climate is the sole competent authority able to give permits for CO ₂ production and storage.
Bio-mass Firing Strategy	2021		This strategy outlines the Netherlands policy on use of Biomass in it's economy for certain activities. Due to questions about the sustainability of biomass, in particular wood pellets, the Dutch government indicated that it would consider biomass' future place in the economy. After discussions the government communicated that despite the controversy over wood pellets, biomass and other bio-fuels will continue to play a significant part in the renewable energy strategy of the country. Though it was indicated that tighter sustainability requirements would be put into place, which is in line with changes to the Renewable Energy Directive and LULUCF directive.	N/A	Supply	Non-binding (strategy)	National (Dutch Gov)	Cabinet of the government	Organizations (firms)	Infrastructure	(Immaterial) Removal & Energy	The policy outlines the role of Biomass to reaching climate neutrality in 2050, which defines a high value for the material. This strategy will have implications for firms that are firing biomass for BECCS. It may mean a higher price of biomass due to the sustainability requirements that biomass imports must follow, though that is largely dependant on the market and may not lead to higher prices if suppliers can fill this gap with biomass that fits the EU's Renewable energy directive.
SDE++			The fund for sustainable energy development (SDE++) was established to provide financial support to experiment of and scaling up of innovative projects in sustainable energy and industry. It is meant to cover the 'losses' from investing in emissions reduction measures. It has recently been expanded in scope to focus on a climate target (50% reduction of GHG), and so has its funding (from the ODE).	Carrot (in relation to CO ₂)	Supply	Binding (economic)	National (firms)	Netherlands Enterprise Agency (RVO)	Organizations (firms)	Infrastructure	Immaterial - Finance	A large percentage of the ODE funds which are used in the SDE++ will come from industry. Actors in the third and fourth tax bracket will pay two thirds towards that effect. None of the firms in Amsterdam working with CO ₂ are likely to do so however. The SDE++ however has been beneficial in the investment for AEB, Portico, ABAMIS and OCCAP, which are critical to Dutch CCS.

Multi-annual Mission-driven innovation program (MMIP)	The mission-driven topsector innovation program (MTIP) is a national government direct topsector innovation program designed to unite innovation efforts around clear societal goals. It covers 13 goals including biomass use, industrial heating and landuse that contribute to national climate goals. Working with networks, research organisations and firms the dutch government aims to create coordination and alignment mechanisms to drive innovation towards the 13 goals. Carbon dioxide removal and CCS are covered by MMIPs 6,7, and 8.	Carrot	Supply & Demand	Under the MMIPs, funding was made available to actors who signed a Knowledge and Innovation Contract. A portion of the funding would then be covered by public sources with the rest being source from private companies. Coordination and networking mechanisms have also been setup to facilitate organisations to make the best use of available instruments.	The MTIP program is aimed at guiding innovation, the products of which may be applied to supply and or demand within the economy.	Aspects are Binding, though the program is not enshrined in the legal system	National	Dutch Government	As a mission-driven innovation program, the aim is to coordinate, connect and support research and development in innovate sectors (around specific goals) by actors including universities, firms, knowledge networks. The average citizen is not impacted by this program.	The aim of the program is to drive innovation in specific sectors towards 13 societal goals. By stimulating the development of knowledge and skills towards new processes and technologies and the coordinated dissemination the MTIP will positively impact those goals.	Several categories	No potential barriers are expected of this program. Any innovation towards the circularity of processes and materials in CCS is a positive benefit. This includes the possibility that CCS becomes unnesecary, though not likely, is possible.
Carbon Levy	To incentivise reduction of emissions in the industrial and waste incineration sectors by setting a progressively rising tax on carbon emissions from fossil sources. A cost of 20 euro's per tonne of CO2 will be applied starting in 2021 and reach 125-150 euros in 2030, for firms this will be taxed on top of the allowances they must pay for to the ETS. The tax is intended to doubly incentivizes companies to reduce emissions quickly and stay within the 10% carbon efficient companies in the EU. Thus companies in the netherlands will avoid paying the higher costs of the ETS allowances overall. A recalibration of the system will be requested in 2025 to adjust benchmarks. Significant effort will be expended by the government to ensure that businesses can avoid the levy through investment in measures to reduce emissions, speeding up permitting processes and providing subsidies in a timely manner.	Stick	Supply	As this policy tax's carbon emissions in similar fashion to the ETS, it is intended as a disincentive to the use of fossil fuels by industry and waste incineration plants.	This policy targets industry (supply side).	Binding	Firm (regional)	The Dutch Government	Industrial firms are the target of this policy, in particular the "big twelve" that represent the majority of industrial emissioned tied to the Netherlands. In addition, waste incineration plants are named due to their importance to municipal energy and heat generation.	By place a carbon tax on national actors aligned with the ETS (including waste-incineration) the policy is intended to further incentivize industrial actors, in particular the big 12, to reduce emissions such that they are within the top 10% of carbon efficient plants. Operators are thus incentivized to invest in carbon efficient materials and processes.	Material by organizations	Waste-incineration plants are not required under the ETS to purchase allowances equal to CO2 emissions over the baseline until 2026. In addition they will not be measured to create a baseline until 2025 when the ETS will re-evaluate baselines for all actors (including aviation and maritime). As a result, there is no baseline for waste-incineration plants to be measured against and so will continue to pay the levy without option to perform. A exchange option for allowances is also planned but not elaborated. This presents a potential barrier if the same issues with the ETS if an overabundance of allocations is created, reducing the efficacy of the levy.
Demonstration of Energy and Climate Innovation Scheme (DEI+)	The DEI+ is a funding scheme for innovation and demonstration projects for energy and climate technology. It can be applied to project on including crular economy, CCSU, local infrastructure, Emissions reductions in industry. Though the scheme is temporarily closed.	Carrot	Supply									Potential restrictions to small waste incineration plants as covered by the
https://www.oecd-ilibrary.org/sites/5d01a32d-en/index.html?itemid=/content/component/5d01a32d-en&lang=en&tablegrp-d1e8342												
Wet Milieubeheer (Environment management law)	This law outlines management rules and regulation for the environment and transposes EU laws regarding waste (WFD). Contains regulations on end-of-waste status criteria and procedures. Regulations are enshrined in the Environmental management law. This law will be incorporated into the environment planning law when it comes into force.	Stick	Supply & Demand	Regulations on management are responsibilities for firms and municipalities. Compliance is enforced by national government.	Rules regarding waste management will eventually influence how consumers interact with products, which may lead to changes in consumption patterns.	Binding	National	Dutch government (ministry of interior and kingdom relations)	The law outlines powers and responsibilities for regional and local government regarding the management of the environment including planning, environmental impact from projects and waste.		Hard (standards and procedures)	The act outlines what rights and duties municipalities have in regards to managing the environment in their jurisdiction. The only powers afforded to municipalities to address CC adaptation/mitigation are the ability to 1) to voluntarily set ambitious 'values' for the env 2) Define env permitting requirements for certain activities and landuses (requires committee approval) if they fall within the municipality 3)Define rules for activities that impact the env (by order of council) 4)levy costs on applicants of environmental permits. However, it falls to provincial gov to manage land use questions that fall across multiple municipalities. This law decentralizes more actions fromt he state level to municipal by principle of subsidiarity without comparable increasing of immaterial and financial infrastructure.
Netherlands CAP strategy	To outline the Netherlands implementation of the policies to reach targets under the EU common agriculture practice policy. The report outlines a vision for action on economic stability and fair pay, sustainable practices, regional support, animal rights, and research and innovation. A total of 4.9 billion euros are earmarked by the EU and Netherlands towards those actions, 79% of which will in some way go towards environmental objectives.	Carrot	Supply			Binding (strategy approved by EUC)	Local & Regional	Dutch government & EUC	Various	Infrastructure & Actors	Immaterial / Various actors	The CAP strategy of the netherlands covers job creation, subsidies for agriculture, research and innovation and sustainability targets.
Environmental Investment Allowance (MIA)	The environmental investment allowance allows firms and individuals to deduct sustainability investments from taxes. Up to 45% of the cost can be deducted based on criteria including firm size, risk, and innovativeness of purchase. The MIA is state aid, and there is a cap on state aid for all actors.	Carrot	Supply & Demand	The fund incentivises sustainable investments by reducing costs to the firm	This fund support firms and individuals alike.	Binding	National	Dutch gov (ministry of economic affairs and climate)	Various (organizations & actors)	Infrastructure	Material	This funding mechanism applies to companies within the netherlands and associated countries. The rebate system covers CCS and CCU, supporting AEB.

C.3: Amsterdam policy

Policy Instrument	Latest in-force document	Direct or indirect?	Primary purpose	Approach type	Market approach	Legal status (binding / non-binding)	Deployed scale (institution)	Deploying institution	Target actor	Sphere of effect (specific)	Barriers: Real and potential
Climate action plan	2022	Direct	The purpose of this document is to outline the municipality of Amsterdam's vision for reaching climate-neutral in 2050 to address the dual disasters of rapid climate change and biodiversity loss. To this effect, the city has outlined strategies that it aims to pursue on a number of sectors including housing, mobility and the Port of Amsterdam. The document was published in 2022 by the Municipality coinciding with the launch of its sustainable amsterdam website.	Sermon - Carrot	Demand	Non-Binding	Municipal (local)	Municipal of Amsterdam	Various (organization & individual) Many of the strategies and plans in the action plan will affect businesses, public organizations and residents of the city.	Infrastructure Both material infrastructure (housing, bike paths, parks) and immaterial infrastructure (permit requirements, awareness campaigns) are targeted by the climate action plan.	There are no barriers to CDR from the climate action plan. In fact the plan explicitly vouches support for AEB and other CCU firms in the port of Amsterdam in legislative discussions with National and EU bodies. The only potential barrier is the goal to reduce use of Biomass for electricity and heating completely, which will impact AEB's business model. Though the municipality has no control over its use and can only advocate for changing national and Eu legislation.
Circular economy strategy		Indirect	The purpose of this document is to outline the municipality of Amsterdam's vision for reducing the cities impact on the climate by way of materials imported into the city, or Scope 3 emissions. These emission do not play a part in the climate action plan as they are not considered Amsterdam's emissions by the UNFCCC's production-based guidelines for emissions accounting. The city is still committed to reducing this inline with circular economy initiatives at the National and EU level.	Carrot	Demand	Non-Binding	Municipal (local)	Municipal of Amsterdam	Various (organization & individual) Many of the strategies and plans in the action plan will affect businesses, public organizations and residents of the city.	Infrastructure The circular economy aims to reduce waste from the city and encourage a circular economy within amsterdam.	No barriers. The success of this strategy can only be positive for CCS & CCU. Although if biomass waste from the city is significantly reduced it may cut into AEB's operation, although not significantly. AEB sources most of its biomass from low-grade woody biomass.
Port Vision Amsterdam 2020-2040		Direct	The purpose of this document is to ouline strategic vision for industry in the port of amsterdam including development of the hydrogen economy, reduction in emissions from maritime and industry, and targets overall.	N/A	None	Non-Binding	Municipal (local)	Port of Amsterdam	Organization (firms)	Infrastructure The strategy for the port targets material infrastructure.	The strategy from the port of amsterdam is in support of AEB and includes implied support for the firm.
Climate Fund (AKEF & Sustainability fund)		Indirect	The climate fund was developed to provide a financial mechanism for encouraging action in the 16 pillars towards climate-neutrality. It represents 150million euros in funding. Five percent of the fund has been earmarked for the harbor and industry. Two funds exist: the Amsterdam Climate & energy fund and sustainability fund. The AKEF supports larger scale commercial projects in energy efficiency and sustainable energy production. The sustainability fund provides small low-risk, non-commercial loans mostly to residents.	Carrot	Supply & Demand	Binding	Municipal (local)	E3 & Municipality of Amsterdam	Commercial buisnesses and residents	Infrastructure The fund supports investment in material infrastructure for energy efficiency, energy production and other sustainability items.	The funds are not earmarked for large scale commercial projects in heat production such as AEB.
Association of Netherlands Municipalities (VNG)		Direct	"The Association of Dutch Municipalities (VNG) is the advocacy organization and knowledge platform for all Dutch municipalities. The goal of the VNG is to unite the strengths of municipalities and collectively act in the interest of local government and its residents. We achieve this by exerting influence at the national level and placing societal challenges at the forefront. Together, we ensure a strong and resilient local government."	none	None	Non-Binding	national	VNG	Municipality of Amsterdam	Actor This is a network that amsterdam uses as a platform to discuss issues with other municipalities and lobby to the national government.	If the concensus from other municipalities is that funding from the national gov should be divested from industry to municipalities it may negatively affect AEB's plans. However, this is not likely given Amsterdam's influence as a G4 municipality.